

Generalities on group schemes

SGA 3, Exposé VIB

English draft

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About this draft

This is an unofficial English translation of Exposé VIB, “Generalities on group schemes” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF Exp6B-13oct24.pdf (112 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

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Exposé VIB. Generalities on group schemes

by J.-E. Bertin

⁽⁰⁾This exposé, which does not correspond to any oral lecture of the seminar, is intended to bring together a number of commonly used technical results concerning group schemes.*

1. Morphisms of groups locally of finite type over a field

1.1. — Let A be an artinian local ring, G and H two A -groups⁽²⁾ and $u : G \rightarrow H$ a morphism of A -groups. Then u induces a morphism of groups $u(A) : G(A) \rightarrow H(A)$.⁽³⁾ Since $H(A)$ acts on H by right translation, $u(A)$ defines by restriction an action of $G(A)$ on H . This action is compatible with the morphism u and the action of $G(A)$ on G defined by right translation. Since $G(A)$ acts transitively on the strictly rational points of G (see VIA 0.4 for the definition), one sees that these points “all behave in the same way with respect to u ”; from this spring the following properties:

Proposition 1.2. *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type over A . Then the set $u(G)$ is closed in H and one has $\dim G = \dim u(G) + \dim \text{Ker } u$.⁽⁴⁾*

Since u commutes with the inversion morphisms of G and H , the image $u(G)$ is invariant under the inversion morphism of H ; the same therefore holds for $\overline{u(G)}$, its closure in H . On the other hand, let L be the set of points of $H \times_A H$ whose two projections belong to $u(G)$; it is clear that L is the image of the morphism $u \times_A u : G \times_A G \rightarrow H \times_A H$; thus the multiplication morphism of H sends L into $u(G)$, in other words $u(G) \cdot u(G) = u(G)$. On the other hand, Lemma 1.2.1 below shows that $\overline{L}$, the closure of L in $H \times_A H$, is the set of points of $H \times_A H$ whose two projections belong to $\overline{u(G)}$; thus $\overline{u(G)} \cdot \overline{u(G)} = \overline{u(G)}$, so that the reduced subscheme of H whose underlying space is $\overline{u(G)}$ is naturally endowed with a group structure in the category $(\text{Sch}/k)_{\text{red}}$, where k is the residue field of A (cf. VIA 0.2).

Let us prove the first assertion of 1.2: replacing A by the algebraic closure of its residue field k if necessary, we may suppose that A is an algebraically closed field k (cf. EGA IV₂, 2.3.12). Replacing u by $u_{\text{red}} : G_{\text{red}} \rightarrow H_{\text{red}}$ if necessary, one may suppose G and H reduced; in this case, as we have just seen, $u(G)$ is the underlying space of a reduced subgroup scheme of G ; we may therefore suppose u dominant. Then $G(k)$ acts transitively on the set of connected components of H and it suffices to show that $u(G) \cap H^0$ is closed: one is reduced to the case where H is connected, hence irreducible and of finite type (VIA 2.4). Then u is of finite type since it is quasi-compact and locally of finite type; since H is noetherian, $u(G)$

⁽⁰⁾**N.D.E.** Version of 13 October 2024.

*This exposé has been fairly substantially revised since its mimeographed edition; in particular, §§10 and 11 have been entirely rewritten.⁽¹⁾

⁽¹⁾**N.D.E.** And also §5, cf. N.D.E. (34) and (46).

⁽²⁾**N.D.E.** Recall the convention adopted at the beginning of VIA: an A -group is an A -group scheme (and such a scheme is separated, cf. VIA, 0.3).

⁽³⁾**N.D.E.** “Homomorphism” has been replaced by “morphism”.

⁽⁴⁾**N.D.E.** This statement also appears in VIA, 2.5.2.

is constructible (EGA IV₁, 1.8.5), hence contains an open subset V of H (EGA 0_{III}, 9.2.2), and then, by VIA 0.5, one has $H = V \cdot V \subset u(G) \cdot u(G) = u(G)$.

Let us prove the second assertion. First recall that the functor $\text{Ker } u$ (cf. I, 2.3.6.1) is representable by $u^{-1}(e)$, where e denotes the identity element of H . When u is locally of finite type, $\text{Ker } u$ is therefore locally of finite type over A . As before, this time by EGA IV₂, 4.1.4, one reduces to the case where A is an algebraically closed field k . One may moreover suppose G and H irreducible and of finite type and u dominant: indeed, since k is algebraically closed, it is clear that the connected components of G , on whose set $G(k)$ acts transitively, all have the same dimension, and that, if u^0 is the restriction of u to G^0 , then $(\text{Ker } u)^0 \subset \text{Ker } u^0$, and $\dim \text{Ker } u^0 = \dim \text{Ker } u$. One sees similarly that u is then of finite type over k . If η denotes the generic point of H , one has $\dim u^{-1}(\eta) = \dim G - \dim H$ (EGA IV₃, 10.6.1 (ii)). By EGA IV₃, 9.2.3 and 9.2.6, the set of $y \in H$ such that $\dim u^{-1}(y) = \dim u^{-1}(\eta)$ contains a nonempty open subset V . Since u is dominant, $U = u^{-1}(V)$ is then a nonempty open subset of G , and contains a closed point x of G , since G is a Jacobson scheme (EGA IV₃, 10.4.7). Then right translation r_x is an isomorphism of $\text{Ker } u$ onto $u^{-1}(u(x))$, so that:

$$\dim \text{Ker } u = \dim u^{-1}(u(x)) = \dim u^{-1}(\eta) = \dim G - \dim H.$$

Lemma 1.2.1. *Let $f : X' \rightarrow X$ and $g : Y' \rightarrow Y$ be two quasi-compact dominant morphisms of schemes over an artinian local ring A ; then $f \times_A g : X' \times_A Y' \rightarrow X \times_A Y$ is dominant (and quasi-compact).*

Indeed, one has $f \times_A g = (\text{id}_X \times_A g) \circ (f \times_A \text{id}_{Y'})$. It therefore suffices to show that $f \times_A \text{id}_{Y'}$ and $\text{id}_X \times_A g$ are dominant. For this one may replace A by its residue field k . In this case, X and Y' are flat over $A = k$, and since $f \times_A \text{id}_{Y'}$ (resp. $\text{id}_X \times_A g$) is obtained from f (resp. g) by the flat base change $Y' \rightarrow A$ (resp. $X \rightarrow A$), it is dominant (and quasi-compact), by EGA IV₂, 2.3.7.

Counterexample 1.2.2. Let k be a field of characteristic 0, G the constant k -group $\mathbf{Z}$ and H the additive k -group $\mathbf{G}_{a,k}$. Let $u : G \rightarrow H$ be a morphism of k -groups. If $u \neq 0$, $u(G)$ is not closed in H .

Proposition 1.3. ⁽⁵⁾ *Let A be an artinian local ring, k its residue field, G an A -group locally of finite type and flat, and $u : G \rightarrow H$ a morphism of A -groups. The following assertions are equivalent:*

- (i) *u is flat (resp. quasi-finite, resp. unramified, resp. smooth, resp. étale) at a point of G .*⁽⁶⁾
- (ii) *u is flat (resp. quasi-finite, resp. unramified, resp. smooth, étale).*

PROOF. It suffices to show that (i) implies (ii). First one has the following lemma:

Lemma 1.3.0. *Let $A \rightarrow B \rightarrow C$ be morphisms of commutative rings and $\mathfrak{n}$ a nilpotent ideal of A . Suppose that $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$ -algebra of finite type.*

⁽⁵⁾**N.D.E.** In the original, 1.3, 1.3.1 and 1.3.1.1 are stated for A a field. For completeness, the case of an artinian local ring has been treated and, to do so, Lemma 1.3.0 has been added.

⁽⁶⁾**N.D.E.** Recall that a morphism of schemes $f : X \rightarrow Y$ is said to be of finite type (resp. of finite presentation, resp. quasi-finite) at x if there are affine open subsets V containing $f(x)$ and U containing x such that $f(U) \subset V$ and $\mathcal{O}_X(U)$ is an $\mathcal{O}_Y(V)$ -algebra of finite type (resp. of finite presentation, resp. and if moreover the fiber $f^{-1}(f(x'))$ is finite for every $x' \in U$ (see also N.D.E. (40) of Exp. V)). One says that f is locally quasi-finite (resp. of finite type, of finite presentation) if it is so at every point x . On the other hand, one says that f is smooth (resp. unramified, resp. étale) at x if there is an open neighborhood U of x such that the morphism $f|_U : U \rightarrow Y$ is smooth (resp. unramified, resp. étale). In view of these definitions, it is clear that the set of points where f is of finite presentation, resp. of finite type, quasi-finite, smooth, unramified, étale, is an open subset of X . Moreover, since the flatness locus of a morphism locally of finite presentation is open (EGA IV₃, 11.3.1), the set of points of X where f is of finite presentation and flat is also open in X . All this will be used on numerous occasions below.

- (i) *Then C is a B -algebra of finite type.*
- (ii) *Moreover, if C is flat over A and $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$ -algebra of finite presentation, then C is a B -algebra of finite presentation.*

Indeed, let $x_1, \dots, x_n$ be elements of C whose images generate $C/\mathfrak{n}C$ as a $(B/\mathfrak{n})$ -algebra. By the nilpotent Nakayama lemma, the x_i generate C as a B -algebra. This proves (i). Let ϕ be the surjective morphism $B[X_1, \dots, X_n] \longrightarrow C$ thus obtained, and let $I = \text{Ker}(\phi)$.

Now suppose that C is flat over A and $C/\mathfrak{n}C$ is of finite presentation over $B/\mathfrak{n}B$. Then, on the one hand, $I/\mathfrak{n}I$ is identified with the kernel of $\bar{\phi} = \phi \otimes_A (A/\mathfrak{n})$. On the other hand, by EGA IV₁, 1.4.4, the kernel of $\bar{\phi}$ is a finitely generated ideal. Then let $P_1, \dots, P_s$ be elements of I whose images generate $I/\mathfrak{n}I$ as an ideal; by the nilpotent Nakayama lemma, they generate I . This proves (ii).

Let us return to the proof of 1.3. Let x be an arbitrary point of G . Since G is flat over A , by the fiberwise flatness criterion, in the form of EGA IV₃, 11.3.10.2, u will be flat at x if $u \otimes_A k$ is. Similarly, by the preceding lemma, one sees that u will be of finite type (resp. of finite presentation) at x if $u \otimes_A k$ is. Since the other properties can then be checked on the fibers (cf. EGA IV₄ 17.4.1, 17.5.1 and 17.6.1 for unramified, smooth and étale), one is reduced to the case where $A = k$.

Now let x be a point of G where one of the conditions of 1.3 (i) holds. Since the properties under consideration are preserved by (fpqc) descent (cf. EGA IV, 2.5.1, 2.7.1 and 17.7.1), replacing k by an algebraic closure of $\kappa(x)$ reduces one to the case where k is algebraically closed and $x \in G(k)$.

Since G is a Jacobson scheme (cf. EGA IV₃, 10.4.7) and since the set W of points of G where u is flat (resp. quasi-finite, unramified, smooth, or étale) is stable under generization⁽⁷⁾ (resp. open), it suffices to show that every point y of $G(k)$ belongs to W . But, for such a point y , the translation $r_y \circ r_{x^{-1}}$ sends x to y , so u has the desired property at y , i.e. $y \in W$. $\square$

Corollary 1.3.1. *Let A be an artinian local ring, k its residue field, G a flat A -group. The following assertions are equivalent:⁽⁸⁾*

- (i) *G is locally quasi-finite (resp. unramified, resp. smooth, resp. étale) over A at a point.*
- (ii) *G is locally quasi-finite (resp. unramified, resp. smooth, resp. étale) over A .*

⁽⁹⁾Indeed, if G satisfies one of the conditions of (i) at a point x , there is an open neighborhood U of x which is of finite type over A . Consequently, it suffices to apply 1.3 to the case where H is the unit A -group, taking account of the following lemma:

Lemma 1.3.1.1. *Let A be an artinian local ring and G an A -group. If there is a nonempty open subset of G of finite type over A , then G is locally of finite type over A .*

By Lemma 1.3.0, one may suppose A equal to its residue field k . Moreover, by (fpqc) descent, one may suppose k algebraically closed (cf. EGA IV₂, 2.7.1). Let V be the open subset of G formed by the points where G is of finite type over k ; by hypothesis, $V \neq \emptyset$. Since G is a Jacobson scheme, V contains a closed point x and, to show that $V = G$, it suffices to show that every closed point y of G belongs to V . But, for such a point y , the translation $r_y \circ r_{x^{-1}}$ sends x to y , whence $y \in V$.

⁽⁷⁾**N.D.E.** The original indicated that W_{flat} is open, which does not seem evident a priori...

⁽⁸⁾**N.D.E.** Note that it is not sufficient to suppose that G is flat over A at a point: for example, supposing $A \neq k$, let H be the constant A -group $(\mathbf{Z}/2\mathbf{Z})_A$ and G the closed A -subgroup of H whose nonidentity component is reduced; then the structure morphism $G \longrightarrow \text{Spec } A$ is a local isomorphism at the identity point e , but is not flat at the point $g \neq e$.

⁽⁹⁾**N.D.E.** The following sentence has been added, and the proof of Lemma 1.3.1.1 simplified by invoking EGA IV, 2.7.1 instead of loc. cit., 17.7.5.

Corollary 1.3.2. *Let A be an artinian local ring, $u : G \rightarrow H$ a morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is universally open,
- (ii) u is open,
- (iii) u is open at a point of G ,
- (iv) the morphism $u^0 : G^0 \rightarrow H^0$ obtained from u is dominant,
- (iv') u^0 is surjective,
- (v) there is a connected component C of G such that, if D denotes the connected component of H containing $u(C)$, the morphism $u' : C \rightarrow D$ obtained from u is dominant.

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) and (iv') $\Rightarrow$ (iv) $\Rightarrow$ (v) are clear. Since G^0 is of finite type over A (VIA 2.4), hence noetherian, u^0 is quasi-compact, so $u^0(G^0)$ is closed in H^0 , by 1.2, hence (iv) $\Rightarrow$ (iv'). On the other hand, since G^0 (resp. C) is open in G (VIA 2.3) and H^0 (resp. D) is irreducible (VIA 2.4.1), one sees that (ii) implies (iv) (resp. that (iii) implies (v)). It therefore remains to show that (v) implies (i).

The open subset C (resp. D) of G (resp. H) will be endowed with its induced scheme structure, and u' will denote the morphism $C \rightarrow D$ obtained from u . Let k be the residue field of A .⁽¹⁰⁾ Since the base change $\text{Spec } k \rightarrow \text{Spec } A$ is a universal homeomorphism, one may suppose $A = k$. By hypothesis, u' is dominant and, since C is of finite type over k (VIA 2.4.1), hence noetherian, u' is quasi-compact. By EGA IV₂, 2.3.7, $u' \otimes_k \bar{k}$ is still quasi-compact and dominant, where $\bar{k}$ denotes an algebraic closure of k . Then, since $C \otimes_k \bar{k}$ is a union of connected components of $G \otimes_k \bar{k}$, the morphism $u \otimes_k \bar{k} : G \otimes_k \bar{k} \rightarrow H \otimes_k \bar{k}$ satisfies assertion (v). We are thus reduced to the case where $A = k$ is an algebraically closed field, taking account of EGA IV₂, 2.6.4.

In this case, we may moreover replace u by u_{red} , and we are reduced to the case where H is reduced. Then let ξ (resp. η) be the generic point of C (resp. D). Since u' is quasi-compact and dominant, $u'(\xi) = \eta$ (cf. EGA IV₁, 1.1.5). On the other hand, since H is reduced, the local ring $\mathcal{O}_{H,\eta}$ is a field, and hence u' is flat at the point ξ .⁽¹¹⁾ Thus, by 1.3, u is flat; moreover, since u is locally of finite type and H is locally noetherian, u is locally of finite presentation. Hence, by EGA IV₂, 2.4.6, u is universally open.

Proposition 1.4. *Let A be an artinian local ring, and $u : G \rightarrow H$ a quasi-compact morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is proper,
- (ii) there is $h \in H$ such that the fiber $u^{-1}(h)$ is nonempty and proper over $\kappa(h)$,
- (iii) $\text{Ker } u$ is proper over A .

It is clear that (i) implies (iii), and that (iii) implies (ii). On the other hand, it follows from the hypotheses that u is of finite type and, since G is separated (VIA 0.3), so is u (EGA I, 5.5.1). It therefore remains to show that assertion (ii) implies that u is universally closed, so we may suppose A equal to its residue field k .⁽¹²⁾ Let k' be an algebraic closure of $\kappa(h)$, $u' : G' \rightarrow H'$ the morphism obtained from u by base change, h' a point of H' above h ; then the fiber $u'^{-1}(h') = u^{-1}(h) \times_{\kappa(h)} k'$ is nonempty and proper, and it suffices to show that u' is proper (EGA IV₂, 2.6.4). One may therefore suppose that k is algebraically closed and $h \in H(k)$.

We have seen (1.2) that $u(G)$ is then the underlying set of a closed reduced subgroup scheme of H ; since every closed immersion is proper (EGA II, 5.4.2), we may suppose u surjective, and H reduced. Since u is surjective, the group $G(k)$ acts transitively on the set of closed points of H ; thus, for every closed point y of H , $u^{-1}(y)$ is proper over $\kappa(y)$. By EGA

⁽¹⁰⁾**N.D.E.** The original has been expanded in what follows.

⁽¹¹⁾**N.D.E.** The original invoked the generic flatness theorem (EGA IV₂, 6.9.1), which is not needed here.

⁽¹²⁾**N.D.E.** The following sentence has been added.

IV₃, 9.6.1, the set of $y \in H$ such that $u^{-1}(y)$ is not proper over $\kappa(y)$ is locally constructible; since it contains no closed point, it is empty (cf. EGA IV₃, 10.3.1 and 10.4.7).

⁽¹³⁾Then consider the generic point η of H^0 ; by the preceding argument, the fiber $u^{-1}(\eta) = G \times_H \text{Spec } \kappa(\eta)$ is proper over $\kappa(\eta)$. On the other hand, since H is reduced, $\kappa(\eta)$ equals $\mathcal{O}_{H,\eta}$. Since $\mathcal{O}_{H,\eta}$ is the inductive limit of the rings $\mathcal{O}_H(V)$, as V ranges over the nonempty open subsets of H^0 , it follows from EGA IV₃, 8.1.2 a) and 8.10.5 (xii), that there is a nonempty open subset V of H^0 such that the restriction of u above the open subset V is proper. It is then clear that the $g \cdot V$, for $g \in G(k)$, form an open covering of H such that, for every $g \in G(k)$, the restriction of u above the open subset $g \cdot V$ is proper; one deduces that u is proper (cf. EGA II, 5.4.1).

Corollary 1.4.1. *Let A be an artinian local ring, and $u : G \longrightarrow H$ a morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is locally quasi-finite,
- (ii) u is quasi-finite at a point,
- (iii) $\text{Ker } u$ is discrete,
- (iv) the restriction of u to every connected component of G is finite.

Finally, if u is quasi-compact, these assertions are equivalent to the following:

- (v) u is finite.

It is clear that (iv) implies (iii), that (iii) implies (ii) (EGA I, 6.4.4), and that, when u is quasi-compact, assertions (iv) and (v) are equivalent. We have already seen in 1.3 that assertions (i) and (ii) are equivalent.

Finally let us show that (i) implies (iv). Let C be a connected component of G ; since C is of finite type over A (VIA 2.4.1) and G and H are separated (VIA 0.2), by EGA I, 5.5.1 and 6.3.4, the restriction u' of u to C is separated and of finite type.⁽¹⁴⁾ Since the fibers of u' are discrete, it follows that u' is quasi-finite (cf. EGA II, 6.2.2). Since every quasi-finite proper morphism is finite (cf. EGA III₁, 4.4.2), it therefore suffices to show that u' is universally closed.

For this we may suppose that A equals its residue field k . Then, by (fpqc) descent (cf. EGA IV₂, 2.6.4), it suffices to show that $u' \otimes_k \bar{k}$ is universally closed, where $\bar{k}$ denotes an algebraic closure of k . Moreover, since C is of finite type over k , $C \otimes_k \bar{k}$ is the sum of a finite number of connected components $C'_1, \dots, C'_n$ of $G \otimes_k \bar{k}$, and it suffices to show the assertion for each C'_i . One is thus reduced to the case where k is algebraically closed.

Then let g be a closed point of C ; if $u^0 : G^0 \longrightarrow H$ is the restriction of u to G^0 , one has $u' = r_{u(g)} \circ u^0 \circ r_{g^{-1}}$, where r_g denotes right translation by g , and hence, to show that u' is proper, it suffices to show that u^0 is. By hypothesis, u is locally quasi-finite, so the fiber $\text{Ker } u$ is discrete (and nonempty); we have seen that u^0 is of finite type, so the fiber $\text{Ker } u^0$ is finite (cf. EGA II, 6.2.2), hence proper, and nonempty; thus u^0 is proper by 1.4.

Corollary 1.4.2. *Let A be an artinian local ring, and $u : G \longrightarrow H$ a quasi-compact morphism between A -groups locally of finite type. The following assertions are equivalent:*⁽¹⁵⁾

- (i) u is a closed immersion,
- (ii) u is a monomorphism,
- (iii) $\text{Ker } u$ is trivial, i.e. isomorphic to the unit k -group.

⁽¹³⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁴⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁵⁾**N.D.E.** The hypothesis that G and H be locally of finite type can be removed, for, by [Per76, 4.2.4], every quasi-compact monomorphism $u : G \longrightarrow H$ between group schemes over a field k is a closed immersion; see also [DG70, III.3.7.2 b)] for the case where G and H are affine (in which case every morphism $G \longrightarrow H$ is affine (EGA II, 1.6.2 (v)), hence quasi-compact).

In particular, every subgroup scheme⁽¹⁶⁾ of H is closed.

It is clear that (i) implies (ii), and, considering the functors represented by G and H respectively, it is immediate that conditions (ii) and (iii) are equivalent. Finally, if $\text{Ker } u$ is the unit k -group, $\text{Ker } u$ is a nonempty proper fiber, so u is a proper monomorphism, by 1.4, and is of finite presentation since H is locally noetherian (EGA IV₁, 1.6.1), hence is a closed immersion (EGA IV₃, 8.11.5).

The last assertion follows from the fact that, since H is locally noetherian, every immersion $G \rightarrow H$ is quasi-compact (EGA I, 6.6.4).

Counterexample 1.4.3. Let k be a field of characteristic 0, G the constant k -group $\mathbf{Z}$ and H the k -group $\mathbf{G}_{a,k}$. Let $u : G \rightarrow H$ be a morphism of k -groups. If $u \neq 0$, $\text{Ker } u = 0$, but u is not a closed immersion (cf. 1.2.2).

We shall use later the following two results which should have appeared in Exposé VIA:

Lemma 1.5. *Let k be a field and G a k -group locally of finite type. Then:*

- (i) *All the irreducible components of G have the same dimension.*
- (ii) *For every $g \in G$, one has $\dim_g G = \dim G$.*

⁽¹⁷⁾Assertion (i) has already been proved in VIA, 2.4.1, and assertion (ii) follows from it. Indeed, let g be a point of G and C the connected component of G containing g . By definition (EGA 0_{IV}, 14.1.2), $\dim_g G$ is the infimum of the integers $\dim U$, as U ranges over the open neighborhoods of g ; one therefore has $\dim_g G = \dim U_0$ for some U_0 , which one may suppose contained in C (since $\dim V \leq \dim U$ if $V \subset U$). Then, since C is irreducible and of finite type over k (VIA, 2.4.1), one has $\dim U_0 = \dim C = \text{tr. deg}_k \kappa(\xi)$, where ξ is the generic point of C , by EGA IV₂, 5.2.1. Thus $\dim_g G = \dim C = \dim G$.

Proposition 1.6. ⁽¹⁸⁾ *Let S be a scheme of characteristic zero and G an S -group scheme, locally of finite presentation over S at every point of the unit section $\varepsilon(S)$. For G to be smooth over S at every point of the unit section, it is necessary and sufficient that the $\mathcal{O}_S$ -module $\omega_{G/S} = \varepsilon^*(\Omega_{G/S}^1)$ (called the conormal module of the unit section of G) be locally free.*

Recall that a scheme S is said to be of characteristic zero if, for every closed point x of S , the field $\kappa(x)$ is of characteristic zero.

Also recall (II 4.11) that, if π denotes the structure morphism $G \rightarrow S$, one has $\Omega_{G/S}^1 = \pi^*(\omega_{G/S})$, so that it is equivalent to say that the $\mathcal{O}_S$ -module $\omega_{G/S}$ is locally free or that the $\mathcal{O}_G$ -module $\Omega_{G/S}^1$ is locally free.

If there is an open neighborhood U of $\varepsilon(S)$ smooth over S , then, by EGA IV₄, 17.2.3, $\Omega_{U/S}^1$ is locally free of finite type, as is $\omega_{G/S} = \varepsilon^*(\Omega_{U/S}^1)$.

Conversely, if $\omega_{G/S}$ is locally free, so is $\Omega_{G/S}^1 = \pi^*(\omega_{G/S})$. Since S is of characteristic 0, the Jacobian criterion (EGA IV₄, 16.12.2) therefore implies that G is differentially smooth over S . Then it follows from EGA IV₄, 17.12.5, that G is smooth over S at every point of the unit section.

Corollary 1.6.1 (Cartier). *Given a field k of characteristic zero, every k -group locally of finite type over k is smooth over k .*

⁽¹⁶⁾**N.D.E.** In this particular case, see also VIA, 0.5.2, valid without finiteness hypotheses.

⁽¹⁷⁾**N.D.E.** The implication (ii) $\Rightarrow$ (i) is a general fact (cf. EGA 0_{IV}, 14.1.6), and (i) $\Rightarrow$ (ii) follows from the fact that, if X is an irreducible scheme of finite type over a field, one has $\dim X = \dim U$ for every nonempty open subset U of X ; the essential point here is therefore to establish assertion (i), which has already been done in an addition to VIA, 2.4.1. The statement and proof of the lemma have been modified accordingly.

⁽¹⁸⁾**N.D.E.** In the statement, “along the unit section” has been replaced by “at every point of the unit section”; on the other hand, at the end of the proof, the results of EGA IV₄ cited as references have been made explicit.

Indeed, it is then clear that the k -module $\omega_{G/k}$ is locally free, so, by 1.6, G is smooth over k at the identity point e , hence smooth over k , by 1.3.1.

2. “Open properties” of groups and morphisms of groups locally of finite presentation

2.0. — Throughout what follows, S will denote any scheme; an S -group scheme will be called an S -group. Given an S -group G , we shall denote by ε the unit section, by $c : G \rightarrow G$ the inversion morphism and by μ the multiplication morphism $G \times_S G \rightarrow G$. For every S -scheme X , we shall denote by π or π_X the structure morphism $X \rightarrow S$.

Given a property $\mathcal{P}(u)$ for a morphism of S -schemes $u : X \rightarrow Y$, we shall say that $\mathcal{P}(u)$ is stable under base change if, whenever u satisfies $\mathcal{P}(u)$, the same holds for the morphism $u_{(Y')}$, for every S -morphism $Y' \rightarrow Y$. One says that $\mathcal{P}(u)$ is local for the topology $\mathcal{T}$ (cf. Exp. IV, §§4 and 6) if $\mathcal{P}(u)$ satisfies the following two conditions:

- (a) $\mathcal{P}(u)$ is stable under base change,
- (b) whenever there is a family of S -morphisms $Y_i \rightarrow Y$ covering for the topology $\mathcal{T}$ and such that each of the morphisms $u_{(Y_i)}$ satisfies $\mathcal{P}(u)$, then u satisfies $\mathcal{P}(u)$.

Proposition 2.1. *Let $\mathcal{P}(u)$ be a property for a morphism of S -schemes, local for the (fpqc) topology. Let $u : G \rightarrow H$ be a morphism of S -groups. Suppose G flat and universally open over S .*

Let W be the largest open subset of H above which u satisfies the property $\mathcal{P}(u)$ and let $V = u^{-1}(W)$. Then $U = \pi_G(V)$ is an open subset of S and V is an open subgroup scheme of $G|_U$.

The existence of a largest open subset W of H above which u satisfies $\mathcal{P}(u)$ follows from the fact that $\mathcal{P}(u)$ is local for the Zariski topology. Since π_G is universally open, $\pi_G(V)$ is an open subset U of S . It suffices to show that V is a subgroup scheme of $G|_U$. We may therefore suppose that $U = S$.

Then put $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$, $W' = W \times_S V$ and $u' = u_{(V)}$; let W'_1 be the largest open subset of H' above which u' satisfies $\mathcal{P}(u)$; since V is flat and universally open over S , the same holds for H' over H , and Lemma 2.1.1 below shows that $W'_1 = W'$. Then consider the automorphism of V -schemes a (resp. b) of G' (resp. H'), right translation by the inverse of the diagonal section δ (resp. by the inverse of $u(\delta)$), defined by

$$a(g, v) = (g \cdot v^{-1}, v), \quad (\text{resp. } (h, v) = (h \cdot u(v^{-1}), v)),$$

for every morphism $T \rightarrow S$, $g \in G(T)$, $v \in V(T)$ and $h \in H(T)$. It is clear that $u' \circ a = b \circ u'$, which shows that W' is stable under b , hence that V' is stable under a , so that V is a subgroup scheme of G .

Lemma 2.1.1. *Let $\mathcal{P}(u)$ be a property for an S -morphism u , local for the (fpqc) topology. Consider a cartesian square of morphisms of S -schemes:*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Y \end{array},$$

where g is flat and open. Let W (resp. W'_1) be the largest open subset of Y (resp. Y') above which f (resp. $f' = f_{(Y')}$) satisfies $\mathcal{P}(u)$. Then $W'_1 = W \times_Y Y'$.

Put $W' = W \times_Y Y'$; since $\mathcal{P}(u)$ is stable under base change, one has $W' \subset W'_1$. Since g is open, $W_1 = g(W'_1)$ is an open subset of Y . Put $V_1 = f^{-1}(W_1)$ and $V'_1 = V_1 \times_{W_1} W'_1$; it is clear that $V'_1 = f'^{-1}(W'_1)$. Since g is flat and open, the morphism $W'_1 \rightarrow W_1 = g(W'_1)$ obtained from g is faithfully flat and open, hence covering for the (fpqc) topology (cf. IV 6.3.1

(iv)). Since the morphism $V'_1 \rightarrow W'_1$ obtained from f' satisfies $\mathcal{P}(u)$, the same holds for the morphism $V_1 \rightarrow W_1$ obtained from f , hence $W_1 \subset W$, and $W'_1 \subset g^{-1}(W_1) \subset g^{-1}(W) = W'$; thus $W' = W'_1$.

Remark 2.1.2. A great many properties for a morphism are local for the (fpqc) topology; let us mention those of being flat, (universally) open, (locally) of finite type, of finite presentation, quasi-finite (cf. EGA IV₂, 2.5.1, 2.6.1 and 2.7.1), smooth, étale, unramified (EGA IV₄, 17.7.3).

The proof of 2.1 in fact uses only base changes by flat morphisms; the proposition will therefore apply to a property satisfying condition b) of 2.0 for the (fpqc) topology, and stable under base changes by flat morphisms (example: that of being quasi-compact and dominant).

Of course, one may state an analogous proposition concerning properties local for a topology $\mathcal{T}$ finer than the Zariski topology, the condition to be checked on G then being that π_G is universally open and covering for the topology $\mathcal{T}$.

In particular, if G is flat and locally of finite presentation over S , one has an analogous statement for properties stable under base changes by flat morphisms locally of finite presentation, and satisfying condition b) of 2.0 for the (fpqc) topology (for example, those of being regular, reduced, Cohen–Macaulay, etc. (EGA IV₂, 6.8)).

Proposition 2.2. *Let G and H be two S -groups and $u : G \rightarrow H$ a morphism of S -groups. Then.⁽¹⁹⁾*

- (i) *Suppose G or H flat over S , and G or H locally of finite presentation over S , and let V be the largest open subset of G such that the restriction of u to V is flat and locally of finite presentation (resp. smooth, resp. étale). Then $U = \pi_V(V)$ is an open subset of S , and V is an open subgroup scheme of $G|_U$.*
- (ii) *Suppose G or H universally open over S , and let V be the largest open subset of G such that the restriction of u to V is universally open. Then $U = \pi_V(V)$ is an open subset of S , and V is an open subgroup scheme of $G|_U$.*

Let us first prove (i). Let us show that the restriction π_V of π_G to V is flat and locally of finite presentation:

- (a) If π_G is flat (resp. locally of finite presentation), so is π_V .
- (b) If π_H is flat (resp. locally of finite presentation), so is π_V , as the composite of the restriction of u to V and π_H .

Thus, in the four cases considered in the statement, π_V is flat and locally of finite presentation, hence universally open (EGA IV₂, 2.4.6). Put $U = \pi_V(V)$; U is therefore an open subset of S . It then suffices to show that V is an open subgroup scheme of $G|_U$; we may therefore suppose that $U = S$.

Then put $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$ and $u' = u|_{V'}$. Then, since V is flat and locally of finite presentation over S , the same holds for H' over H . By EGA IV₄, 17.7.4, V' is then the largest open subset of G' such that the restriction of u' to V' is flat and locally of finite presentation (resp. smooth, resp. étale). With the automorphisms a and b defined as in the proof of 2.1, it is then clear that V' is stable under a , hence that V is a subgroup scheme of G .

Let us prove (ii). The restriction π_V of π_G to V is a universally open morphism, either because this holds for π_G , or as the composite of the restriction of u to V and π_H in the case where π_H is universally open. Put $U = \pi_V(V)$; U is then an open subset of S . It suffices to show that V is an open subgroup scheme of $G|_U$. We may therefore suppose that $U = S$.

⁽¹⁹⁾**N.D.E.** In case (i), the open subset V consists of all the points of G at which u is smooth, resp. étale, resp. of finite presentation and flat, cf. N.D.E. (6). On the other hand, in case (ii), V is the largest open subset contained in the set E of points of G at which u is universally open, but E is not necessarily open, as the following example shows (EGA IV₃, 14.1.3 (i)): let k be a field, $H = S = \text{Spec } A$, where $A = k[x]$, and G the S -group $\text{Spec } A[y]/(xy)$; then E equals the unit section of G , which is not open.

As before put $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$ and $u' = u_{(V)}$. Then $\pi_V : V \rightarrow S$ is surjective and universally open, and so is $G' \rightarrow G$, so that V' is the largest open subset of G' such that the restriction of u' to V' is universally open, by (EGA IV₃, 14.3.4 (i) and (ii)). With the automorphisms a and b defined as before, it is then clear that V' is stable under a , hence that V is a subgroup scheme of G .

Corollary 2.3. *Let G be an S -group and V the largest open subset of G flat and locally of finite presentation (resp. smooth, étale, universally open) over S . Then $U = \pi_V(V)$ is an open subset of S , and V is an open subgroup scheme of $G|_U$.*

It suffices to apply 2.2 to the case where H is the unit S -group and u is the unique morphism of S -groups $G \rightarrow H$, for then π_H is an isomorphism and $\pi_G = \pi_H \circ u$.

Corollary 2.4. *Let G be an S -group; if there is a neighborhood X of the unit section having one (or several) of the following properties:*

X is flat and locally of finite presentation (resp. smooth, étale, universally open) over S ,

then there is an open subgroup scheme of G having the same properties.

It suffices to apply 2.3, observing that here, with the notation of 2.2, one has $\varepsilon(S) \subset V$, hence $U = S$.

Proposition 2.5. ⁽²⁰⁾ *Let $u : G \rightarrow H$ be a morphism of S -groups.*

- (i) *Suppose that G (resp. H) is of finite presentation and flat (resp. of finite type) over S at the points of its unit section. Then the sets*

$$U_{\text{flat}} \supset U_{\text{smooth}} \supset U_{\text{ét.}},$$

formed by the points $s \in S$ such that u_s is flat (resp. smooth, étale), are open in S .

If moreover G (resp. H) is locally of finite presentation and flat (resp. locally of finite type) over S , then the set V_{flat} (resp. V_{smooth} , $V_{\text{ét.}}$) of points of G where u is flat (resp. smooth, resp. étale) is the inverse image under π_G of U_{flat} (resp. U_{smooth} , $U_{\text{ét.}}$).

- (ii) *Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$ (a condition satisfied if G is of finite type over S at the points of the unit section (1.3.1.1)), and that u is locally of finite type (resp. locally of finite presentation) at the points of the unit section of G . Then the sets*

$$U_{\text{l.q.f.}} \supset U_{\text{unr.}}$$

formed by the $s \in S$ such that u_s is locally quasi-finite (resp. unramified) are open in S .

If moreover u is locally of finite type (resp. locally of finite presentation), then the set $V_{\text{l.q.f.}}$ (resp. $V_{\text{unr.}}$) of points of G where u is quasi-finite (resp. unramified) is the inverse image under π_G of $U_{\text{l.q.f.}}$ (resp. $U_{\text{unr.}}$).

Let us prove (i). First note (1.3.1.1) that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Let Y be the open subset of H formed by the points where π_H is of finite type, and let X_1 be the open subset of $u^{-1}(Y)$ formed by the points where π_G is of finite presentation. By EGA IV₃, 11.3.1, the set X of points of X_1 where π_G is of finite presentation and flat is open in X_1 , hence in G .

Let $u_X : X \rightarrow Y$ be the morphism obtained from u . Since Y (resp. X) contains the unit section of H (resp. G), the restriction π_X of π_G to X is surjective, and the morphism $S \rightarrow X$ obtained from ε_G is an S -section of X .

⁽²⁰⁾**N.D.E.** The statement and proof have been expanded, and in (i) the hypothesis on H has been weakened by replacing “of finite presentation” by “of finite type”.

Consider the sets $W_{\text{flat}} \supset W_{\text{smooth}} \supset W_{\text{ét.}}$, formed by the points of X where u_X is flat (resp. smooth, resp. étale). It is clear that W_{smooth} and $W_{\text{ét.}}$ are open in X , cf. N.D.E. (6). On the other hand, since π_Y is locally of finite type and $\pi_X = \pi_Y \circ u_X$ locally of finite presentation, by EGA IV₁, 1.4.3 (v), u_X is locally of finite presentation. Consequently, the flatness locus W_X of u_X is open in X (EGA IV₃, 11.3.1).

Let $x \in X$ and put $s = \pi_X(x)$; then, by EGA IV₂, 11.3.10 (combined with EGA IV₄, 17.5.1, resp. 17.6.1), x belongs to W_{flat} (resp. W_{smooth} , $W_{\text{ét.}}$) if and only if u_s is flat (resp. smooth, étale) at the point x , or, equivalently by 1.3, if and only if u_s is flat (resp. smooth, étale).

Consequently, for “ $\flat$ = flat, smooth or étale”, one has $U_{\flat} = \varepsilon_G^{-1}(W_{\flat})$ and $W_{\flat} = \pi_X^{-1}(U_{\flat})$. Since $W_{\flat}$ is open in X , hence in G , the first equality shows that $U_{\flat}$ is open in S .

The second assertion of (i) follows from what has just been seen, for then $Y = H$, $X = G$, and $V_{\flat} = W_{\flat} = \pi_G^{-1}(U_{\flat})$.

Let us prove assertion (ii). Let $V_{\text{l.f.t.}} \supset V_{\text{l.q.f.}}$ be the open subsets of G formed by the points at which u is of finite type (resp. quasi-finite). Put $X = V_{\text{l.f.t.}}$ and denote by u_X the restriction of u to X . By hypothesis, X contains $\varepsilon(S)$. Let $x \in X$ and put $s = \pi_X(x)$.

If u is quasi-finite at x , then, by base change, u_s is quasi-finite at x , and hence, since G_s is supposed locally of finite type, u_s is locally quasi-finite, by 1.3.1.

Conversely, if u_s is locally quasi-finite, then $u_X^{-1}(u_X(x)) \subset u_s^{-1}(u_s(x))$ is a finite set. Since $u_X : X \rightarrow H$ is locally of finite type, it follows from Chevalley’s semicontinuity theorem (EGA IV₃, 13.1.3) that there is an open neighborhood W of x in X such that the fiber $u_X^{-1}(u_X(x'))$ is discrete for every $x' \in W$. Thus, by EGA II, 6.2.2 (and EGA III₂, ErrIII 20), u_X is quasi-finite at x .

Consequently, one has $U_{\text{l.q.f.}} = \varepsilon_G^{-1}(V_{\text{l.q.f.}})$ and $V_{\text{l.q.f.}} = \pi_X^{-1}(U_{\text{l.q.f.}})$, and the first equality shows that $U_{\text{l.q.f.}}$ is open in S .

If moreover G is locally of finite type over S , then $X = G$, and hence $V_{\text{l.q.f.}}$ is the inverse image under π_G of $U_{\text{l.q.f.}}$. This proves the first half of (ii).

Now consider the open subsets $V_{\text{l.f.p.}} \supset V_{\text{unr.}}$, formed by the points where u is of finite presentation (resp. unramified), and suppose that $Z = V_{\text{l.f.p.}}$ contains the unit section of G . Let $z \in Z$; put $s = \pi_Z(z)$ and $h = u_Z(z)$.

If u is unramified at z , then, by base change, u_s is unramified at z , and hence, since G_s is supposed locally of finite type, u_s is unramified, by 1.3.1.

Conversely, if u_s is unramified at the point z , then the fiber $u_s^{-1}(h) = u^{-1}(h)$ is unramified over $\kappa(h)$ at the point z . Since Z is an open subset of G , the fiber $u_Z^{-1}(h) = u_s^{-1}(h) \cap Z$ is unramified over $\kappa(h)$ at the point z . Since u_Z is locally of finite presentation, this implies, by EGA IV₄, 17.4.1, that u_Z is unramified at the point z .

Thus one has $U_{\text{unr.}} = \varepsilon_G^{-1}(V_{\text{unr.}})$ and $V_{\text{unr.}} = \pi_Z^{-1}(U_{\text{unr.}})$, and the first equality shows that $U_{\text{unr.}}$ is open in S .

If moreover G is locally of finite presentation over S , then $Z = G$, and hence $V_{\text{unr.}}$ is the inverse image under π_G of $U_{\text{unr.}}$. This completes the proof of assertion (ii).

Corollary 2.6. *Let $u : G \rightarrow H$ be a morphism of S -groups which is a radicial morphism (as is the case if u is a monomorphism (EGA I, 3.5.4)). Suppose G (resp. H) locally of finite presentation and flat (resp. locally of finite type) over S . Then the set U of $s \in S$ such that u_s is an open immersion is open in S , and the restriction of u above U is an open immersion.*

By 2.5 (i), the set U' of points $s \in S$ such that u_s is étale is open in S . Since u is radicial, so is u_s , for every $s \in S$, hence, by EGA IV₄, 17.9.1, one has $U = U'$, which shows that U is open. Finally, by 2.5 (i), the restriction of u above U is étale; since u is radicial, this restriction is an open immersion (EGA IV₄, 17.9.1).

Proposition 2.7. *Let G be an S -group. The following conditions are equivalent:*

- (i) *G is unramified over S at the points of the unit section,*

- (ii) *the unit section is an open immersion,*
- (iii) *G is of finite presentation over S at the points of the unit section, and, for every $s \in S$, G_s is unramified over $\kappa(s)$.*

If moreover G is supposed locally of finite presentation over S , each of the preceding three conditions is equivalent to the following:

- (iv) *G is unramified over S .*

The equivalence of assertions (i) and (ii) follows from the more general Lemma 2.7.1 below. Observe (1.3.1.1) that either of conditions (i) or (iii) implies that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Then (EGA IV₄, 17.4.1), assertion (i) is equivalent to the fact that, for every $s \in S$, G_s is unramified over $\kappa(s)$ at the point e_s , the identity of G_s , or again (1.3.1), to the fact that G_s is unramified over $\kappa(s)$, so assertions (i) and (iii) are equivalent. Finally, if G is locally of finite presentation over S , assertions (iii) and (iv) are equivalent (EGA IV₄, 17.4.1).

Lemma 2.7.1. *Let G be an S -scheme endowed with a section ε . For G to be unramified over S at the points of this section, it is necessary and sufficient that ε be an open immersion.*

The condition is necessary, by EGA IV₄, 17.4.1 a) $\Rightarrow$ b''). Conversely, if ε is an open immersion, the restriction to $\varepsilon(S)$ of the structure morphism $G \rightarrow S$ is an isomorphism, so G is unramified over S at the points of $\varepsilon(S)$.

Corollary 2.8. *Let $u : G \rightarrow H$ be a morphism of S -groups. Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$.⁽²¹⁾*

- (a) *If u is locally of finite type, the following conditions are equivalent:*
 - (i) *u is locally quasi-finite,*
 - (ii) *for every $s \in S$, $u_s : G_s \rightarrow H_s$ is locally quasi-finite,*
 - (iii) *$\text{Ker } u$ is locally quasi-finite over S ,*
 - (iv) *the fibers of $\text{Ker } u$ are discrete.*
- (b) *If u is locally of finite presentation, the following conditions are equivalent:*
 - (v) *u is unramified,*
 - (vi) *for every $s \in S$, $u_s : G_s \rightarrow H_s$ is unramified,*
 - (vii) *$\text{Ker } u$ is unramified,*
 - (viii) *the unit section $S \rightarrow \text{Ker } u$ is an open immersion.*

The equivalences (i) $\Leftrightarrow$ (ii) and (v) $\Leftrightarrow$ (vi) follow from 2.5 (ii), and the reminder 2.8.1 below shows that (i) $\Rightarrow$ (iii) and (v) $\Rightarrow$ (vii).

For every $s \in S$, denote by e_s the identity element of H_s . Then (iii) (resp. (vii)) implies that, for every $s \in S$,

$$(\text{Ker } u)_s = \text{Ker } u_s = u_s^{-1}(e_s)$$

is locally quasi-finite (resp. unramified) over $\kappa(s) = \kappa(e_s)$, hence that u_s is quasi-finite (resp. unramified) at the identity point of G_s , hence, by 1.3, that u_s is locally quasi-finite (resp. unramified). Thus (iii) $\Rightarrow$ (ii) and (vii) $\Rightarrow$ (vi). Finally, (ii) $\Leftrightarrow$ (iv) by 1.4.1, and (vii) $\Leftrightarrow$ (viii) by 2.7.

Reminder 2.8.1. Recall (cf. I 2.3.6.1) that, given a morphism $u : G \rightarrow H$ of S -groups, one calls the kernel of u , and denotes by $\text{Ker } u$, the subgroup functor of G defined by putting, for every morphism $f : T \rightarrow S$,

$$(\text{Ker } u)(T) = \{a \in G(T) \mid u \circ a = \varepsilon_H \circ f\}.$$

⁽²¹⁾**N.D.E.** The presentation has been modified, separating the assertions concerning the “quasi-finite” case from those concerning the “unramified” case.

By loc. cit. (or EGA I, 4.4.1), this functor is representable by the S -group $G \times_H S = u^{-1}(\varepsilon_H(S))$, denoted simply by $\text{Ker } u$. In particular, the structure morphism $\text{Ker } u \rightarrow S$ is obtained from u by base change.

Lemma 2.9. *Let $\pi : X \rightarrow S$ be a morphism admitting an S -section ε .*

- (i) *If π is injective, it is integral.**
- (ii) *If π is locally of finite type, and if, for every $s \in S$, π_s is an isomorphism, then π is an isomorphism.***

First observe that, by Lemma 2.9.1 below, π , being a homeomorphism, is an affine morphism.

If π is injective, ε is surjective. Since ε is a surjective immersion, $\varepsilon(S)$ is isomorphic to the closed subscheme of X defined by a nilideal $\mathcal{I}$ of $\mathcal{O}_X$. Since ε is an S -section of the morphism π , one has a direct sum decomposition $\mathcal{O}_X = \mathcal{O}_S \oplus \mathcal{I}$ of $\mathcal{O}_S$ -modules. Since $\mathcal{I}$ is a nilideal of $\mathcal{O}_X$, $\mathcal{I}$ is evidently integral over $\mathcal{O}_S$, so $\mathcal{O}_X$ is integral over $\mathcal{O}_S$, and π is integral.

Now suppose π locally of finite type. Since $\pi \circ \varepsilon = \text{id}_S$, ε is locally of finite presentation (cf. EGA IV₁, 1.4.3 (v)), so $\mathcal{I}$ is a finitely generated ideal of $\mathcal{O}_X$ (EGA IV₁, 1.4.1). For every $s \in S$, one has $\mathcal{O}_{X_s} = \kappa(s) \oplus \mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s)$. By hypothesis, ε_s is an isomorphism, so $\mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s) = 0$ for every $s \in S$, hence a fortiori $\mathcal{I} \otimes_{\mathcal{O}_X} \kappa(x) = 0$ for every $x \in X$, which implies, by Nakayama's lemma, that $\mathcal{I} = 0$, hence that π is an isomorphism.

Lemma 2.9.1. *Let $f : X \rightarrow Y$ be a morphism of schemes which is a homeomorphism; then f is an affine morphism.****

It suffices to show that every point $y \in Y$ has an open neighborhood W such that the restriction of f above W is an affine morphism. Then let $y \in Y$, and let V be an affine open neighborhood of y in Y . Let $V' = f^{-1}(V)$. Then V' is an open neighborhood of $x = f^{-1}(y)$ in X . There is an affine open neighborhood W' of x in X contained in V' . Then put $W = f(W')$. Then W is an open neighborhood of y in Y contained in the affine scheme V , so W is separated. Since W' is an affine scheme, the restriction of f above W is then an affine morphism (EGA II, 1.2.3).

Corollary 2.10. *Let G be an S -group locally of finite type. Suppose that, for every $s \in S$, G_s is the unit $\kappa(s)$ -group; then G is the unit S -group.*

More generally:

Corollary 2.11. *Let $f : X \rightarrow Y$ be an S -morphism locally of finite type. For f to be a monomorphism, it is necessary and sufficient that, for every $s \in S$, f_s be a monomorphism.*

It is clear that the condition is necessary; let us show that it is sufficient. If, for every $s \in S$, f_s is a monomorphism, a fortiori for every $y \in Y$, f_y is a monomorphism; we may therefore suppose that $Y = S$.

By EGA I, 5.3.8, to show that f is a monomorphism, it suffices to show that $\Delta_f : X \rightarrow X \times_S X$ is an isomorphism, or, equivalently, that the first projection $p : X \times_S X \rightarrow X$ is an isomorphism. But if f_s is a monomorphism, it similarly follows from EGA I, 5.3.8 that the first projection $X_s \times_{\kappa(s)} X_s \rightarrow X_s$ (which is identified with p_s) is an isomorphism. Now p has the S -section Δ_f , so Lemma 2.9 asserts that if, for every $s \in S$, p_s is an isomorphism, then so is p .

Corollary 2.12. *Let G be an S -scheme having an S -section ε and such that the structure morphism $\pi : G \rightarrow S$ is closed. Let $s \in S$ be such that π is of finite presentation at the point $\varepsilon(s)$, and $\varepsilon_s : \text{Spec}(\kappa(s)) \rightarrow G_s$ is an isomorphism (or, equivalently, $\pi_s : G_s \rightarrow \text{Spec } \kappa(s)$ is*

*This is also a special case of EGA IV₄, 18.12.11, for π is evidently a universal homeomorphism.

**This is also an immediate consequence of EGA IV₄, 18.12.6.

***Cf. EGA IV₄, 18.12.7.1 for a slightly more general result, proved by the same argument.

an isomorphism). Then there is an open neighborhood U of s in S such that $\varepsilon|_U : U \rightarrow G \times_S U$ is an isomorphism.

Let V be the set of points of G where π is unramified; one knows that V is open (cf. N.D.E. (6)) and contains $\varepsilon(s)$. Thus $W = \varepsilon^{-1}(V)$ is an open subset of S containing s , and such that, for every $t \in W$, π is unramified at $\varepsilon(t)$. Since π is closed, so is $\pi|_W$, so we may suppose that $W = S$.

Then, by 2.7.1, $X = G - \varepsilon(S)$ is a closed subset of G not meeting G_s ; thus, since π is closed, $F = \pi(X)$ is a closed subset of S not meeting s ; put $U = S - F$; then U is an open subset of S such that $\varepsilon|_U$ is an isomorphism of U onto $G|_U$.

3. Identity component of a group locally of finite presentation

3.0. — Given a subset A (resp. B) of an S -scheme X (resp. Y), by abuse of notation, $A \times_S B$ will denote the subset of $X \times_S Y$ formed by the points whose first projection belongs to A and second to B . Given a subset A of an S -group G , we shall say that A is stable under the group law of G if one has $c(A) \subset A$ and $\mu(A \times_S A) \subset A$.

Definition 3.1. Let G be an S -group functor satisfying the following condition:

(+) for every $s \in S$, the functor $G_s = G \otimes_S \kappa(s)$ is representable.

One then denotes by G_s^0 the connected component of the identity element of the $\kappa(s)$ -group G_s .⁽²²⁾ One defines a subgroup S -functor of G , called the *identity component of G* , denoted by G^0 , by putting, for every morphism $T \rightarrow S$,

$$G^0(T) = \{u \in G(T) \mid \forall s \in S, u_s(T_s) \subset G_s^0\}.$$

One has thus defined the functor $G \mapsto G^0$ from $(\widehat{\mathbf{Sch}}/S)$ -gr. to $(\widehat{\mathbf{Sch}}/S)$ -gr.

Remark 3.2. (i) Let G be an S -group functor satisfying condition (+); then $\underline{\mathrm{Lie}}(G^0/S) = \underline{\mathrm{Lie}}(G/S)$, by Exposé II.⁽²³⁾ Indeed, for every S -scheme T , denote by I_T the T -scheme of dual numbers over T and by $\tau : T \rightarrow I_T$ the zero section (cf. II, 2.1). By definition, one has

$$\underline{\mathrm{Lie}}(G/S)(T) = \{u \in G(I_T) \mid u \circ \tau = e\},$$

where $e : T \rightarrow G$ denotes the composite of $T \rightarrow S$ and the unit section $S \rightarrow G$, and similarly

$$\begin{aligned} \underline{\mathrm{Lie}}(G^0/S)(T) &= \{u \in G^0(I_T) \mid u \circ \tau = e\} \\ &= \{u \in G(I_T) \mid u \circ \tau = e \text{ and } u_s((I_T)_s) \subset G_s^0, \forall s \in S\}. \end{aligned}$$

Now, for every $s \in S$, T_s and $(I_T)_s = I_{T_s}$ have the same underlying set, so if $u \in \underline{\mathrm{Lie}}(G/S)(T)$, one has $u_s(I_{T_s}) = e_s$, where e_s denotes the identity point of G_s , whence $u \in \underline{\mathrm{Lie}}(G^0/S)(T)$. Thus the inclusion $\underline{\mathrm{Lie}}(G^0/S)(T) \subset \underline{\mathrm{Lie}}(G/S)(T)$ is an equality for every T , whence $\underline{\mathrm{Lie}}(G^0/S) = \underline{\mathrm{Lie}}(G/S)$.

(ii) Let G and G' be two S -group functors satisfying (+); then:

- (a) if $G \subset G'$, then $G^0 \subset G'^0$,
- (b) if $G \subset G'$ and $G'^0 \subset G$, then $G^0 = G'^0$,
- (c) if, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$, then G^0 satisfies property (+), by VIA.2.3, and one has $(G^0)^0 = G^0$.

⁽²²⁾N.D.E. This is a slightly abusive notation, but is compatible with the notation of VIA, 2.3 when this connected component is the underlying topological space of an open subgroup scheme G^0 of G , cf. VIA, 2.2.bis.

⁽²³⁾N.D.E. What follows has been added.

Proposition 3.3. *Let G be an S -group functor satisfying condition (+) and let S' be an S -scheme. Then $(G \times_S S')^0 = G^0 \times_S S'$; in other words, the functor $G \mapsto G^0$ commutes with base changes, i.e. the following diagram is commutative:*

$$\begin{array}{ccc} (\widehat{\mathbf{Sch}}/S)\text{-gr.} & \xrightarrow{G \mapsto G^0} & (\widehat{\mathbf{Sch}}/S)\text{-gr.} \\ \downarrow & & \downarrow \\ (\widehat{\mathbf{Sch}}/S')\text{-gr.} & \longrightarrow & (\widehat{\mathbf{Sch}}/S')\text{-gr.} \end{array}$$

Indeed, it suffices to check that, for every $s' \in S'$, whose image in S is denoted by s , $((G \times_S S') \otimes_{S'} \kappa(s'))^0 = (G_s \otimes_{\kappa(s)} \kappa(s'))^0$ equals $G_s^0 \otimes_{\kappa(s)} \kappa(s')$; this follows from VIA, 2.1.2. Note, for later use in 4.2, that the group structure of G_s has not been used, but only the fact that G_s^0 has a rational point, namely $\varepsilon(s)$, hence is geometrically connected (see also EGA IV₂, 4.5.14).

Special case 3.4. Let G be an S -group scheme; denote by $\underline{G^0}$ the subset of G which is the union of the $\underline{G_s^0}$ as s ranges over S . Then $\underline{G^0}$ is a subset of G stable under the group law of G (cf. 3.0), and, for every morphism $S' \rightarrow S$, one has:

$$G^0(S') = \{u \in G(S') \mid u(S') \subset \underline{G^0}\}.$$

When $\underline{G^0}$ is an open subset of G , it is clear that G^0 is representable by the subscheme of G induced on the open subset $\underline{G^0}$.

Proposition 3.5. *Let S be a quasi-compact quasi-separated scheme, and G an S -group with fibers locally of finite type. There is then a quasi-compact open subset U of G containing $\underline{G^0}$.*

Since the unit section ε is an immersion, $\varepsilon(S)$ is a quasi-compact subspace of G , so there is a quasi-compact open subset V of G containing $\varepsilon(S)$. Since S is quasi-separated and V is quasi-compact, V is quasi-compact over S (EGA IV₁, 1.2.4), hence $V \times_S V$ is quasi-compact over S , hence quasi-compact. Thus $V \cdot V = \mu(V \times_S V)$ is quasi-compact. Put $V_s = V \cap G_s$ and $V_s^0 = V \cap G_s^0$. Then V_s^0 is an open subset of G_s^0 , dense in G_s^0 since G_s^0 is irreducible (VIA 2.4), so $V_s^0 \cdot V_s^0 = G_s^0$ (VIA 0.5), which shows that $V_s \cdot V_s \supset G_s^0$, hence that $V \cdot V \supset \underline{G^0}$. Finally, since $V \cdot V$ is quasi-compact, there is a quasi-compact open subset U of G containing $V \cdot V$ and a fortiori $\underline{G^0}$.

Corollary 3.6. *Let G be an S -group with fibers locally of finite type and connected. Then G is quasi-compact over S .*

Our assertion being local on S (EGA I, 6.6.1), one is reduced to the case where S is affine. By 3.5, there is then a quasi-compact open subset U of G containing $\underline{G^0} = G$, so G is quasi-compact, hence quasi-compact over the affine scheme S (EGA I, 6.6.4 (v)).

Proposition 3.7. ⁽²⁴⁾ *Let G be an S -group locally of finite presentation; then:*

- (i) $\underline{G^0}$ is ind-constructible in G .

⁽²⁴⁾**N.D.E.** Recall the following definitions and results (cf. EGA 0_{III}, §9.1 and EGA IV₁, §§1.8 and 1.9). Let X be a topological space. (i) An open subset U of X is said to be retrocompact if the inclusion $U \hookrightarrow X$ is quasi-compact, i.e. if $U \cap V$ is quasi-compact for every quasi-compact open subset $V \subset X$. (ii) A subset C of X is said to be constructible if it is a finite union of subsets of the form $U - V$, where U and V are retrocompact open subsets of X . (iii) A subset L of X is said to be locally constructible if, for every $x \in X$, there is an open neighborhood U of x in X such that $L \cap U$ is constructible in U . (N.B. If X is quasi-compact and quasi-separated, L is then constructible.) (iv) A subset I of X is said to be ind-constructible if, for every $x \in X$, there is an open neighborhood U of x in X such that $I \cap U$ is a union of locally constructible subsets in U . Now let $f : X \rightarrow Y$ be a morphism of schemes. By Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.4 and 1.9.5 (viii)), if f is of finite presentation (resp. locally of finite presentation), then the image under f of every locally constructible subset is locally constructible (resp. ind-constructible).

- (ii) *If moreover G is quasi-separated over S , and S quasi-compact and quasi-separated, then $\underline{G}^0$ is constructible.*
- (iii) *Consequently, if G is quasi-separated over S , $\underline{G}^0$ is locally constructible.*

Let us first prove the first assertion. Since $\pi : G \rightarrow S$ is locally of finite presentation, given $s \in S$, there is an open subset U of G containing $\varepsilon(s)$ and an open subset V of S containing s such that $\pi(U) \subset V$ and the morphism $\pi' : U \rightarrow V$ obtained from π is of finite presentation. Then $T = \varepsilon^{-1}(U)$ is an open subset of S and, writing $W = \pi'^{-1}(T)$ and $\pi'' = \pi'|_W$, $\pi'' : W \rightarrow T$ is of finite presentation, and admits as section the morphism $\varepsilon'' : T \rightarrow W$ obtained from ε .

For every $t \in T$, since G_t^0 is irreducible (VIA 2.4), $W \cap G_t^0$ is dense in G_t^0 , hence irreducible, hence connected: it is therefore the connected component of $\pi''^{-1}(t)$ containing $\varepsilon''(t)$. It then follows from EGA IV₃, 9.7.12 that the union W^0 of the $W \cap G_t^0$, for $t \in T$, is locally constructible in W . On the other hand, it follows from VIA 0.5 that $\underline{G}^0 \cap \pi^{-1}(T) = W^0 \cdot W^0$, i.e. $\underline{G}^0 \cap \pi^{-1}(T)$ is the image of $W^0 \times_T W^0$ under the morphism $\mu'' : W \times_T W \rightarrow \pi^{-1}(T)$ obtained from μ .⁽²⁵⁾ Since $W \times_T W$ (resp. $\pi^{-1}(T)$) is of finite presentation (resp. locally of finite presentation) over T , μ'' is locally of finite presentation and quasi-separated, by EGA IV₁, 1.4.3 and 1.2.2; if moreover $\pi^{-1}(T)$ is quasi-separated over T , μ'' is quasi-compact (loc. cit. 1.2.4), hence of finite presentation. Since $W^0 \times_T W^0$ is locally constructible in $W \times_T W$ (since W^0 is so in W), it follows from Chevalley's constructibility theorem (loc. cit., 1.8.4 and 1.9.5 (viii)) that $\underline{G}^0 \cap \pi^{-1}(T)$ is ind-constructible in $\pi^{-1}(T)$, and is locally constructible in $\pi^{-1}(T)$ if G (and hence $\pi^{-1}(T)$) is quasi-separated over T . This proves assertions (i) and (iii).

Now suppose G quasi-separated over S , and S quasi-compact and quasi-separated. Then, by 3.5, there is a quasi-compact open subset U of G containing $\underline{G}^0$. Since G is quasi-separated over S , G is quasi-separated, hence the open subset U is retrocompact (EGA IV₁, 1.2.7), and it suffices to show that $\underline{G}^0$ is constructible in U (EGA 0_{III}, 9.1.8). Moreover, since U is quasi-compact, hence quasi-compact over S (EGA IV₁, 1.2.4), and quasi-separated over S , the restriction of π to U is of finite presentation, so, by EGA IV₃, 9.7.12, $\underline{G}^0$ is locally constructible in U , hence constructible in U , since U is quasi-compact and quasi-separated (EGA IV₁, 1.8.1). This proves (ii), and it follows that, for every quasi-compact quasi-separated open subset T of S (for example, for every affine open subset of S), $\underline{G}^0 \cap \pi^{-1}(T)$ is constructible.

Corollary 3.8. *Let S_0 be a quasi-compact quasi-separated scheme, I an increasingly filtering preordered set, $(\mathcal{A}_i)_{i \in I}$ an inductive system of commutative quasi-coherent $\mathcal{O}_{S_0}$ -algebras, $\mathcal{A} = \varinjlim \mathcal{A}_i$, $S_i = \operatorname{Spec} \mathcal{A}_i$ for $i \in I$, and $S = \operatorname{Spec} \mathcal{A}$ (cf. EGA II, 1.3.1).*

Let G be an S_0 -group scheme locally of finite presentation. Then the canonical map $\varinjlim G^0(S_i) \rightarrow G^0(S)$ is bijective.

Since G is locally of finite presentation over S , the canonical map $\varinjlim G(S_i) \rightarrow G(S)$ is bijective, by EGA IV₂, 8.14.2 c). It follows immediately that the canonical map $\varinjlim G^0(S_i) \rightarrow G^0(S)$ is injective. Let us show that it is surjective. Let $g \in G^0(S) \subset G(S)$. There is $i \in I$ such that g factors by means of $g_i \in G(S_i)$ through $S \rightarrow S_i$; by hypothesis, $g^{-1}(\underline{G}^0) = S$. But, by 3.7, $\underline{G}^0$ is ind-constructible in G , so $g_i^{-1}(\underline{G}^0)$ is so in S_i . It then follows from EGA IV₂, 8.3.4 that there is an index $j \geq i$ such that $g_j^{-1}(\underline{G}^0) = S_j$, where g_j is the map obtained from g_i by the base change $S_j \rightarrow S_i$. This shows that $g_j \in G^0(S_j)$, hence that g comes from an element of $\varinjlim G^0(S_i)$.

Proposition 3.9. *Let G be an S -group locally of finite presentation. Suppose G^0 representable; then the canonical morphism $i : G^0 \rightarrow G$ is an open immersion; moreover, G^0 is quasi-compact over S .*

⁽²⁵⁾**N.D.E.** The original has been expanded in what follows.

Since G^0 is a subfunctor of the functor G , the morphism i is a monomorphism, hence is radicial. By the definition of the functor G^0 , one checks immediately from the definition (EGA IV₄, 17.1.1) that i is a formally étale morphism (observing that $\underline{G^0}$ is the image of i in G). Finally, it follows from the characterization (EGA IV₃, 8.14.2 c)) of S -schemes locally of finite presentation by means of the functor they represent, and from 3.8, that, since G is locally of finite presentation over S , so is G^0 . Thus i is locally of finite presentation (cf. EGA IV₁, 1.4.3); it is therefore a radicial étale morphism, hence an open immersion (EGA IV₄, 17.9.1).

Finally, the last assertion follows from 3.6.

Theorem 3.10. *Let G be an S -group. The following conditions are equivalent:*

- (i) G is smooth over S at the points of the unit section.
- (ii) G is flat and locally of finite presentation over S at the points of the unit section, and, for every $s \in S$, G_s is smooth over $\kappa(s)$.
- (iii) There is an open subgroup scheme G' of G , smooth over S .
- (iv) G^0 is representable by an open subscheme of G , smooth over S .

It is clear that (iv) $\Rightarrow$ (iii) $\Rightarrow$ (i) and, by 1.3.1 and 2.4, (i) implies (ii) and (iii). Moreover, (ii) $\Rightarrow$ (i) by EGA IV₄, 17.5.1.

Finally let us show that (iii) implies (iv). Lemma 3.10.1 below shows that G' contains $\underline{G^0}$, and that $\underline{G'^0} = \underline{G^0}$. It therefore suffices to show that $\underline{G^0}$ is open in G , for we have already seen (3.4) that G^0 will then be representable by the smooth subgroup scheme induced by G' on the open subset $\underline{G'^0} = \underline{G^0}$. One may therefore suppose that $G' = G$.

To show that $\underline{G^0}$ is open, it suffices to show that every $s \in S$ has a neighborhood T in S such that $\underline{G^0} \cap \pi^{-1}(T)$ is open in $\pi^{-1}(T)$. Let $s \in S$. Since $G = G'$, π is locally of finite presentation, so one may construct, as in the proof of 3.7, an open subset T of S containing s , and an open subset W of G containing $\varepsilon(s)$, such that the morphism $\pi'' : W \rightarrow T$ obtained from π is of finite presentation and admits as section the morphism $\varepsilon'' : T \rightarrow W$ obtained from ε . For every $t \in T$, $W \cap G_t^0$ is then the connected component of $\pi''^{-1}(t)$ containing $\varepsilon''(t)$. Since π is smooth, so is π'' , which is therefore smooth and of finite presentation. Then, by EGA IV₃, 15.6.5, the union W^0 of the $W \cap G_t^0$ for $t \in T$ is open in W .

On the other hand, by VIA 0.5, one has $W^0 \cdot W^0 = \underline{G^0} \cap \pi^{-1}(T)$, and one must show that this is open in $\pi^{-1}(T)$. We may therefore now suppose that $T = S$; it remains to show that $W^0 \cdot W^0$ is open in G . Since π is universally open, so is μ (VIA 0.1). Thus, since W^0 is open in G , so is $W^0 \cdot W^0 = \mu(W^0 \times_S W^0)$.

This result will be improved in 4.4.

Lemma 3.10.1. *Let G be an S -group with fibers locally of finite type. Then every open subgroup scheme H of G contains $\underline{G^0}$, and satisfies $\underline{G^0} = \underline{H^0}$.*

Let $s \in S$; put $G'_s = H_s \cap G_s^0$; then G'_s is an open subset of G_s^0 , dense in G_s^0 since G_s^0 is irreducible (VIA 2.4), so $G'_s \cdot G'_s = G_s^0$ (VIA 0.5), which shows that $G_s^0 = G'_s \cdot G'_s \subset H_s \cdot H_s = H_s$. Thus one has $G_s^0 = H_s^0$ for every s , whence $\underline{G^0} \subset H$ and $\underline{G^0} = \underline{H^0}$.

Proposition 3.11. *Let $u : G \rightarrow H$ be a morphism between S -groups locally of finite presentation. If u is flat, the map $u^0 : \underline{G^0} \rightarrow \underline{H^0}$ obtained from u is surjective; the converse holds if G is flat over S and H has reduced fibers.⁽²⁶⁾*

Suppose u flat; then, for every $s \in S$, u_s is flat and locally of finite presentation, hence open (EGA IV₂, 2.4.6), so the morphism $u_s^0 : G_s^0 \rightarrow H_s^0$ is surjective, by 1.3.2.⁽²⁷⁾ Thus the map $u^0 : \underline{G^0} \rightarrow \underline{H^0}$ is surjective.

⁽²⁶⁾N.D.E. Compare with VIA, 5.6.

⁽²⁷⁾N.D.E. The original has been shortened by referring here to 1.3.2. On the other hand, the original has been simplified below by removing a reference to the generic flatness theorem (EGA IV₂, 6.9.1), which is not necessary here.

Conversely, suppose the map $u^0 : \underline{G}^0 \rightarrow \underline{H}^0$ surjective, G flat over S and H with reduced fibers. Then, for every $s \in S$, the morphism $u_s^0 : G_s^0 \rightarrow H_s^0$ is surjective, hence flat at every point above the generic point η of H_s^0 (since $\mathcal{O}_{H_s^0, \eta}$ is a field), so u_s is flat by 1.3. Thus u is flat, by the “fiberwise flatness criterion” (EGA IV₃, 11.3.11).

4. Dimension of the fibers of groups locally of finite presentation

Proposition 4.1. *Let G be an S -scheme locally of finite type, endowed with an S -section ε , and such that, for every $s \in S$, one has $\dim G_s = \dim_{\varepsilon(s)} G_s$ (as is the case if G is an S -group (1.5)).*

- (i) *The function $s \mapsto \dim G_s$ is upper semicontinuous on S .*
- (ii) *If moreover G is locally of finite presentation over S , this function is locally constructible.*

Let $\pi : G \rightarrow S$ be the structure morphism. Chevalley’s semicontinuity theorem (EGA IV₃, 13.1.3) asserts that the function $x \mapsto \dim_x \pi^{-1}(\pi(x))$ is upper semicontinuous on G . Now, for every $s \in S$, one has

$$\dim G_s = \dim \pi^{-1}(s) = \dim_{\varepsilon(s)} \pi^{-1}(\pi(\varepsilon(s)));$$

and, since the function $s \mapsto \varepsilon(s)$ is continuous on S , the composite function $s \mapsto \dim G_s$ is upper semicontinuous on S .

Suppose G locally of finite presentation over S . To show that the function $s \mapsto \dim G_s$ is locally constructible, arguing as before one sees that it suffices to show that the function $x \mapsto \dim_x \pi^{-1}(\pi(x))$ is locally constructible on G , which follows from EGA IV₃, 9.9.1.

Proposition 4.2. *Let $\pi : G \rightarrow S$ be an S -scheme locally of finite presentation, endowed with an S -section ε and satisfying the following two conditions (which hold if G is an S -group, by 1.5 and VIA 2.4.1):*

- (a) *For every $s \in S$ and every $x \in G_s$, one has $\dim G_s = \dim_x G_s$ (or, equivalently (1.5), for every $s \in S$, all the irreducible components of G_s have the same dimension).*
- (b) *For every $s \in S$, denoting by G_s^0 the connected component of G_s containing $\varepsilon(s)$, G_s^0 is geometrically irreducible.*

Let $s \in S$. The following conditions are equivalent:

- (i) *G is universally open over S at the points of G_s^0 .*
- (i bis) *G is universally open over S at every point of a neighborhood of $\varepsilon(s)$ in G_s^0 .⁽²⁸⁾*
- (ii) *The function $t \mapsto \dim G_t$ is constant in a neighborhood of s in S .*
- (iii) *$\underline{G}^0$ is “universally open over S at the points of G_s^0 ”, that is, given $S' \rightarrow S$, $s' \in S'_s$, $g \in G_{s'}^0$ and an open neighborhood V of g in $G' = G_{S'}$, $\pi(V \cap \underline{G}^0)$ is an open neighborhood of s' in S' .⁽²⁹⁾*

Evidently (i) $\Rightarrow$ (i bis). By EGA IV₃, 14.3.3.1 (ii), the set of points of G_s^0 where π_G is universally open is closed in G_s^0 . Thus, since G_s^0 is irreducible, one has (i bis) $\Rightarrow$ (i).

Let us show that (i) implies (ii). Let T be the set of $t \in S$ such that $\dim G_t = \dim G_s$. By 4.1, T is locally constructible, so, by EGA IV₁, 1.10.1, to show that T is a neighborhood of s , it suffices to show that every generization s' of s belongs to T .

Let x be the generic point of G_s^0 and U an open subset of G containing x . Since π_G is universally open at ξ , by EGA IV₃, 14.3.13, for every generization s' of s , one has $\dim(U \cap G_{s'}) \geq \dim_x(U \cap G_s)$. In view of our hypothesis a), this implies $\dim G_{s'} \geq \dim G_s$. Now, since the function $s \mapsto \dim G_s$ is upper semicontinuous, by 4.1, one also has $\dim G_{s'} \leq \dim G_s$, whence $s' \in T$. This proves that (i) $\Rightarrow$ (ii).

⁽²⁸⁾**N.D.E.** The original reads: “at the points of $\varepsilon(s)$ ”; it is not sufficient to suppose that G is universally open over S at $\varepsilon(s)$, as the example given in N.D.E. (19) shows. The proof has been corrected accordingly.

⁽²⁹⁾**N.D.E.** Condition (iii), pointed out by O. Gabber, has been added; it will be useful later (5.7).

It is clear that (iii) $\Rightarrow$ (i); let us show that (ii) $\Rightarrow$ (iii). Since the dimension of the fibers is unchanged by extension of the base field and the formation of $\underline{G}^0$ commutes with base change (cf. the proof of 3.3), one may suppose that $S' = S$ and $s' = s$. Moreover, since every open subset V of G meeting G_s^0 contains the generic point η of G_s^0 , one may suppose that $g = \eta$.

One may moreover suppose S affine. Let U be an affine open neighborhood of $\varepsilon(s)$; it is then of finite presentation over S . Replacing S by $\varepsilon^{-1}(U)$ and then U by $U \cap \pi^{-1}(S)$, one is reduced to the case where $\pi : U \rightarrow S$ is of finite presentation and admits $\varepsilon : S \rightarrow U$ as section. Then, by EGA IV₃, 9.7.12, $\underline{G}^0 \cap U$ is constructible in U . Then, replacing V by an affine open subset contained in $V \cap U$, one obtains that $\underline{G}^0 \cap V$ is constructible in V . Since $\pi : V \rightarrow S$ is of finite presentation, $\pi(\underline{G}^0 \cap V)$ is locally constructible in S , by Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.4).

Thus, by loc. cit., 1.10.1, to show that $\pi(\underline{G}^0 \cap V)$ is an open neighborhood of s , it suffices to show that, for every generization t of s , there is a generization ξ of η belonging to $\underline{G}^0$ (and hence to $\underline{G}^0 \cap V$). Now the generic point ξ of G_t^0 is a generization of η . Indeed, $\varepsilon(s)$ belongs to the closure X of $\{\xi\}$ in G , so, by Chevalley's semicontinuity theorem (cf. EGA IV₃, 13.1.3), one has $\dim_{\varepsilon(s)} X_s \geq \dim_{\xi} X_t$; on the other hand, hypothesis (ii) implies that $\dim_{\xi} G_t^0 = \dim_{\varepsilon(s)} G_s$. It follows that one of the irreducible components of X_s containing $\varepsilon(s)$ equals G_s^0 , whence $\eta \in \overline{\{\xi\}}$. This proves that (ii) $\Rightarrow$ (iii), which proves the proposition.

⁽³⁰⁾One may also prove the implication (ii) $\Rightarrow$ (i) as follows. Since π is locally of finite presentation, there is an open subset U of G containing $\varepsilon(s)$ and an open subset V of S containing s such that $\pi(U) \subset V$ and the morphism $\pi' : U \rightarrow V$ obtained from π is of finite presentation. Then put $T = \varepsilon^{-1}(U)$ and $W = \pi'^{-1}(T) = U \cap \pi^{-1}(T)$. Then the morphism $\pi'' : W \rightarrow T$ obtained from π' is of finite presentation and admits as section the morphism $\varepsilon'' : T \rightarrow W$ obtained from ε . Moreover, for every $t \in T$, since G_t^0 is irreducible, $W \cap G_t^0$ is dense in G_t^0 , hence irreducible, hence connected: it is therefore the connected component of $\pi''^{-1}(t)$ containing $\varepsilon''(t)$.

Since $W \cap G_t^0$ is a dense open subset of G_t^0 , by 1.5 and EGA IV₂, 5.2.1, one has $\dim(W \cap G_t^0) = \dim G_t^0 = \dim G_t$, so the function $t \mapsto \dim W \cap G_t^0$ is constant in a neighborhood of s in T . Finally let us show that, for every $t \in T$, $W \cap G_t^0$ is geometrically irreducible. Let K be an extension of $\kappa(t)$; then $(W \cap G_t^0) \otimes_{\kappa(t)} K = (W \otimes_{\kappa(t)} K) \cap (G_t^0 \otimes_{\kappa(t)} K)$ is a nonempty open subset of $G_t^0 \otimes_{\kappa(t)} K$, hence is irreducible since $G_t^0 \otimes_{\kappa(t)} K$ is.

We are then in the situation to apply EGA IV₃, 15.6.6 (ii), which asserts that π'' (hence π) is universally open at the points of $W \cap G_s^0$. But, by EGA IV₃, 14.3.3.1 (ii), the subset of G_s formed by the points where π is universally open is closed in G_s ; since it contains $W \cap G_s^0$, it therefore contains its closure G_s^0 .

Corollary 4.3. *Let G be a flat S -group locally of finite presentation. Then the function $s \mapsto \dim G_s$ is locally constant on S .*

This follows immediately from 4.2, for every flat morphism locally of finite presentation is universally open (EGA IV₂, 2.4.6).

Corollary 4.4. *Let G be an S -group locally of finite presentation over S at the points of the unit section. Consider the conditions:*

- (i) *G is smooth over S at the points of the unit section (cf. 3.10).*
- (ii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and the function $s \mapsto \dim G_s$ is locally constant on S .*
- (iii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and there is a neighborhood V of the unit section such that $\pi_V : V \rightarrow S$ is universally open.*
- (iv) *For every $s \in S$, G_s^0 is smooth over $\kappa(s)$, and G^0 is representable by an open subgroup scheme of G , universally open over S .*

⁽³⁰⁾**N.D.E.** The preceding argument was communicated to us by O. Gabber; the proof of the implication (ii) $\Rightarrow$ (i) given in the original has also been retained.

Then one has the following implications: (i) $\Rightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv).

If moreover S is supposed reduced, conditions (i) to (iv) are equivalent, and imply that G^0 is smooth over S .⁽³¹⁾

Let us show that (i) implies (ii). For every $x \in G$, one has $\dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$, by 1.5. Consequently, by EGA IV₄, 17.10.2, the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$$

is continuous in a neighborhood of the unit section; thus the function $s \longmapsto \dim G_s$ is continuous on S , hence locally constant on S . On the other hand, for every $s \in S$, G_s is smooth over $\kappa(s)$, by 1.3.1.

Let us show that (ii) implies (iv). It suffices to show that $\underline{G}^0$ is open in G , for then, by 3.4, G^0 will be representable by the subgroup scheme induced on $\underline{G}^0$, and the properties of G^0 cited in the statement will follow from 2.4 and 4.2. Given $s \in S$, construct W , T , π'' , ε'' and W^0 as in the proof of 3.10. Then, by EGA IV₃, 15.6.7, W^0 is open in W . On the other hand, under hypothesis 4.4 (ii), it follows from 4.2 that π is universally open at every point of W^0 , hence (VIA 0.1) μ is universally open at every point of $W^0 \times_S W^0$, which shows that $W^0 \cdot W^0$ is open, and one concludes as in the proof of Theorem 3.10.

It is clear that (iv) $\Rightarrow$ (iii), and (iii) $\Rightarrow$ (ii) follows from 4.2 applied to V .

Finally, suppose (ii)–(iv) satisfied and let us show that G^0 is smooth over S if S is reduced.⁽³²⁾ For this one may suppose $G = G^0$. Then G is of finite presentation over S by 5.5 below. Thus π_G is of finite presentation, with geometrically integral fibers of locally constant dimension on S . Then, by EGA IV₃, 15.6.7, the morphism $G \times_S S_{\text{red}} \rightarrow S_{\text{red}}$ obtained from π_G is flat, hence π_G is flat if S is reduced. In this case, $G = G^0$ is smooth over S , by Theorem 3.10.

5. Separation of groups and homogeneous spaces

Proposition 5.1. *For an S -group G to be separated, it is necessary and sufficient that the unit section of G be a closed immersion.*

The condition is necessary (EGA I, 5.4.6); it is sufficient by the following cartesian diagram (cf. VIA 0.3):

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\mu \circ (\text{id}_G \times c)} & G \\ \Delta_{G/S} \uparrow & & \uparrow \varepsilon \\ G & \xrightarrow{\pi} & S \end{array} .$$

Proposition 5.2. *If S is discrete, every S -group is separated.*

Indeed, S then equals $\coprod_{s \in S} \text{Spec } \mathcal{O}_{S,s}$, and, by EGA I, 5.5.5, it suffices to show that, for every $s \in S$, $G \times_S \text{Spec } \mathcal{O}_{S,s}$ is separated, which follows from VIA 0.3, since $\mathcal{O}_{S,s}$ is a zero-dimensional local ring.⁽³³⁾

Theorem 5.3. *Let S be a scheme, G an S -group scheme locally of finite presentation over S and universally open over S in a neighborhood of the unit section, X an S -scheme on which*

⁽³¹⁾**N.D.E.** One cannot dispense here with the hypothesis that S be reduced: if k is a field, $S = \text{Spec } A$, where $A = k[\delta]$ with $\delta^2 = 0$, and G is the closed subgroup $\text{Spec } A[X]/(\delta X)$ of $\mathbf{G}_{a,S}$ (which associates to every A -algebra R the subgroup of $r \in R$ such that $\delta r = 0$), then G is an S -group satisfying (ii)–(iv), but is not flat, hence not smooth, over S .

⁽³²⁾**N.D.E.** The original has been expanded in what follows.

⁽³³⁾**N.D.E.** “Artinian” has been replaced by “zero-dimensional” (and this generalization has been mentioned in VIA 0.3).

G acts so that the morphism

$$\Phi : G \times_S X \longrightarrow X \times_S X, \quad (g, x) \longmapsto (gx, x)$$

is surjective. Suppose moreover that, for every $s \in S$:

- (i) there is an open subscheme U of X , separated over S , such that U_s is dense in X_s ,
- (ii) the fiber X_s is locally of finite type over $\kappa(s)$.⁽³⁴⁾

Then X is separated over S .

Corollary 5.4. *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover X has fibers locally of finite type and connected. Then:*

- (i) X is separated over S .
- (ii) If there is an open subset V of X , quasi-compact over S and meeting every nonempty fiber X_s ,⁽³⁵⁾ then X is quasi-compact over S .

PROOF. (i) Indeed, let $s \in S$ be such that $X_s \neq \emptyset$. Since the morphism $\Phi_s : G_s \times_{\kappa(s)} X_s \longrightarrow X_s \times_{\kappa(s)} X_s$ obtained from Φ by base change is surjective, and since X_s is connected, X_s is irreducible, by VIA, 2.5.4. Thus, if U is an affine open subset of X such that U_s is nonempty, U_s is dense in X_s , and the theorem applies.

To prove (ii), one may suppose S affine. Then V is quasi-compact and, by 3.5, there is a quasi-compact open subset U of G containing G^0 . Let $s \in S$ be such that $X_s \neq \emptyset$. Then X_s is irreducible (VIA, 2.6.6) and hence, since U_s contains G_s^0 , the morphism $U_s \times_{\kappa(s)} V_s \longrightarrow X_s$, $(g, x) \longmapsto gx$ is surjective (VIA, 2.6.4). Consequently the morphism $U \times_S V \longrightarrow X$ is surjective and hence X is quasi-compact (since U and V are); since we have supposed S affine, hence separated, it follows that X is quasi-compact over S (cf. EGA I, 6.6.4 (v)). $\square$

Remark 5.4.1. ⁽³⁶⁾ One will see in the course of the proof that the conclusion of Theorem 5.3 holds if one makes only the hypothesis: (i') there is an open subscheme U of X , separated over S , such that U_s is dense in every irreducible component of X_s . (This follows from (i) if X_s locally has a finite number of irreducible components, as is the case under hypothesis (ii).) On the other hand, one will see later that 5.4 also holds under the hypothesis that each fiber X_s is quasi-separated and connected.

Corollary 5.5. *Let S be a scheme, G an S -group scheme, locally of finite presentation, with connected fibers, and universally open over S . Then G is separated and of finite presentation over S .⁽³⁷⁾*

⁽³⁴⁾N.D.E. The original has been corrected by supposing G universally open over S in a neighborhood of the unit section and adding hypothesis (ii); see below for examples due to O. Gabber, which show that statements 5.3 and 5.4 of the original are false without additional hypotheses. On the other hand, let us point out that Theorem 5.3 is a revised version of Theorem 5.3A below, which appears in the 1965 edition of SGAD, and is due to M. Raynaud, cf. the Notes (*) in Exp. X, 8.5 and 8.8.

Theorem 5.3A (Raynaud). *Let G be an S -group locally of finite presentation, universally open over S , and with connected fibers. Then G is separated over S .*

More generally, every S -scheme X locally of finite presentation over S , endowed with an action of G such that the morphism $\Phi : G \times_S X \longrightarrow X \times_S X$, $(g, x) \longmapsto (gx, x)$ is surjective, is separated over S .

⁽³⁵⁾N.D.E. Without this hypothesis, one has the following counterexample, pointed out by O. Gabber: let $G = S$ be a local scheme with closed point s , such that $S^* = S - \{s\}$ is not quasi-compact, and X the disjoint union of $\{s\}$ and S^* .

⁽³⁶⁾N.D.E. This remark has been added, cf. N.D.E. (34).

⁽³⁷⁾N.D.E. Let us point out here the following result ([Ray70a], VI 2.5): if S is normal, G smooth with connected fibers, and X is an (fppf) homogeneous G -space (i.e. the morphisms $X \longrightarrow S$ and $G \times_S X \longrightarrow X \times_S X$ are covering for the (fppf) topology), locally of finite type over S , then X is locally quasi-projective over S . In particular, G is quasi-projective over S . See also N.D.E. (35) in VIA.

Indeed, by 3.6 and 5.4, G is quasi-compact and separated over S , and, since G is locally of finite presentation over S , it is therefore of finite presentation over S .

5.6. — Proof of Theorem 5.3. Before establishing 5.3, let us prove a few lemmas.

Lemma 5.6.0. ⁽³⁸⁾ (i) *Let $A \subset B$ be integral domains, with B integral over A , and let $\mathfrak{q} \in \operatorname{Spec}(B)$ be such that A is unibranch at the point $\mathfrak{p} = \mathfrak{q} \cap A$. Then the morphism $\pi : \operatorname{Spec}(B) \rightarrow \operatorname{Spec}(A)$ is open at the point $\mathfrak{q}$.*

(ii) *Let X, Y be two irreducible preschemes, $f : X \rightarrow Y$ a dominant morphism, x a point of X such that f is quasi-finite at x and $y = f(x)$ is a unibranch point of Y . Then f is open at the point x . In particular, if $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$ is a local prescheme with closed point y , then $f(U) = Y$ for every neighborhood U of x .*

(i) Let K (resp. L) be the field of fractions of A (resp. B), A' the normalization of A , and B' the subring of L generated by A' and B . Then B' is integral over A' . Put $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(A')$, $X' = \operatorname{Spec}(B')$, so that one has a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\pi'} & Y' \\ \downarrow & & \downarrow \phi \\ X & \xrightarrow{\pi} & Y \end{array}$$

in which all the morphisms are integral and surjective.

Since A is unibranch at $\mathfrak{p}$, Y' has a unique point $\mathfrak{p}'$ above $\mathfrak{p}$; consequently, if U is an open neighborhood of $\mathfrak{p}'$ in Y' , $\phi(Y' - U)$ is a closed subset not containing $\mathfrak{p}$, so that its open complement is contained in $\phi(U)$. This shows that $\phi : Y' \rightarrow Y$ is open at $\mathfrak{p}'$, and it therefore suffices to show that π' is open. One is thus reduced to the case where $A = A'$ is normal.

Let N be a quasi-Galois extension of K containing L , let $\tilde{X} = \operatorname{Spec}(\tilde{B})$, where $\tilde{B}$ is the integral closure of B in N , and let $G = \operatorname{Aut}(N/K)$. Since $\tilde{X} \rightarrow X$ is surjective, it suffices to show that $\tilde{\pi} : \tilde{X} \rightarrow Y$ is open. Let U be an open subset of $\tilde{X}$ and $U' = \bigcup_{g \in G} gU$. Since G acts transitively on the fibers of $\tilde{X} \rightarrow Y$ (cf. [BAC], Chap. V, §2.3, Prop. 6), $\tilde{\pi}(U)$ equals $\tilde{\pi}(U')$, and the latter equals the open complement of the closed subset $\tilde{\pi}(\tilde{X} - U')$. This proves (i).

(ii) One may suppose Y and X reduced. By “Zariski’s main theorem” (cf. EGA IV₃, 8.12.9), there are affine open neighborhoods U of x and $V = \operatorname{Spec}(A)$ of y , such that $f(U) \subset V$, and a factorization:

$$\begin{array}{ccc} U & \xrightarrow{j} & V' \\ & \searrow f & \downarrow u \\ & & V \end{array},$$

where j is an open immersion and u is finite. Replacing V' by the closure of $j(U)$, one may suppose V' irreducible, hence $V' = \operatorname{Spec}(B)$, where B is an integral A -algebra, finite over A . Moreover, since f is dominant, so is u , so that the morphism $A \rightarrow B$ is injective. Since, by hypothesis, A is unibranch at the point y , it follows from (i) that u is open at the point $j(x)$, and hence $f = u \circ j$ is open at the point x . This proves the first assertion of (ii). The second follows from it, for, if Y is a local prescheme with closed point y , every open subset containing y equals Y . (In the case where $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$, one may also use, instead of EGA IV₃, 8.12.9, the local form of Zariski’s main theorem, found for example in [Pes66], or [Ray70b], Ch. IV, Th. 1.)

⁽³⁸⁾**N.D.E.** This lemma, communicated by O. Gabber, has been added; it improves EGA IV₃, 14.4.1.2 and corrects the proof of loc. cit., 14.4.1.3 without changing its hypotheses (compare with the erratum (ErrIV, 38) in EGA IV₄).

Lemma 5.6.1.0. ⁽³⁹⁾ *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $s \in S$, and $x \in X_s$. Let $n = \dim_x(X_s)$ and let q be the structure morphism $\mathbb{A}_S^n \rightarrow S$.*

Suppose given a quasi-finite morphism $u : X \rightarrow \mathbb{A}_S^n$ such that $f = q \circ u$, and suppose f universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s , containing x and of dimension n . Then u is universally open at the point $x \in X$.

First observe that (†) the hypotheses are preserved by every base change $\pi : S' \rightarrow S$ covering s (i.e. such that $\pi^{-1}(s) \neq \emptyset$). Indeed, let $S' \rightarrow S$ be such a morphism, let

$$\begin{array}{ccc} X' & \xrightarrow{u'} & \mathbb{A}_{S'}^n \\ & \searrow f' & \downarrow q' \\ & & S' \end{array}$$

be the diagram obtained by base change, and let s' be a point of S' above s and x' a point of $X'_{s'}$ above x . Since $X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s')$, by lifting of generizations and invariance of dimension under field extension (EGA IV₂, 2.3.4 (i) and 4.1.4), x' is contained in an irreducible component $\mathfrak{z}'$ of $X'_{s'}$ whose generic point z' lies above z , and one has $n \leq \dim \mathfrak{z}' \leq \dim_{x'} X'_{s'} \leq n$, whence $\dim \mathfrak{z}' = n = \dim_{x'} X'_{s'}$. Since f is universally open at x , f' is universally open at x' .

Put $Y = \mathbb{A}_S^n$. By EGA IV₃, 14.3.3.1 (i), to prove that u is universally open at x , it suffices to prove that, for every integer $r \geq 0$ and every point x' of $X' = X[T_1, \dots, T_r]$ above x , the morphism $u' : X' \rightarrow Y' = \mathbb{A}_S^n[T_1, \dots, T_r]$ is open at the point x' . Now, writing $S' = S[T_1, \dots, T_r]$ and q' for the projection $Y' \simeq \mathbb{A}_{S'}^n \rightarrow S'$, one is in the situation obtained by the base change $S' \rightarrow S$. Thus, by the preceding argument, one is reduced to showing that u is open at the point x .

Put $y = u(x)$. Since u is locally of finite presentation (since f and q are, cf. EGA IV₁, 1.4.3), by EGA IV₁, 1.10.3, it suffices to show that $u(\text{Spec } \mathcal{O}_{X,x}) = \text{Spec } \mathcal{O}_{Y,y}$. For this one may suppose S affine and integral. Then let $\pi : S' \rightarrow S$ be its normalization; denote by $u' : X' \rightarrow Y'$ the morphism obtained from u by base change. Since the morphism $\pi_Y : Y' \rightarrow Y$ is integral and surjective, one has

$$\text{Spec}(\mathcal{O}_{Y,y}) = \bigcup_{y'} \pi_Y(\text{Spec}(\mathcal{O}_{Y',y'})),$$

the union being taken over all the points of Y' above y ; it therefore suffices to show that, for each such y' , and every $x' \in X'_{y'}$ above x , one has $u'(\text{Spec } \mathcal{O}_{X',x'}) = \text{Spec } \mathcal{O}_{Y',y'}$. Since the hypotheses are preserved by the base change $\pi : S' \rightarrow S$, one is thus reduced to the case of S' , i.e. one may suppose S an integral normal scheme.

Now the hypotheses on f imply, by EGA IV₃, 14.3.13, that there is an irreducible component Z of X containing z (and hence x), dominating S and such that

$$\dim_z(Z_s) = n = \dim(Z_\eta),$$

where η is the generic point of S . Let ξ be the generic point of Z . Since u , hence also $u_\eta : Z_\eta \rightarrow \mathbb{A}_\eta^n$, is quasi-finite, the closure in $\mathbb{A}_\eta^n$ of the point $u_\eta(\xi) = u(\xi)$ has dimension n , hence $u(\xi)$ is the generic point of $\mathbb{A}_\eta^n$, which is also the generic point of $Y = \mathbb{A}_S^n$. Consequently, denoting by g the restriction of u to Z , the morphism $g : Z \rightarrow Y$ is quasi-finite and dominant. Since Y is normal, it follows from Lemma 5.6.0 that g is open, so u is open at every point of Z , in particular at the point x . Consequently, one has $u(\text{Spec } \mathcal{O}_{X,x}) = \text{Spec } \mathcal{O}_{Y,y}$, and this completes the proof of Lemma 5.6.1.0.

The following lemma advantageously replaces EGA IV₃, 14.5.10,⁽⁴⁰⁾ in that it is independent of noetherian hypotheses.

⁽³⁹⁾**N.D.E.** This lemma, used in the proof of Lemma 5.6.1, has been made explicit.

⁽⁴⁰⁾**N.D.E.** The original, which indicated 19.5.10, has been corrected.

Lemma 5.6.1. *Let S be a scheme, $f : X \rightarrow S$ an S -scheme locally of finite presentation, s a point of S , x a closed point of X_s . Suppose f universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s , containing x and such that $\dim_x(X_s) = \dim \mathfrak{z}$. Then there is a commutative diagram:*

$$\begin{array}{ccc} X & \xrightarrow{f} & S \\ h \uparrow & \nearrow \phi & \uparrow w \\ S'' & \xrightarrow{\pi} & S' \end{array},$$

where S' is an affine scheme, w an étale morphism, π a finite surjective morphism of finite presentation, and $\phi^{-1}(s)$ consists of a single point s'' , such that $h(s'') = x$ and $\kappa(s'') = \kappa(x)$.⁽⁴¹⁾

PROOF. First one may suppose $S = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$, where B is an A -algebra of finite presentation. Let $\mathfrak{p}$ and $\mathfrak{q}$ be the prime ideals of A and B corresponding to s and x , respectively, so that $\mathfrak{p} = A \cap \mathfrak{q}$. Put $n = \dim_x X_s$, and let $t_1, \dots, t_n$ be elements of B whose images in $\mathcal{O}_{X_s, x} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ form a system of parameters. Then $\mathcal{O}_{X_s, x}/(t_1, \dots, t_n)$ is finite-dimensional over $\kappa(x)$ and hence also over $\kappa(s)$, since x is a closed point of the algebraic $\kappa(s)$ -scheme X_s . Consequently, the S -morphism

$$u : X \rightarrow \mathbb{A}_S^n = \operatorname{Spec}(A[T_1, \dots, T_n]),$$

defined by $T_i \mapsto t_i$, is of finite presentation, x is isolated in its fiber $u^{-1}(u(x))$, and one has $u(x) = \tau_0(s)$, where $\tau_0 : S \rightarrow \mathbb{A}_S^n$ denotes the “zero section” of $\mathbb{A}_S^n \rightarrow S$, corresponding to the morphism of A -algebras $A[T_1, \dots, T_n] \rightarrow A$ which sends every T_i to 0. Since the set of points of X isolated in their fiber above $\mathbb{A}_S^n$ is open (EGA IV₃, 13.1.4), one may suppose, shrinking X if necessary, that u is quasi-finite and $u^{-1}(u(x)) = \{x\}$. By Lemma 5.6.1.0, u is universally open at the point x .

Let $B_0 = B/(t_1, \dots, t_n)$ and let $X_0 = X \times_{\mathbb{A}_S^n} \tau_0(S) = \operatorname{Spec} B_0$ and $u_0 : X_0 \rightarrow S$ be the morphism obtained from u by the base change $\tau_0 : S \hookrightarrow \mathbb{A}_S^n$. Then u_0 is quasi-finite and of finite presentation, universally open at the point x , and x is the unique point of X_0 above s .

Let A' be the henselization of the local ring $A_{\mathfrak{p}} = \mathcal{O}_{S, s}$, and let $S' = \operatorname{Spec}(A')$, $B'_0 = B_0 \otimes_A A'$, and $X'_0 = X_0 \times_S S' = \operatorname{Spec}(B'_0)$. Then the closed point s' of S' is the unique point of S' above s , one has $\kappa(s') = \kappa(s)$, and X'_0 has a unique point x' above s' ; one has $\kappa(x') = \kappa(x)$ and x' is also the unique point of X'_0 above s and x . Since A' is henselian, by EGA IV₄, 18.5.11, X'_0 is the disjoint sum of two open and closed subsets:

$$(*) \quad X'_0 = V \sqcup W, \quad \text{where } V = \operatorname{Spec}(\mathcal{O}_{X'_0, x'});$$

and the local ring $\mathcal{O}_{X'_0, x'}$ is finite and of finite presentation over A' . The restriction π of u'_0 to V is therefore finite and of finite presentation. Moreover, since $u'_0 : X'_0 \rightarrow S'$ is open at x' , one has $u'_0(V) = \operatorname{Spec}(\mathcal{O}_{S', s}) = S'$, so π is surjective.

This proves the desired result when $S = S'$. In the general case, A' is a filtering inductive limit of subalgebras A_i étale over A , and such that $S_i = \operatorname{Spec}(A_i)$ has a unique point s_i above s (and one has $\kappa(s_i) = \kappa(s)$). Then $B'_0 = \varinjlim B_i$, where $B_i = B_0 \otimes_A A_i$. Put $X_i = \operatorname{Spec}(B_i) = X_0 \times_S S_i$ and $C = \mathcal{O}_{X'_0, x'}$. By (*) above, one has $C \simeq B'_0/fB'_0$, for some idempotent $f \in B'_0$, and there is an index i and $f_i \in B_i$ such that f is the image of f_i in B'_0 . Put $C_i = B_i/f_i B_i$ and $V_i = \operatorname{Spec}(C_i)$.

⁽⁴¹⁾**N.D.E.** Consequently, h induces a section σ of $X \times_S S'' \rightarrow S''$ such that $\sigma(s'')$ lies above x ; compare with EGA IV₃, 14.5.10.

Then $C = C_i \otimes_{A_i} A'$, whence $V = V_i \times_{S_i} S'$, and C_i is an A_i -algebra of finite presentation (since this holds for B_i). Consequently, the morphisms

$$\begin{array}{ccc} X'_0 & & \\ \uparrow h' & \searrow u'_0 & \\ V & \xrightarrow{\pi} & S' \end{array}$$

come, by the base change $S' \rightarrow S_i$, from morphisms of finite presentation:

$$\begin{array}{ccc} X_i & & \\ \uparrow h_i & \searrow u_i & \\ V_i & \xrightarrow{\pi_i} & S_i \end{array}.$$

For every $j \geq i$, let $V_j = V_i \times_{S_i} S_j$ and let $\pi_j : V_j \rightarrow S_j$ be the morphism (of finite presentation) obtained from π_i by base change. Since $\pi : V \rightarrow S'$ is finite and surjective, by EGA IV₂, 8.10.5, there is an index j such that $\pi_j : V_j \rightarrow S_j$ is finite and surjective. Then $w : S_j \rightarrow S$ is affine étale, S_j has a unique point s_j above s , and V_j has a unique point x_j above s_j (since x' is the unique point of $V = V_j \times_{S_j} S'$ above s'):

$$\begin{array}{ccccccc} X_j & \longrightarrow & X_0 & \longrightarrow & X & & \\ \uparrow h_j & \searrow u_j & & \searrow u_0 & \downarrow f & \searrow u & \\ V_j & \xrightarrow{\pi_j} & S_j & \xrightarrow{w} & S & \xrightarrow{\tau_0} & \mathbb{A}_S^n \end{array}.$$

Thus x_j is the unique point of V_j above s , its image under the morphism $V_j \rightarrow X_j \rightarrow X$ is x , and one has $\kappa(x_j) = \kappa(x)$. $\square$

Lemma 5.6.2.0. ⁽⁴²⁾ *Let k be a field and G a nonempty k -scheme of finite type.*

(i) *Let K be an extension of k and W a dense open subset of G_K . Denote by π the projection $G_K \rightarrow G$; then*

$$U = \{g \in G \mid W \text{ contains every maximal point of } \pi^{-1}(g)\}$$

is a dense open subset of G . (N.B. If g is a closed point of G belonging to U , one therefore has $\pi^{-1}(g) \subset W$.)

(ii) *Suppose moreover G geometrically irreducible. Let $\mu : G \times X \rightarrow Y$ be a morphism of k -schemes, x a point of X , and Ω an open subset of Y such that $\mu_x^{-1}(\Omega_{\kappa(x)}) \neq \emptyset$, where μ_x denotes the morphism $G_{\kappa(x)} \rightarrow Y_{\kappa(x)}$, $g \mapsto \mu(g, x)$. Then*

$$U = \{g \in G \mid \mu \text{ sends every maximal point of } g \times x \text{ into } \Omega\}$$

is a dense open subset of G , and, for every closed point g of G belonging to U , one has $\mu(g \times x) \subset \Omega$ and hence $\mu(g', x) \in \Omega_{\kappa(x)}$ (resp. $\mu(g, x') \in \Omega_{\kappa(g)}$), for every point $g' \in G_{\kappa(x)}$ above g (resp. $x' \in X_{\kappa(g)}$ above x).

(iii) *Suppose that $G = H^0$, where H is a k -group scheme locally of finite type, acting on a nonempty k -scheme X , so that the morphism $H \times_S X \rightarrow X \times_S X$, $(h, x) \mapsto (hx, x)$ is surjective. Let U be an open subset of X . Then:*

- (a) *$G \cdot U$ is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .*
- (b) *Suppose U dense in every irreducible component of X . Then, for every finite subset A of X , there is a closed point $g \in G$ such that $g \cdot A' \subset U_{\kappa(g)}$, where A' is the inverse image of A in $X_{\kappa(g)}$. In particular, the morphism $G \times U \rightarrow X$ is surjective.*

⁽⁴²⁾**N.D.E.** This lemma, communicated by O. Gabber, has been added. It makes it possible to simplify the proof of 5.6.2, and to prove Theorem 5.3, as well as 5.4, in a more general form; see 5.7 and 5.8 below.

PROOF. (i) Let $\bar{k}$ be a separable closure of k and let L be an extension of k containing a copy of $\bar{k}$ and of K . Denote by π_L (resp. ϕ) the projection $G_L \rightarrow G$ (resp. $G_L \rightarrow G_K$). Since, for every $g \in G$, ϕ sends the maximal points of $\pi_L^{-1}(g)$ surjectively onto those of $\pi^{-1}(g)$, one sees that $U = \{g \in G \mid W_L \text{ contains every maximal point of } \pi_L^{-1}(g)\}$. Thus, replacing K by L , one is reduced to the case where K contains $\bar{k}$.

Since the projection $p : G_K \rightarrow G_{\bar{k}}$ is surjective and open, $V = p(W)$ is a dense open subset of $G_{\bar{k}}$, and, since $p^{-1}(g) = \text{Spec}(\kappa(g) \otimes_{\bar{k}} K)$ is irreducible for every $g \in G_{\bar{k}}$ (cf. EGA IV₂, 4.3.3), for every $g \in V$ the generic point of $p^{-1}(g)$ belongs to W . On the other hand, let $\mathcal{G} = \text{Gal}(\bar{k}/k)$; since the projection $q : G_{\bar{k}} \rightarrow G$ is surjective and $\mathcal{G}$ acts transitively on the fibers of q , $U = q(V')$, where V' is the intersection of the $\mathcal{G}$ -conjugates of V .

Now let Z be the closed subset $G_{\bar{k}} - V$, endowed with the reduced closed subscheme structure. Since $G_{\bar{k}}$, and hence Z , is of finite presentation over $\bar{k}$, Z comes by base change from a reduced closed subscheme Z_1 of $G \otimes_k k_1$, for some finite Galois extension k_1 of k , so the $\mathcal{G}$ -conjugates of Z are finite in number, so their union is still a nowhere dense closed subset F of $G_{\bar{k}}$, and $V' = G_{\bar{k}} - F$ is a dense open subset of $G_{\bar{k}}$. Thus, since the projection $q : G_{\bar{k}} \rightarrow G$ is surjective and open, $U = q(V')$ is a dense open subset of G . Moreover, for every closed point g of G , the fiber $\pi^{-1}(g)$ consists of a finite number of closed points of G_K , so if $g \in U$, then $\pi^{-1}(g) \subset W$. This proves (i), and (ii) follows from it.

On the other hand, (iii)(a) has been proved in VIA, 2.6.4. Finally, if U is dense in every irreducible component of X , $X = G \cdot U$, so, for every $x \in X$, $\mu_x^{-1}(U_{\kappa(x)})$ is a nonempty open subset of $G_{\kappa(x)}$, and then (iii)(b) follows from (ii). $\square$

Corollary 5.6.2. *Let S, G, X be as in the preliminary hypotheses of 5.3, and let U be an open subset of X , $s \in S$, and A a finite subset of X_s . Suppose U_s dense in X_s and X_s locally of finite type over $\kappa(s)$.⁽⁴³⁾ Then there is a morphism $f : S'' \rightarrow S$, the composite of a finite surjective morphism $S'' \rightarrow S'$ and an étale morphism $S' \rightarrow S$, and a morphism $h : S'' \rightarrow G$, such that the inverse image A'' of A in $X \times_S S''$ (i.e. in $X_s \times_{\text{Spec } \kappa(s)} S''_s$) is contained in $\ell_h^{-1}(U_{S''})$, where ℓ_h denotes the translation of $X_{S''} = X \times_S S''$ defined by the element $h \in G(S'')$.*

PROOF. Since X_s is locally of finite type over $\kappa(s)$, the connected components of X_s are open, and irreducible (cf. VIA, 2.5.4), so U_s is dense in every irreducible component of X_s .

Thus, by Lemma 5.6.2.0, there is a closed point $g \in G_s^0$ such that $g \cdot a' \in U_s \otimes_{\kappa(s)} \kappa(g)$ for every $a' \in X_{\kappa(g)}$ above a point of A .

By Lemma 5.6.1, there is a morphism $f : S'' \rightarrow S$, the composite of a finite surjective morphism $S'' \rightarrow S'$ and an étale morphism $S' \rightarrow S$, and a morphism $h : S'' \rightarrow G$, such that $f^{-1}(s)$ consists of a single point s_0 , and $h(s_0) = g$ and $\kappa(s_0) = \kappa(g)$. Then, denoting by A'' the inverse image of A in $X_s \times_{\text{Spec } \kappa(s)} S''_s = X_s \otimes_{\kappa(s)} \kappa(s_0) = X_s \otimes_{\kappa(s)} \kappa(g)$, the translation ℓ_h of $X_{S''}$ sends A'' into $U_s \otimes_{\kappa(s)} \kappa(s_0)$. $\square$

Lemma 5.6.3. *Let X be an S -scheme. The following conditions are equivalent:*

- (i) X is separated over S .
- (ii) For every S -scheme T , every section $\sigma : T \rightarrow X_T$ is a closed immersion.
- (iii) For every reduced S -scheme T , two S -morphisms f_1 and $f_2 : T \rightarrow X$ coinciding on a dense open subset U of T are equal.
- (iv) For every $s \in S$ and every pair of points x_1, x_2 of X_s , there is a morphism $\phi : S'' \rightarrow S'$ and an open subscheme V of $X_{S''}$, separated over S'' , such that:
 - (a) $S' \rightarrow S$ is open, $S'' \rightarrow S'$ closed and surjective, and $\phi^{-1}(s) \neq \emptyset$.
 - (b) The inverse image of $\{x_1, x_2\}$ in $X_{S''}$ is contained in V .

⁽⁴³⁾**N.D.E.** The hypothesis on X_s has been added and the proof simplified (and corrected) by taking account of the addition 5.6.2.0. On the other hand, the proof shows that the conclusion holds if one supposes only that U_s is dense in every irreducible component of X_s .

(iv') For every $s \in S$, every pair of points $x_1, x_2 \in X_s$ is contained in an open subset V of X , separated over S .⁽⁴⁴⁾

The implication (ii) $\Rightarrow$ (i) is clear (take $T = X$ and σ = the diagonal section), as is (i) $\Rightarrow$ (iv') $\Rightarrow$ (iv). On the other hand, one has (i) $\Rightarrow$ (ii) by EGA I, 5.4.6.

(iii) $\Rightarrow$ (ii). Let $\sigma : T \rightarrow X_T$ be a section of $p : X_T \rightarrow T$. By EGA I, 5.3.13, σ is an immersion, i.e. an isomorphism of T onto a locally closed subscheme E of X_T . To show that E is closed, one may suppose T and E reduced. Let $\bar{E}$ be the reduced closed subscheme of X_T with the closure of E as underlying space, so that E is a dense open subscheme of $\bar{E}$. Then the immersion $i : \bar{E} \hookrightarrow X_T$ and $\sigma \circ p \circ i$ coincide on E , hence on $\bar{E}$ by hypothesis (iii). Thus every point of $\bar{E}$ belongs to $\sigma(T) = E$, whence $\bar{E} = E$.

(i) $\Rightarrow$ (iii). Suppose X separated over S , and let T be a reduced S -scheme, f_1, f_2 two S -morphisms $T \rightarrow X$ coinciding on a dense open subset U of T , and g the morphism $T \rightarrow X \times_S X$ with components f_1 and f_2 . Since $D = \Delta_{X/S}(X)$ is closed, its inverse image under g is a closed subset of T containing U , hence equal to T , and, since T is reduced, g factors through D (cf. EGA I, 5.2.2); consequently $f_1 = p_1 \circ g$ equals $p_2 \circ g = f_2$.

(iv) $\Rightarrow$ (iii). Let T be a reduced S -scheme and f_1, f_2 two S -morphisms $T \rightarrow X$ coinciding on a dense open subset U . Since T is reduced, to see that $f_1 = f_2$, it suffices to see that $f_1 = f_2$ as maps of sets. Indeed, suppose this established, and let $t \in T$, V an affine open subset of X containing $f_1(t) = f_2(t)$, and W the open neighborhood of t equal to the inverse image of V under the continuous map underlying f_1 and f_2 ; then the morphisms $f_i|_W : W \rightarrow V$ coincide on the dense open subset $U \cap W$ of W . Since V is separated and W reduced, this implies that $f_1|_W = f_2|_W$, whence $f_1 = f_2$.

Then let $t \in T$ and s be its image in S ; let us show that the points $x_1 = f_1(t)$ and $x_2 = f_2(t)$ of X_s are equal. Let $\phi : S'' \rightarrow S' \rightarrow S$ and V be an open subset of $X \times_S S''$ as in (iv); put $T' = T \times_S S'$ and $T'' = T \times_S S''$ and denote by $g : T'' \rightarrow T$ and $f'_i : T'' \rightarrow X_{S''}$ ($i = 1, 2$) the morphisms obtained by base change.

Since U is dense in T and $T' \rightarrow T$ is open, the inverse image U' of U in T' is dense in T' . Let U'' be the inverse image of U' in T'' and let F be the reduced subscheme of T'' with $\bar{U''}$ as underlying space. Since $T'' \rightarrow T'$ is surjective and closed, the image of F contains U' and is closed, hence equals T' . Consequently, $F \cap g^{-1}(t)$ contains a point u .

For $i = 1, 2$, denote by h_i the restriction of f'_i to F . Then $h_i(u)$ is a point of $X_{S''}$ above $f_i(t) = x_i$, hence belongs to V , since V contains the inverse image of $\{x_1, x_2\}$ in $X_{S''}$.

Then $W = h_1^{-1}(V) \cap h_2^{-1}(V)$ is an open subset of F containing u , and the S'' -morphisms $h_i|_W : W \rightarrow V$ coincide on the dense open subset $U'' \cap W$ of W . Since V is separated over S'' , one has $h_1|_W = h_2|_W$, whence $h_1(u) = h_2(u)$, hence $f_1(t) = f_2(t)$. This proves (iv) $\Rightarrow$ (iii).

Theorem 5.3 then follows from 5.6.2 and the implication (iv) $\Rightarrow$ (i) of 5.6.3.

Counterexample 5.6.4. Not every S -group G is separated. Let S be a scheme having a nonisolated closed point s ; let G be the scheme obtained by gluing two copies of S along the open subset $S - \{s\}$; one easily sees that G is not separated over S , and is endowed with a natural S -group structure, all of whose fibers are trivial except the fiber G_s , which is isomorphic to the two-element group $\mathbf{Z}/2\mathbf{Z}$.⁽⁴⁵⁾

⁽⁴⁴⁾**N.D.E.** The formulation of condition (iv) has been simplified and condition (iv') added. On the other hand, the proof of the implication (i) $\Rightarrow$ (iii), used in the proof of (iv) $\Rightarrow$ (iii), has been added. Observe moreover that if $T = \text{Spec } k[\varepsilon, x]/(\varepsilon^2, \varepsilon x)$ (k a field), $X = T_{\text{red}} = \text{Spec } k[x]$, the morphisms $\phi_\lambda : T \rightarrow X$ defined by $x \mapsto x + \lambda\varepsilon$ ($\lambda \in k$) coincide on the dense open subset $\text{Spec } B_x$ but are not equal.

⁽⁴⁵⁾**N.D.E.** This example has been reproduced in VIA 0.3, N.D.E. (5).

Theorem 5.7. ⁽⁴⁶⁾ *Let S be a scheme, G an S -group locally of finite presentation over S such that the function $s \mapsto \dim G_s$ is locally constant on S , X an S -scheme on which G acts, and U an open subset of X , separated over S . Then $\underline{G}^0 \cdot U$ is an open subset of X , separated over S .*

PROOF. Denote by $\mu : G \times_S X \rightarrow X$ the action of G on X ; it is the composite of the automorphism $(g, x) \mapsto (g, gx)$ of $G \times_S X$ and the projection onto X . Since $G \times_S U$ is an open subset of $G \times_S X$, it follows from 4.2 (iii) that $V = \underline{G}^0 \cdot U$ is open in X .

Then, proceeding as in the proof of 5.6.2, one deduces from the implication (iv) $\Rightarrow$ (i) of 5.6.3 that V is separated over S . $\square$

Corollary 5.7.1. *Let S and G be as in 5.7 and let σ, τ be two S -sections of G^0 (i.e. $\sigma, \tau \in G^0(S)$). Then the coincidence subscheme $S(\sigma, \tau) \subset S$ of σ and τ (i.e. the inverse image of the diagonal of $G \times_S G$ under the morphism (σ, τ)) is closed.*

Indeed, for every $s \in S$, let U be an affine open subset of G containing $\varepsilon(s)$; then $V = \varepsilon^{-1}(U)$ is an open neighborhood of s in S , and, since $G^0 U$ is separated, $S(\sigma, \tau) \cap V$ is closed in V .

Remark 5.7.2. Gabber points out to us that one can show that if S is henselian local with closed point s , the intersection of the open subsets $G^0 U$, as U ranges over the affine open neighborhoods of $\varepsilon(s)$, is an open subgroup scheme of G , separated over S .

5.8. — Complements. ⁽⁴⁷⁾ Let us begin by recalling Proposition VIA 2.6.6:

Proposition 5.8.1. *Let k be a field, G a k -group locally of finite type acting on a k -scheme X so that the morphism $G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$ is surjective. Suppose X quasi-separated. Then the connected components of X are irreducible.*

Corollary 5.8.2. *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that every fiber X_s is quasi-separated and connected. Then X is separated and quasi-compact over S .*

Indeed, by the preceding proposition, every fiber X_s is irreducible, and the rest of the proof is identical to that of 5.4.

Example 5.8.3. Fix an algebraically closed field k . First recall that every “locally Boolean” topological space X (i.e. having a basis of compact open subsets), endowed with the sheaf of locally constant functions with values in k , is a k -scheme (cf. [DG70], I §1, 2.12). One then says that X is a locally Boolean k -scheme.

Observe on the other hand that every topological space E admitting a basis of separated open subsets (and similarly every scheme X) admits a dense separated open subset. Indeed, since every increasing union of separated open subsets is a separated open subset (for a scheme X this follows from 5.6.3), there is such an open subset U which is maximal. But then U is dense, for if there were a nonempty open subset V such that $U \cap V = \emptyset$, V would contain a separated open subset $W \neq \emptyset$, and $U \cup W$ would still be separated, contradicting the maximality of U .

Now let C be the Cantor space, which one may regard as the space underlying the group $(\mathbf{Z}/2\mathbf{Z})^{\mathbf{N}}$ endowed with the product topology. For every point p of C , let $C(p)$ be another

⁽⁴⁶⁾**N.D.E.** On the one hand, Corollary 5.6.5, which repeated 5.5, has been removed. On the other hand, the original stated Corollary 5.7.1 below as Remark 5.7, referring for the proof to 4.7, a number which does not exist in Lect. Notes 151 (but which appeared in the 1965 edition of SGAD, whose nos. 4.5 and 4.6 have become 5.6.1 and 5.6.2); Theorem 5.7, communicated by O. Gabber, has been added; it makes the aforementioned SGAD statement more precise.

⁽⁴⁷⁾**N.D.E.** This paragraph of complements and counterexamples, all communicated by O. Gabber, has been added.

copy of C , and let X be the space obtained by gluing each $C(p)$ to C along $C - \{p\}$; then X is a nonseparated locally Boolean k -scheme, and C is a dense open subset of X .

Let G be the group of automorphisms of the k -scheme X (i.e. of homeomorphisms of X). Then G acts transitively on X . Indeed, X is the union of C and, for each point $p \in C$, of a second point p' , which can be characterized as the unique point x of $X - \{p\}$ such that $(C - \{p\}) \cup \{x\}$ is separated. It follows that every automorphism ϕ of C extends to an automorphism ϕ_X of X such that $\phi_X(p') = \phi_X(p)$ for every p . On the other hand, the map $\theta_p : X \rightarrow X$ which exchanges p and p' and fixes the other points is an automorphism of X . Finally, the group of automorphisms of C acts transitively on C : for example, using the group structure of C , it suffices to consider translations.

Thus the discrete k -group G (hence locally of finite type) acts on the k -scheme X so that the morphism $G \times X \rightarrow X \times X$, $(g, x) \mapsto (gx, x)$ is surjective, but X is not separated (although C is a dense separated open subset).

Example 5.8.4. Retain the notation of the preceding example. Using the description of C as the set of sequences $(u_n)_{n \in \mathbf{N}}$ of elements of $\mathbf{Z}/2\mathbf{Z}$, one sees that C with one point p removed is homeomorphic to a countable disjoint union of copies of C . Indeed, using for example the group structure of C , one reduces by translation to the case where p is the element 0, i.e. the zero sequence; then $C - \{0\}$ is the disjoint union of the subspaces $C_i = \{(u_n)_{n \in \mathbf{N}} \mid u_i = 1 \text{ and } u_j = 0 \text{ for } j < i\}$, for $i \in \mathbf{N}$, each homeomorphic to C . For every nonempty finite subset F of C , proceeding by induction on $|F|$ one deduces that $C - F$ is homeomorphic to a countable disjoint union of copies of C , hence to C with one point removed.

For each F of cardinality 2, let $C(F)$ be another copy of C , denote by q_F the point 0 of $C(F)$ and choose a homeomorphism $\phi_F : C(F) - \{q_F\} \xrightarrow{\sim} C - F$; then let X' be the space obtained by gluing each $C(F)$ to C by means of ϕ_F . Then X' is a nonseparated locally Boolean k -scheme. Moreover, it follows from the construction that every locally constant function $f : X' \rightarrow k$ is constant. Indeed, if $x, y \in C$ and $F = \{x, y\}$, every neighborhood of q_F meets every neighborhood of x or y , so if $f : X \rightarrow k$ is locally constant, one has $f(x) = f(q_F) = f(y)$, and if $F' = \{z, t\}$, with $z, t \in C$, one similarly has $f(q_{F'}) = f(z) = f(q_{\{z, x\}}) = f(x)$. Consequently, X' is connected.

Moreover, every point $x \in X'$ has a pointed neighborhood homeomorphic to $(C, 0)$. More precisely, fix for each F a homeomorphism of pointed topological spaces $h_F : (C, 0) \xrightarrow{\sim} (C(F), q_F)$ and denote by T the group of translations of C . Then, if $x = q_F$, one has the homeomorphism q_F , and if $x \in C$, the translation $t_x \in T$ is a homeomorphism $(C, 0) \xrightarrow{\sim} (C, x)$ (and it is the unique element of T with this property).

Denote by L the free group generated by the h_F and let $G = T * L$ be the “free product” (= coproduct) of T and L . For every $h \in \{h_F\}_{|F|=2} \cup T$, let $\sigma(h)$ be the generator of G corresponding to h and let $S(h)$ (resp. $B(h)$) be the source (resp. target) of h . It is convenient also to put $\sigma(h_F^{-1}) = \sigma(h_F)^{-1}$ and $S(h_F^{-1}) = B(h_F)$ (resp. $B(h_F^{-1}) = S(h_F)$), and to denote by E the set consisting of T , and the h_F and h_F^{-1} .

On the product space $P = G \times X'$ (where G is endowed with the discrete topology), consider the equivalence relation generated by the relations:

$$(g\sigma(h), x) \sim (g, h(x)) \quad \text{when } x \in S(h)$$

for every $h \in E$, and let Z be the quotient space. Then Z is obtained from the disjoint union $\coprod_{g \in G} \{g\} \times X'$ by gluing open subsets, and hence, for every open subset Ω of P , its saturation $\tilde{\Omega}$ is open (cf. [BTop], I §5.1, Example 2). Explicitly, since every open subset of P is the union of its intersections with the “slices” $\{g\} \times X'$, it suffices to consider an open subset of the form $\{1\} \times W$, where W is an open subset of X' . In this case, the saturation is the union

of $\{1\} \times W$ and of

$$\begin{aligned} & \{\sigma(h_1)\} \times h_1^{-1}(W \cap B(h_1)), \\ & \{\sigma(h_1)\sigma(h_2)\} \times h_2^{-1}(h_1^{-1}(W \cap B(h_1)) \cap B(h_2)), \end{aligned}$$

etc., for all finite sequences $h_1, \dots, h_n$ of elements of E , hence is open. Thus the projection $\pi : P \longrightarrow Z$ is open.

Note moreover that the word $\sigma(h_1) \cdots \sigma(h_n)$ is a reduced word of G , unless one of the h_i is the identity element of T , or two consecutive h_i belong to T , or (h_i, h_{i+1}) equals (h_F, h_F^{-1}) or (h_F^{-1}, h_F) . Thus, if $x \in X'$ and an element $\beta = (\sigma(h_1) \cdots \sigma(h_n), h_n^{-1} \cdots h_1^{-1}(x))$ belongs to $\{1\} \times X'$, one may suppose that every h_i is a translation t_i , and in this case the equality $\sigma(t_1 \cdots t_n) = 1$ implies that $t_1 \cdots t_n = 1$, hence $\beta = (1, x)$. Since the equivalence relation is compatible with the action of G (acting on $P = G \times X'$ by left translations on the first factor), one deduces that, for every $g \in G$, the restriction of π to $\{g\} \times X'$ is injective.

Then let $z \in Z$ be arbitrary and let $(g, x) \in P$ be a representative of z . By the preceding argument, $U = \pi(\{g\} \times X')$ is an open neighborhood of z , and the continuous map $\{g\} \times X' \longrightarrow U$ induced by π is open and bijective, hence a homeomorphism. This shows that Z is locally isomorphic to X' (hence also to C), so is again a locally Boolean k -scheme.

Finally, G acts transitively on Z . Indeed, since every $z \in Z$ is G -conjugate to an element of the form $\pi((1, x))$, it suffices to see that every element $(1, x) \in P$ is equivalent to an element $(\sigma(h), 0)$; if $x \in C$, one may take for h the translation t_x , and if $x = q_F$, one may take $h = h_F$. Thus Z is a locally Boolean k -scheme endowed with a transitive action of the discrete group G , but Z is not separated.

6. Subfunctors and subgroup schemes

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Definition 6.1. (i) Let X be an S -functor (that is, a functor from $(\mathbf{Sch}/S)^\circ$ to $(\mathbf{Ens})$), G an S -group functor, u and v two S -morphisms from X to G . One calls the *transporter of u into v* , and denotes by $\underline{\text{Transp}}(u, v)$, the sub- S -functor of G defined as follows:

$$\begin{aligned} \underline{\text{Transp}}(u, v)(S') &= \{g \in G(S') \mid (\text{int } g) \circ u_{S'} = v_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} u_{S''}(x) g_{S''}^{-1} = v_{S''}(x), \quad \forall x \in X(S''), S'' \longrightarrow S'\}. \end{aligned}$$

In particular, $\underline{\text{Transp}}(u, u)$ is a subgroup S -functor of G ; one calls it the *centralizer of u* and denotes it by $\underline{\text{Centr}}(u)$.

(ii) Let G be an S -group functor, X and Y two sub- S -functors of G ; one calls the *transporter of X into Y* (resp. *strict transporter of X into Y*), and denotes by $\underline{\text{Transp}}_G(X, Y)$ (resp. $\underline{\text{Transp}}_G^{\text{str}}(X, Y)$), the sub- S -functors of G defined as follows:

$$\begin{aligned} \underline{\text{Transp}}_G(X, Y)(S') &= \{g \in G(S') \mid (\text{int } g)(X_{S'}) \subset Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} \subset Y(S''), \quad \forall S'' \longrightarrow S'\}, \\ \text{resp. } \underline{\text{Transp}}_G^{\text{str}}(X, Y)(S') &= \{g \in G(S') \mid (\text{int } g)(X_{S'}) = Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} = Y(S''), \quad \forall S'' \longrightarrow S'\}. \end{aligned}$$

Note that one has

$$\underline{\text{Transp}}_G^{\text{str}}(X, Y) = \underline{\text{Transp}}_G(X, Y) \cap c(\underline{\text{Transp}}_G(Y, X)),$$

*On this same subject, see also the results of XI 6, whose natural place would be in the present Exposé VIB. In particular one will find there representability criteria for certain subgroup functors of a given group scheme.⁽⁴⁸⁾

⁽⁴⁸⁾**N.D.E.** These results have been inserted in what follows (nos. 6.5.2 to 6.5.5).

where c denotes the inversion morphism of G .⁽⁴⁹⁾

(iii) Let G be an S -group functor, H a sub- S -functor of G , i the canonical S -morphism $H \rightarrow G$; one calls the *centralizer* and *normalizer* of H in G the following subgroup S -functors of G :

$$\underline{\text{Centr}}_G H = \underline{\text{Centr}}(i) = \underline{\text{Transp}}(i, i), \quad \underline{\text{Norm}}_G H = \underline{\text{Transp}}^{\text{str}}_G(H, H).$$

Finally, one calls the *center* of G the S -group functor $\underline{\text{Centr}}(\text{id}_G) = \underline{\text{Centr}}_G G$; it will be denoted by $\underline{\text{Centr}}G$.

Remark 6.1.1. ⁽⁵⁰⁾ It follows from the definitions that the functors $\underline{\text{Transp}}(u, v)$ and $\underline{\text{Transp}}_G(X, Y)$ (and hence also $\underline{\text{Transp}}^{\text{str}}_G(X, Y)$) commute with base change: for every $S' \rightarrow S$, if G', X', Y', u', v' are obtained from G, X, Y, u, v by base change, one has:

$$\underline{\text{Transp}}(u, v)_{S'} = \underline{\text{Transp}}(u', v') \quad \text{and} \quad \underline{\text{Transp}}(X, Y)_{S'} = \underline{\text{Transp}}(X', Y').$$

Proposition 6.2. ⁽⁵¹⁾ Let G be an S -group. Consider, for a subfunctor T of the functor G , the following property:

(+_f) for every $s \in S$, T_s is representable by a closed subscheme of G_s .

Let $u, w : X \rightarrow G$ and $v : Y \rightarrow G$ be morphisms of S -schemes. Then:

- (i) $\underline{\text{Transp}}(u, w)$ and $\underline{\text{Centr}}(u) = \underline{\text{Transp}}(u, u)$ satisfy condition (+_f).
- (ii) $\underline{\text{Transp}}_G(u(X), v(Y))$ and $\underline{\text{Norm}}_G v(Y)$ satisfy condition (+_f) if, for every $s \in S$, v_s is a closed immersion.
- (iii) $\underline{\text{Transp}}^{\text{str}}_G(X, Y)$ satisfies condition (+_f) if, for every $s \in S$, u_s and v_s are closed immersions.

This follows from Remark 6.1.1 and Corollary 6.2.5 below.⁽⁵²⁾

Definition 6.2.1. Let $f : X \rightarrow S$ be a morphism of schemes. One says that f is *essentially free*, or that X is *essentially free over S* , if one can find a covering of S by affine open subsets S_i , for every i an S_i -scheme S'_i affine and faithfully flat over S_i , and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine open subsets X'_{ij} , such that, for every (i, j) , the ring of X'_{ij} is a projective module⁽⁵³⁾ over the ring of S'_i .

Proposition 6.2.2.

- (a) If X is essentially free over S , it is flat over S , the converse holding if S is artinian.
- (b) If S is the spectrum of a field, every S -scheme is essentially free over S .

⁽⁴⁹⁾**N.D.E.** If $u : X \rightarrow G$ and $v : Y \rightarrow G$ are two arbitrary morphisms of S -functors, let $u(X)$ be the image functor of u , defined by $u(X)(S') = u(S')(X(S')) \subset G(S')$; it is a subfunctor of G , as is the image functor $v(Y)$; one may then consider the transporter of the image of u into the image of v , $\underline{\text{Transp}}_G(u(X), v(Y))$. One thus sees that, in definition (ii), it is not necessary to restrict oneself to subfunctors, i.e. to the case where u and v are monomorphisms. This restriction sometimes imposed repetitions in the hypotheses in the original, such as: “Let $u, v : X \rightarrow G$ be morphisms of S -functors, $w : H \rightarrow G$ and $w' : K \rightarrow G$ monomorphisms; then $\underline{\text{Transp}}(u, v)$, $\underline{\text{Centr}}_G(w) = \underline{\text{Centr}}_G H$ and $\underline{\text{Transp}}_G(H, K)$ satisfy...”, which can be avoided by considering $\underline{\text{Transp}}_G(u(X), v(Y))$. Such modifications have been made in 6.2 and, later, in 10.11.

⁽⁵⁰⁾**N.D.E.** This remark, implicitly used in Proposition 6.2, has been added.

⁽⁵¹⁾**N.D.E.** The statement has been modified as indicated in N.D.E. (49).

⁽⁵²⁾**N.D.E.** The original referred to the results of Exp. VIII, §6. For the reader's convenience, these results (except VIII, 6.3 and 6.8) have been reproduced in nos. 6.2.1 to 6.2.5 below. Moreover, this was suggested by A. Grothendieck in a Note at the beginning of VIII.6: “The present number is independent of the theory of diagonalizable groups; its natural place would be in VIB.”

⁽⁵³⁾**N.D.E.** On the one hand, the word “free” has been replaced by “projective”, as indicated in VIII, Remark 6.8. On the other hand, the notion of an essentially free S -scheme is intimately related to the geometric notion of a flat and pure S -scheme, introduced and studied by M. Raynaud and L. Gruson ([RG71]), cf. addition 6.9 below.

- (c) If X is essentially free over S , $X' = X \times_S S'$ is essentially free over S' , for every $S' \rightarrow S$. The converse holds if $S' \rightarrow S$ is faithfully flat and quasi-compact.

The proof is immediate, using for the converse in a) the fact that a flat module over an artinian local ring is free.⁽⁵⁴⁾

The introduction of Definition 6.2.1 is justified by the following

Theorem 6.2.3. *Let S be a scheme, Z an essentially free S -scheme, Y a closed subscheme of Z . Consider the functor $F = \prod_{Z/S} Y/Z : (\mathbf{Sch})_S^\circ \rightarrow (\mathbf{Ens})$ defined by the following condition: $F(S') = \emptyset$ when $Y_{S'} \neq Z_{S'}$, $F(S')$ consists of one element in the opposite case.⁽⁵⁵⁾*

- (i) *This functor is representable by a closed subscheme T of S .*
- (ii) *If moreover $Y \rightarrow Z$ is of finite presentation, so is $T \rightarrow S$.*

First note that the functor under consideration is a sheaf for the (fpqc) topology: since $F(S') = \emptyset$ or $\{\text{pt}\}$ for every S' , this reduces to checking that if (S_i) is an open covering of S (resp. $S' \rightarrow S$ a faithfully and quasi-compact morphism), and each $Y_{S_i} \rightarrow Z_{S_i}$ (resp. $Y_{S'} \rightarrow Z_{S'}$) is an isomorphism, the same holds for $Y \rightarrow Z$; this is clear (resp. follows from SGA 1, VIII 5.4 or EGA IV₂, 2.7.1).

Moreover, by SGA 1, VIII 1.9, faithfully flat quasi-compact morphisms are of effective descent for the fibered category of closed-immersion arrows. With the notation of 6.2.1, this allows us to restrict ourselves to the case where $S = S'_i$.

Then let (Z_j) be a covering of Z by affine open subsets such that $\mathcal{O}(Z_j)$ is a projective module over $A = \mathcal{O}(S)$, and let $Y_j = Y \cap Z_j$ and $F_j : (\mathbf{Sch})_S^\circ \rightarrow (\mathbf{Ens})$ be the functor defined in terms of (Z_j, Y_j) as F is in terms of (Z, Y) . It is a subfunctor of the final functor, and one evidently has $F = \bigcap_j F_j$, which reduces us to proving that every F_j is representable by a closed subscheme T_j of S (for then F will be representable by the closed subscheme T which is the intersection of the T_j). One may therefore suppose Z also affine, $Z = \text{Spec}(B)$, where B is a projective A -module. Let L be a free A -module with basis $(e_\lambda)_{\lambda \in \Lambda}$, of which B is a direct summand as an A -module, and let $\varphi_\lambda : L \rightarrow A$ be the “coordinate” forms with respect to this basis. Let E be a set of generators of the ideal J of B defining the subscheme Y of Z , and let I be the ideal in A generated by the coordinates $\varphi_\lambda(x)$, for $x \in E$. For every A -algebra C , one then sees that the morphism $B \otimes_A C \rightarrow (B/J) \otimes_A C$ is an isomorphism if and only if the image of $x \otimes 1$ in $L \otimes_A C$ is zero for every $x \in E$, which is equivalent to saying that the kernel of $A \rightarrow C$ contains the ideal I . This shows that $T = \mathcal{V}(I) = \text{Spec}(A/I)$ satisfies the desired condition, which proves (i).

Moreover, if $Y \rightarrow Z$ is of finite presentation, one may take E finite, and then I is a finitely generated ideal of A , i.e. the closed immersion $T \rightarrow S$ is of finite presentation.

Examples 6.2.4. Let us give important examples of functors which reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.2.3 and for which it is useful subsequently to have representability criteria. We denote by S a scheme, by X, Y, Z , etc. schemes over S .

a) Suppose given an S -morphism

$$(x) \quad q : X \rightarrow \underline{\text{Hom}}_S(Y, Z),$$

(“ X acts on Y , with values in Z ”), i.e. a morphism

$$(xx) \quad r : X \times_S Y \rightarrow Z.$$

⁽⁵⁴⁾**N.D.E.** Indeed, let $(A, \mathfrak{m})$ be an artinian local ring, k its residue field, M an arbitrary A -module, $(x_i)_{i \in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k . Let F be the free A -module with basis $(e_i)_{i \in I}$, and $\phi : F \rightarrow M$ the A -morphism defined by $\phi(e_i) = x_i$. Then $Q = \text{Coker } \phi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose moreover M flat over A ; then $K = \text{Ker } \phi$ satisfies $K \otimes_A k = 0$, i.e. $K = \mathfrak{m}K$, whence $K = 0$.

⁽⁵⁵⁾**N.D.E.** Cf. Exp. II §1, where this functor is denoted by $\prod_{Z/S} Y$; for every $S' \rightarrow S$, $F(S') = \Gamma(Y_{S'}/Z_{S'})$, which here equals $\{\text{id}_{Z_{S'}}\}$ if $Y_{S'} = Z_{S'}$, and is empty otherwise. On the other hand, assertion (ii), used in the proof of 6.5.3, has been added.

Consider a subscheme Z' of Z , whence a monomorphism

$$\underline{\mathrm{Hom}}_S(Y, Z') \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z)$$

which makes the first functor a subfunctor of the second; let X' be the inverse image of this subfunctor under the morphism q ; it is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \longrightarrow Z_T$ factors through Z'_T . This functor X' can be described as follows: put $P = X \times_S Y$, and let P' be the inverse image of Z' under $r : P \longrightarrow Z$; then one has an evident isomorphism

$$(xxx) \quad X' \simeq \prod_{P/X} P'/P.$$

One therefore obtains: if Y is essentially free over S and Z' closed in Z , the subfunctor X' of X is representable by a closed subscheme of X .⁽⁵⁶⁾ If moreover $Z' \longrightarrow Z$ is of finite presentation, so is $X' \longrightarrow X$.

b) Suppose given two ways of making X act on Y with values in Z , i.e. two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\mathrm{Hom}}_S(Y, Z),$$

and put $X' = \mathrm{Ker}(q_1, q_2)$: it is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms $q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$ are equal. Now giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Z \times_S Z),$$

or a morphism $r : X \times_S Y \longrightarrow Z \times_S Z$; then put $U = Z \times_S Z$ and let U' be the diagonal subscheme of $Z \times_S Z$; then X' is none other than the inverse image of the subfunctor $\underline{\mathrm{Hom}}_S(Y, U')$ of $\underline{\mathrm{Hom}}_S(Y, U)$ under q , hence can be put in the form (xxx), with $P = X \times_S Y$, and $P' =$ inverse image of the diagonal under r , i.e. kernel of $X \times_S Y \xrightarrow[r_2]{r_1} Z$. One is therefore under the conditions of (a).

It follows that: if Y is essentially free over S and Z separated over S , the subfunctor X' of X is representable by a closed subscheme of X .⁽⁵⁶⁾ If moreover $Z \longrightarrow S$ is locally of finite type, $X' \longrightarrow X$ is of finite presentation.

c) Suppose given a morphism

$$q : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y),$$

i.e. “ X acts on Y ”. Let X' be the “kernel” of this morphism, i.e. the subfunctor X' of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \longrightarrow Y_T$ is the identity. This functor falls under b), as one sees by introducing a second homomorphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

“by making X act trivially on Y ”. Thus: if Y is essentially free and separated over S , the subfunctor X' which is the kernel of q is representable by a closed subscheme of X .⁽⁵⁶⁾ If moreover $Y \longrightarrow S$ is locally of finite type, $X' \longrightarrow X$ is of finite presentation.

d) Under the conditions of c), consider the subfunctor Y' of Y “of invariants under X ”; thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $q(y) : X_T \longrightarrow Y_T$ is “the constant T -morphism with value y ”. Introducing q' as in c), and the homomorphisms corresponding to q and q' :

$$q, q' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that Y' is precisely $\mathrm{Ker}(q, q')$, and therefore again falls under b) (with the roles of X, Y reversed and $Z = Y$).

Consequently, if X is essentially free over S , Y separated over S , the subfunctor Y' of Y of invariants under X is representable by a closed subscheme of Y .⁽⁵⁶⁾ If moreover $Y \longrightarrow S$ is locally of finite type, $Y' \longrightarrow Y$ is of finite presentation.

⁽⁵⁶⁾**N.D.E.** The following sentence has been added.

e) Constructions of the type made explicit in the preceding examples are particularly frequent in group theory. Thus, when G is an S -group scheme acting on the S -scheme X ,

$$q : G \longrightarrow \underline{\text{Aut}}_S(X),$$

the kernel of q (“the subgroup of G acting trivially”) is a closed subscheme of G provided X is essentially free and separated over S (example c)), and the subobject X^G of invariants is a closed subscheme of X provided G is essentially free over S and X separated over S (example d)).

Let Y, Z be subschemes of X ; consider the subfunctor $\underline{\text{Transp}}_G(Y, Z)$ of G (“transporter of Y into Z ”), whose points with values in a T over S are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subset Z_T$, i.e. induces a morphism $Y_T \longrightarrow X_T$ factoring through $Y_T \longrightarrow Z_T$. Thus: if Y is essentially free over S , and Z closed in X , $\underline{\text{Transp}}_G(Y, Z)$ is a closed subscheme of G (example a)).

One may also consider the strict transporter of Y into Z ,⁽⁵⁷⁾ whose points with values in a T over S are the $g \in G(T)$ such that $g(Y_T) = Z_T$, which is none other than $\underline{\text{Transp}}_G(Y, Z) \cap \sigma(\underline{\text{Transp}}_G(Z, Y))$, where σ is the inversion morphism of G . Consequently, if Y and Z are essentially free over S and closed in X , the strict transporter of Y into Z is a closed subscheme of G .

An important case is that where $X = G$, G acting on itself by inner automorphisms. If H is a subscheme of G , the strict transporter of H into H is also called the *normalizer of H in G* , and denoted by $\underline{\text{Norm}}_G H$. Thus: if H is a closed subgroup scheme of G , essentially free over S , $\underline{\text{Norm}}_G H$ is representable by a closed subgroup scheme of G .

Finally let Z be a subscheme of G ; then its centralizer $\underline{\text{Centr}}_G(Z)$ in G is the subgroup functor of G defined by the procedure of d), when one considers that “ Z acts on G ” by the actions induced by those of G ; thus, if Z is essentially free over S and G is separated over S , $\underline{\text{Centr}}_G(Z)$ is a closed subgroup scheme of G . In particular, if G is essentially free and separated over S , the center C of G , which is none other than $\underline{\text{Centr}}_G(G)$, is a closed subgroup scheme of G .

When S is the spectrum of a field, 6.2.2 b) shows that in examples a) to e) above the conditions “essentially free” are automatically satisfied; only separation conditions remain. Recalling that a group scheme over a field is necessarily separated (VIA, 0.3), one finds for example:

Corollary 6.2.5. *Let G be a group scheme over a field k and let Y, Y' be two subschemes of G . Then:*

- (i) *The centralizer of Y in G is a closed subgroup scheme of G ; this is in particular the case for the center $\underline{\text{Centr}}_G(G)$ of G .*
- (i') *More generally, if $u, v : X \longrightarrow G$ are morphisms of schemes, $\underline{\text{Transp}}_G(u, v)$ is representable by a closed subscheme of G .*
- (ii) *If Y is closed, the transporter $\underline{\text{Transp}}_G(Y', Y)$ is a closed subscheme of G . If Y' is also closed, one has the same conclusion for $\underline{\text{Transp}}_G^{\text{str}}(Y', Y)$.*
- (iii) *For every subgroup scheme⁽⁵⁸⁾ H of G , $\underline{\text{Norm}}_G(H)$ is a closed subgroup scheme of G .*

Corollary 6.2.6.⁽⁵⁹⁾ *Let k be a field, G a connected algebraic k -group. Then $\underline{\text{Centr}}_G(G)$ is representable by a closed subgroup scheme Z of G , and G/Z is an affine algebraic k -group.*

⁽⁵⁷⁾**N.D.E.** Denoted by $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.

⁽⁵⁸⁾**N.D.E.** Indeed, over a field k , every subgroup scheme of G is closed, cf. VIA, 0.5.2. On the other hand, 6.2.5 completes the insertion of the results taken from VIII §6.

⁽⁵⁹⁾**N.D.E.** This corollary (cf. Exp. XII, 6.1), which will be useful later, has been inserted here. Moreover, in 6.3 one returns to the original text of VIB.

PROOF. Of course, the first assertion is contained in 6.2.5 (i), but one will see that it also follows from the proof of the second assertion. Indeed, G acts by the adjoint representation on the finite-dimensional k -vector spaces $V_n = \mathcal{O}_{G,e}/\mathfrak{m}_e^{n+1}$ (where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$); denote by K_n the kernel of the morphism $\rho_n : G \rightarrow \mathrm{GL}(V_n)$. By VIA 5.4.1, ρ_n induces a closed immersion $G/K_n \hookrightarrow \mathrm{GL}(V_n)$, so every G/K_n is affine. Since G is noetherian, the intersection K of the K_n equals one of the K_n , hence G/K is affine.

On the other hand, denoting by Z the center of G , it is clear that $Z \subset K$. Let us show that $Z = K$. Denote by $\widehat{\mathcal{O}}_{G,e}$ the completion of $\mathcal{O}_{G,e}$ for the $\mathfrak{m}_e$ -adic topology and by $\widehat{S}$ its spectrum (resp. $S = \mathrm{Spec} \mathcal{O}_{G,e}$). Since $\widehat{S} \rightarrow S$ is faithfully flat and the two morphisms

$$K \times_k S \rightarrow S, \quad (g, x) \mapsto gxg^{-1} \quad \text{resp.} \quad (g, x) \mapsto x$$

coincide after the base change $\widehat{S} \rightarrow S$, they coincide, i.e. K acts trivially on $\mathcal{O}_{G,e}$. Now, by 6.2.4 e), the subobject G^K of invariants of G under K (which is none other than $\mathrm{Centr}_G(K)$) is a closed subscheme of G , hence defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_G$. Since G^K contains $S = \mathrm{Spec} \mathcal{O}_{G,e}$ and $\mathcal{I}$ is of finite type (since G is noetherian), there is an open neighborhood U of e such that $\mathcal{I}|_U = 0$. Then the subgroup $G^K = \mathrm{Centr}_G(K)$ contains U , hence also $U \cdot U$, which equals G since G is irreducible (VIA 0.5). Thus $\mathrm{Centr}_G(K) = G$, whence $K \subset Z$, hence $Z = K$. $\square$

Remark 6.3. Let k be an algebraically closed field, G a k -group and H a subgroup scheme of G ; suppose G and H of finite type over k and reduced; then $\mathrm{Norm}_G H$ (resp. $\mathrm{Centr}_G H$) is representable by a subgroup scheme of G , whose associated reduced subscheme is none other than the normalizer (resp. centralizer) of H in G in the sense of the *Bible*.

Proposition 6.4. *Let G be an S -group and $u : X \rightarrow G$ a monomorphism of S -schemes. Put $T = \mathrm{Transp}_G(X, X)$. The following conditions are equivalent:*

- (i) T is a subgroup functor of G .
- (ii) $T = \mathrm{Transp}_G^{\mathrm{str}}(X, X) = \mathrm{Norm}_G X$.

These conditions hold in each of the following two cases:

- (a) X is of finite presentation over S .
- (b) T is representable by a scheme of finite presentation over S .

The equivalence of conditions (i) and (ii) follows from the fact that, for every morphism $S' \rightarrow S$ and every $t, t' \in T(S')$, one has $tt' \in T(S')$, and from the fact that $\mathrm{Transp}_G^{\mathrm{str}}(X, X) = T \cap c(T)$ (cf. 6.1 (ii)).

Consider case a). Let $t \in T(S)$; then $\mathrm{int}(t)$ is a monomorphism of X into X , hence an S -automorphism of X (EGA IV₄, 17.9.6), so t belongs to $\mathrm{Transp}_G^{\mathrm{str}}(X, X)$, whence a).

In case b), it is clear that $\mu(T \times_S T) \subset T$, and the assertion follows from the following lemma:

Lemma 6.4.2. ⁽⁶⁰⁾ *Let G be an S -scheme of finite presentation, endowed with an associative law (in the sense of EGA 0_{III} 8.2.5). Suppose that, for every S -scheme S' and every $g \in G(S')$, right and left translations by g in the set $G(S')$ are injective and $G(S) \neq \emptyset$. Then G is an S -group.*

It suffices to show that, for every S -scheme S' , the set $G(S')$ is a group; now it follows at once from the hypothesis that right and left translations by every element $g \in G(S')$ in $G_{S'}$ are S -monomorphisms from $G_{S'}$ to $G_{S'}$. They are therefore S -automorphisms, since G is of finite presentation over S (EGA IV₄, 17.9.6), so right and left translations by g in the set $G(S')$ are bijective, and one is reduced to the following lemma:

⁽⁶⁰⁾**N.D.E.** The numbering of the original has been retained: there is no no. 6.4.1.

Lemma 6.4.3. *Let G be a nonempty set endowed with an associative law such that right and left translations are bijective. Then G is a group.*

The proof is left to the reader.

Definition 6.5. Let G be an S -group, H an S -functor, and $u : H \rightarrow G$ a monomorphism.

- (i) One calls the *connected centralizer of H in G* , and denotes by $\underline{\text{Centr}}_G^0 H$, the identity component of the functor $\underline{\text{Centr}}_G H$ (cf. 3.1 and 6.1 (iii)).
- (i') For every morphism $u : X \rightarrow G$, one similarly defines the functor $\underline{\text{Centr}}^0(u)$ (cf. 6.2 (iv)).
- (ii) When, for every $s \in S$, u_s is a closed immersion, one calls the *connected normalizer of H in G* , and denotes by $\underline{\text{Norm}}_G^0 H$, the identity component of the functor $\underline{\text{Norm}}_G H$.

N.B. By 1.4.2, the hypothesis of (ii) holds when H is an S -group scheme, G and H are locally of finite type over S and u is a quasi-compact morphism of S -groups.

Proposition 6.5.1. *Let G be an S -group locally of finite presentation and quasi-separated over S , H a smooth S -group with connected fibers and $u : H \rightarrow G$ a monomorphism. Let N be the normalizer of H in G (cf. 6.1). By 6.5.5 below, N is representable by a closed subgroup scheme of G and of finite presentation over G . This being so, the following conditions are equivalent:*

- (i) *The canonical morphism $H \rightarrow N$ is an open immersion.*
- (ii) *$N^0 = H$ (cf. 3.10).*
- (iii) *For every $s \in S$, one has $H_s = (N_s)^0$.*

Condition (i) implies (ii) by Lemma 3.10.1, since $H^0 = H$. On the other hand, it is clear that (ii) implies (iii), for $H_s = (N^0)_s = (N_s)^0$.

Finally let us show that (iii) implies (i). Since $H_s = (N^0)_s$ for every $s \in S$, $H_s \rightarrow N_s$ is an open immersion. Moreover, H and N are locally of finite presentation over S , and H is flat over S , so (EGA IV₄, 17.9.5), $H \rightarrow N$ is an open immersion.

⁽⁶¹⁾For the reader's convenience, the results 6.8 to 6.11 of Exp. XI have been included below.

Theorem 6.5.2. *Let X be a smooth scheme over S , with geometrically irreducible fibers.⁽⁶²⁾*

- (i) *For every closed subscheme Y of X , the functor $\prod_{X/S} Y/X$ is representable by a closed subscheme T of S .*
- (ii) *Moreover, if $Y \rightarrow X$ is of finite presentation, so is $T \rightarrow S$.*

Since $f : X \rightarrow S$ is faithfully flat and locally of finite presentation, it is covering for the (fpqc) topology. Since, on the other hand, $T = \prod_{X/S} Y/X$ is evidently a subsheaf of S for the (fpqc) topology, it follows that the question of the representability of T by a closed subscheme of S is local on S for the (fpqc) topology,⁽⁶³⁾ and the same holds for the question of deciding whether T is of finite presentation over S (cf. EGA IV₂, 2.7.1). Making the base change $S' \rightarrow S$ with $S' = X$, one is then reduced to the case where X admits a section ε over S . One may moreover suppose S affine and a fortiori quasi-compact. One then has:

Corollary 6.5.3. *Under the conditions of 6.5.2, suppose that S is quasi-compact, $X \rightarrow S$ admits a section ε , and $Y \rightarrow X$ is of finite presentation. Then there is an integer $n \geq 0$*

⁽⁶¹⁾**N.D.E.** Nos. 6.5.2 to 6.5.5, taken from Exp. XI, 6.8 to 6.11, have been inserted here. This was moreover suggested by A. Grothendieck in a Note at the beginning of XI 6: "The present number does not use the results of nos. 3, 4, 5 (of XI); its natural place would be in VIB."

⁽⁶²⁾**N.D.E.** On the one hand, the original has been corrected by replacing "connected" by "irreducible" (cf. the proof). On the other hand, by [RG71] I, 3.3.4 (iii) and 4.1.1, it suffices to suppose that X is flat over S , with geometrically irreducible fibers and without embedded components.

⁽⁶³⁾**N.D.E.** See for example addition 1.7 in Exp. VIII.

such that one has

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n is the n -th infinitesimal neighborhood of the immersion $\varepsilon : S \rightarrow X$, and $Y_n = Y \cap X_n$.

When Y is of finite presentation over X , this corollary does imply 6.5.2, for, since X is smooth over S , X_n is finite and locally free over S , hence a fortiori “essentially free” over S (cf. 6.2.1), so $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subscheme T of S , of finite presentation over S , by 6.2.3.

Let us first prove 6.5.3 (and hence 6.5.2) when S is noetherian. Let $T_n = \prod_{X_n/S} Y_n/X_n$; then the T_n form a decreasing sequence of closed subschemes of S , and, since S is noetherian, this sequence is stationary. Let $R = \bigcap_{n \geq 0} \prod_{X_n/S} Y_n/X_n$ be their common value for large n ; one evidently has $T \subset R$, and it suffices to establish that $R \subset T$. Making the base change $R \rightarrow S$, one is reduced to the case where $R = S$, i.e. $Y_n = X_n$ for every n , i.e. $Y \supset X_n$ for every n , and one must then prove that $T = S$, i.e. $Y = X$.

Now $Y \supset X_n$ for every n implies (thanks to the fact that X is locally noetherian) that Y is, in a neighborhood of each point of $\varepsilon(S)$, an induced open subscheme of X ,⁽⁶⁴⁾ so there is an induced open subset U of X , containing $\varepsilon(S)$, such that $U \subset Y$. By EGA IV₃, 11.10.10, the fibers of X/S being integral, U is schematically dense in X , hence (Y being a closed subscheme containing U) one has $Y = X$. This proves 6.5.3, hence 6.5.2, when S is noetherian. The general case proceeds by reduction to the preceding case. For every $s \in S$, there is an affine open neighborhood U of s and an affine open neighborhood V of $\varepsilon(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U , and if S' is an affine open neighborhood of s contained in $\varepsilon^{-1}(V) \cap f(V)$, and $X' = V \cap f^{-1}(S')$, then X' and S' are affine open subsets of X resp. S , and X'/S' admits a section. Since 6.5.2 and 6.5.3 are local, one may suppose $S = S'$. Then X' is an affine open subset of X containing $\varepsilon(S)$. Since every fiber X_s is supposed irreducible, X'_s is schematically dense in X_s , hence, by EGA IV₃, 11.10.10, X' is schematically dense in X ; similarly, for every base change $S_1 \rightarrow S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$.

It follows that $\prod_{X/S} Y/X = \prod_{X'/S} Y'/X'$, where $Y' = Y \cap X'$. This reduces us to the case where $X = X'$, so one may suppose S and X affine. Moreover, if $X = \text{Spec}(B)$ and J is the ideal of B defining Y , then J is the inductive limit of its finitely generated subideals, so Y is the intersection of closed subschemes Y_i of X of finite presentation over X , and consequently $\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X$, which reduces us, to prove 6.5.2, to the case where Y is of finite presentation over X , with S and X affine.

But then X and Y over S come, by base change $S \rightarrow S_0$, from an analogous situation X_0 and Y_0 over S_0 , with S_0 noetherian, which reduces us to the case where S is noetherian, already treated. This completes the proof of 6.5.2 and 6.5.3.

Corollary 6.5.4. *Let X be a smooth S -group scheme of finite presentation with connected fibers, Y a group scheme of finite presentation over S , $i : Y \rightarrow X$ a monomorphism of S -group schemes.*

- (i) *Then $\prod_{X/S} Y/X$ is representable by a closed subscheme of finite presentation of S .*
- (ii) *If moreover S is quasi-compact, for sufficiently large n one has:*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n denotes the n -th infinitesimal neighborhood of the unit section $\varepsilon : S \rightarrow X$, and $Y_n = X_n \cap Y$.

⁽⁶⁴⁾**N.D.E.** See, for example, the proof of 6.2.6.

The proof is essentially that of 6.5.3.⁽⁶⁵⁾ On the one hand, i is locally of finite presentation (cf. EGA IV₁, 1.4.3). On the other hand, the unit sections of Y and X induce bijective immersions $S \rightarrow Y_n$ and $S \rightarrow X_n$, hence isomorphisms of S_{red} with $(Y_n)_{\text{red}}$ and $(X_n)_{\text{red}}$. Consequently, i_n is quasi-compact, hence of finite type, and one has a commutative diagram:

$$\begin{array}{ccc} (Y_n)_{\text{red}} & \xrightarrow{\tau} & Y_n \\ & \searrow \sigma & \downarrow i_n \\ & & X_n \end{array}$$

where σ, τ are closed immersions and τ is surjective. Since i_n is separated (being a monomorphism), it follows that i_n is proper (cf. EGA II, 5.4.3). Thus i_n is a proper monomorphism of finite presentation, hence a closed immersion (cf. EGA IV₃, 8.11.5). Consequently, by 6.2.3, already used, $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subscheme T_n of S of finite presentation over S , and it therefore remains to prove the last assertion of 6.5.4 in the case where S is moreover supposed affine.

One again immediately reduces to the case where S is noetherian, and one is reduced to proving that one then has $R = T$ (with the notation of the proof of 6.5.3), or that $Y \supset X_n$ for every n implies $Y = X$. Now the hypothesis implies that $i : Y \rightarrow X$ is étale at the points of the unit section of Y over S , so Y is smooth over S at the points of the unit section, whence it follows that the open subset U of points of Y at which Y is smooth over S is an induced open subgroup of Y (cf. 2.3). Then $\tau : U \rightarrow X$ is an étale monomorphism by 2.5, hence an open immersion; now, since the fibers of X are connected and every open subgroup of an algebraic group is also closed, it follows that τ is a surjective open immersion, i.e. an isomorphism. Thus $U = X$ and a fortiori $Y = X$, which completes the proof of 6.5.4.

Proceeding as in 6.2.4 e), one concludes from 6.5.4:

Corollary 6.5.5. *Let G be an S -group scheme locally of finite type and quasi-separated over S , H a smooth group scheme of finite presentation over S with connected fibers, $i : H \rightarrow G$ a monomorphism of S -groups. Then:*

- (a) $\underline{\text{Centr}}_G(H)$ and $\underline{\text{Norm}}_G(H)$ are representable by closed subschemes of G , of finite presentation over G .
- (a') Similarly, for every monomorphism $j : K \rightarrow G$ of finite presentation of S -group schemes, $\underline{\text{Transp}}_G(H, K)$ is representable by a closed subscheme of G , of finite presentation over G .
- (b) If G is quasi-compact, there is an integer $n \geq 0$ such that (if H_n denotes the n -th infinitesimal neighborhood of the unit section of H) one has:

$$\begin{aligned} \underline{\text{Centr}}_G(H) &= \underline{\text{Centr}}_G(H_n), & \underline{\text{Norm}}_G(H) &= \underline{\text{Norm}}_G(H_n), \\ \underline{\text{Transp}}_G(H, K) &= \underline{\text{Transp}}_G(H_n, K) = \underline{\text{Transp}}_G(H_n, K_n). \end{aligned}$$

PROOF. ⁽⁶⁶⁾First note that the hypothesis on G implies that the monomorphism $H \rightarrow G$ is of finite presentation (EGA IV₁, 1.2.4 and 1.4.3), as is the diagonal immersion $\Delta_{G/S} : G \rightarrow G \times_S G$ (loc. cit. 1.4.3.1). The case of $\underline{\text{Norm}}_G(H)$ therefore reduces to that of the transporter by taking $H = K$. Taking account of 6.2.4 e), we shall apply 6.5.4 to the group scheme $X = H_G = H \times_S G$ over the base scheme G , and to the following subgroup scheme Y .

In the case of $\underline{\text{Transp}}_G(H, K)$, take for Y the inverse image of K_G under the morphism of G -groups $H \times_S G \rightarrow G \times_S G$, $(h, g) \mapsto (ghg^{-1}, g)$. In the case of $\underline{\text{Centr}}_G(H)$, take for Y the inverse image of the diagonal subgroup $\Delta_{G/S}(G)_G$ of $(G \times_S G)_G$ under the morphism of G -groups

$$H \times_S G \rightarrow G \times_S G \times_S G, \quad (h, g) \mapsto (h, ghg^{-1}, g).$$

⁽⁶⁵⁾N.D.E. The original has been expanded in what follows, adding references to EGA IV.

⁽⁶⁶⁾N.D.E. What follows has been expanded. On the other hand, this completes the insertion of XI, 6.8 to 6.11, i.e. in 6.6 below one returns to the original text of VIB.

□

Definition 6.6. Let G be an S -group functor, H a subgroup S -functor; one says that H is *invariant* (resp. *central*, resp. *characteristic*) in G if $\text{Norm}_G H = G$ (resp. if $\text{Centr}_G H = G$, resp. if, for every S -scheme T and every automorphism $a \in \text{Aut}_{T\text{-gr.}}(G_T)$, one has $a(H_T) \subset H_T$), in other words, if, for every S -scheme T , the subgroup $H(T)$ of $G(T)$ is invariant in $G(T)$ (resp. central in $G(T)$, resp. invariant under every automorphism of G_T).

N.B. If H is central (resp. characteristic), it is invariant.

Remark 6.7. Let G and H be two S -groups and $u : H \rightarrow G$ a monomorphism. For H to be invariant (resp. central) in G , it is necessary and sufficient that the morphism

$$\mu \circ (c \circ \text{pr}_2, \mu \circ (u \times \text{id}_G)) : H \times_S G \rightarrow G$$

(defined by $(h, g) \mapsto g^{-1}hg$ for every $g \in G(S')$ and $h \in H(S')$) factor through u (resp. equal $u \circ \text{pr}_1$), and, for H to be characteristic in G , it is necessary and sufficient that, for every S -scheme T and every T -group automorphism a of G_T , $a \circ u_{(T)}$ factor through $u_{(T)}$.

Example 6.8. Let G be an S -group functor. Then Centr_G is characteristic and central. If, for every $s \in S$, G_s is representable, G^0 is characteristic. This follows from the definitions and 3.3.

6.9. — ⁽⁶⁷⁾In [RG71], I 3.3.3, the authors introduce the geometric notion of a pure S -scheme, which is local on S for the étale topology; we refer to loc. cit. for the precise definition. Let us simply point out the following:

- (a) (loc. cit., Th. 3.3.5) If B is a flat A -algebra of finite presentation, B is a projective A -module if and only if $\text{Spec}(B)$ is pure over $\text{Spec}(A)$.
- (b) Consequently, if $X \rightarrow S$ is locally of finite presentation, flat and pure, X is essentially free over S .
- (c) (loc. cit., 3.3.4 (iii)) If $X \rightarrow S$ is locally of finite type, flat, with geometrically irreducible fibers and without embedded components, X is pure over S .

Since every group scheme locally of finite type over a field is Cohen–Macaulay (cf. VIA, 1.1.1), hence without embedded components, one obtains in particular:

- (d) Every S -group scheme G locally of finite presentation, flat and with connected fibers is pure over S , hence essentially free over S .

One may then take up all the statements of 6.2.4 e), taking account of results (b) and (d) above.

7. Generated subgroups; commutator group

In this number, k denotes a fixed field.

Proposition 7.1. Let G be a k -group, $(X_i)_{i \in I}$ a family of geometrically reduced k -schemes;⁽⁶⁸⁾ for every $i \in I$, let $f_i : X_i \rightarrow G$ be a k -morphism.

- (i) There is a smallest closed subgroup k -scheme of G containing each of the f_i , denoted by $\Gamma_G((f_i)_{i \in I})$. It is a geometrically reduced k -scheme, hence smooth if G is supposed locally of finite type over k (1.3.1).
- (ii) Put $X = \coprod_{i \in I} X_i$ and let $f : X \rightarrow G$ be the morphism whose restriction to X_i is f_i for every $i \in I$. Put $X^1 = X \amalg X$, and let $f^1 : X^1 \rightarrow G$ be the morphism whose restrictions to X are respectively f and $c \circ f$. For every $n > 1$, put

$$X^n = X^1 \times_k X^{n-1} \quad \text{and} \quad f^n = \mu \circ (f^1 \times_k f^{n-1}) : X^n \rightarrow G.$$

⁽⁶⁷⁾N.D.E. This subsection has been added.

⁽⁶⁸⁾N.D.E. Here and below, the little-used terminology “separable” has been replaced by the usual terminology “geometrically reduced”, cf. EGA IV₂, 4.6.2.

- Then $\Gamma_G((f_i)_{i \in I})$ is the reduced subscheme of G with underlying space the closure of the union of the $f^n(X^n)$, for $n \geq 1$.
- (iii) For every k -scheme S , $\Gamma_G((f_i)_{i \in I})_S$ is the smallest closed subgroup scheme of G_S containing each of the $f_{i,S} : X_S \rightarrow G_S$.
- (iii') Moreover, $\Gamma_G((f_i)_{i \in I})_S$ is the smallest subgroup scheme of G_S containing each of the $f_{i,S}$.⁽⁶⁹⁾

First observe that, to prove (i), (iii) and (iii'), defining X and f as in (ii) reduces one to the case where I consists of one element.

Let H be the reduced subscheme of G with underlying set $\overline{\bigcup_{n \geq 1} f^n(X^n)}$. Then the family of morphisms $f^n : X^n \rightarrow H$ is schematically dominant (cf. EGA IV₃, 11.10.4), so every closed subscheme of G containing the f^n also contains H . Moreover, by loc. cit., 11.10.7, H is geometrically reduced. Thus, to prove (i) and (ii), it suffices to show that H is a subgroup scheme of G , hence that the restriction of c to H and the restriction of μ to $H \times_k H$ factor through the injection $H \rightarrow G$.

Since H is geometrically reduced, $H \times_k H$ is reduced, and it suffices to check that $c(H) \subset H$ and $\mu(H \times_k H) \subset H$ (as sets). But, by EGA IV₃, 11.10.6, the union of the $f^n_{(H)}(X^n \times_k H)$ is schematically dense in $H \times_k H$. Similarly, for every $n \geq 1$, the union of the $f^n_{(X^n)}(X^n \times_k X^m)$, for $m \geq 1$, is schematically dense in $X^n \times_k H$. Thus it suffices to show that $\mu(f^n_{(H)}(f^n_{(X^n)}(X^n \times_k X^m))) \subset H$ and $c(f^n(X^n)) \subset H$. Now

$$\begin{aligned} \mu(f^n_{(H)}(f^n_{(X^n)}(X^n \times_k X^m))) &= \mu((f^n \times_k f^m)(X^n \times_k X^m)) \\ &= f^{n+m}(X^{n+m}) \subset H; \end{aligned}$$

and, since $c(f^1(X^1)) \subset f^1(X^1)$, one has, for every n , $c(f^n(X^n)) \subset f^n(X^n) \subset H$. This proves (i) and (ii).

Now let us prove (iii). Let G' be a closed subgroup scheme of G_S containing f_S ; one must show that G' contains H_S , or, equivalently, that $H_S = H_S \times_{G_S} G'$. Put $H'_S = H_S \times_{G_S} G' = G' \cap H_S$. Since G' and H_S both contain the f'_S , so does H'_S . Now (EGA IV₃, 11.10.6), since the family of $f^n : X^n \rightarrow H$ is schematically dominant, so is the family of $f'_S : X'_S \rightarrow H_S$, so H'_S , containing each of the f'_S , equals H_S by EGA IV₃, 11.10.1 c). This proves (iii).

Finally let us show that H_S is the smallest (not necessarily closed) subgroup scheme of G_S containing f_S .

Let G' be a subgroup scheme of G_S containing f_S . One must similarly show that, putting $H'_S = H_S \times_{G_S} G'$, one has $H'_S = H_S$. For this it suffices to show that H'_S is closed in H_S and apply (iii). It therefore suffices to show that H_S and H'_S have the same underlying set; a fortiori it suffices to show that, for every $s \in S$, H_s equals

$$H'_s := H'_S \times_S \kappa(s) = H_s \times_{G_s} G'_s.$$

Now, by VIA, 0.5.2, the subgroup $\kappa(s)$ -scheme G'_s is closed in G_s . Thus H'_s is closed in H_s , and then the preceding argument, applied to H'_s , H_s and the f'_s , shows that $H'_s = H_s$.

Corollary 7.1.1. ⁽⁷⁰⁾ *Let G be a k -group locally of finite type and let A, B be two k -subgroups of G , smooth and of finite type, and i_A (resp. i_B) the inclusion of A (resp. B) into G . Suppose B normalizes A ; then $A \cdot B = \Gamma_G(i_A, i_B)$.*

Indeed, let $H = A \rtimes B$ be the semidirect product of A and B (cf. I, 2.3.5); it is a smooth k -group of finite type. Then the morphism of groups $u : H \rightarrow G$, $(a, b) \mapsto ab$, is quasi-compact, so, by 1.2, $u(H) = A \cdot B$ is a reduced closed subscheme of G , which is a group in the category $(\mathbf{Sch}/k)_{\text{red}}$. Now, by EGA IV₃, 11.10.7, $A \cdot B = \overline{u(H)}$ is geometrically reduced

⁽⁶⁹⁾**N.D.E.** The original stated (iii') under the additional hypothesis that G be locally of finite type over k , but this can be omitted, by VIA, 0.5.2.

⁽⁷⁰⁾**N.D.E.** This corollary has been added to point out this special case of 7.1.

(since H is), so it is a closed subgroup of G . Since one evidently has $A \cdot B \subset \Gamma_G(i_A, i_B)$, the corollary follows.

Definitions and remarks 7.2. ⁽⁷¹⁾ (i) Given a k -group G , a family $(X_i)_{i \in I}$ of geometrically reduced k -schemes, and, for each $i \in I$, a k -morphism $f_i : X_i \rightarrow G$, one calls the *closed subgroup scheme of G generated by the family $(f_i)_{i \in I}$* , and in this number we shall denote by $\Gamma_G((f_i)_{i \in I})$, the smallest closed subgroup scheme of G containing each of the f_i . If X is a subscheme of G , geometrically reduced over k , and f is the immersion $X \hookrightarrow G$, one writes $\Gamma_G(X)$ instead of $\Gamma_G(f)$.

(i') With the notation of 7.1 (ii), we shall sometimes put $\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n)$. Note that $\Gamma'_G(f)$ is a subset of G stable under the group law (in the sense of 3.0).

(ii) It is clear that if X_1 and X_2 are two geometrically reduced k -schemes and $f_1 : X_1 \rightarrow G$ and $f_2 : X_2 \rightarrow G$ two k -morphisms such that the sets $\overline{f_1(X_1)}$ and $\overline{f_2(X_2)}$ are equal, $\Gamma_G(f_1) = \Gamma_G(f_2)$.

(iii) Let E be a subset of G such that the reduced subscheme $\overline{E}$ of G is geometrically reduced. One calls the *closed subgroup scheme of G generated by E* , and denotes by $\Gamma_G(E)$, the subgroup scheme $\Gamma_G(i)$, where i is the injection of the reduced subscheme $\overline{E}$ of G into G .

(iv) Since every subgroup scheme of G is closed, by VIA, 0.5.2, ⁽⁷²⁾ one will speak of the “generated subgroup scheme” instead of the “generated closed subgroup scheme”.

(v) Let X be a geometrically reduced k -scheme and $f : X \rightarrow G$ a k -morphism. Suppose $f(X)$ contains the identity element e of G . Put $X'^1 = X \times_k X$ and $f'^1 = \mu \circ (f \times_k (c \circ f))$, and, for $n > 1$,

$$X'^n = X'^1 \times_k X'^{n-1} \quad \text{and} \quad f'^n = \mu \circ (f'^1 \times_k f'^{n-1}).$$

Then $\Gamma_G(f)$ is the reduced subscheme of G whose underlying space is the closure of the union of the $f'^n(X'^n)$, for $n \geq 1$.

Indeed, recalling the notation of 7.1 (ii), $X^1 = X \sqcup X$ and $X^n = X^1 \times_k X^{n-1}$ for $n \geq 2$, one has the following inclusions, where the first follows from the hypothesis $e \in f(X)$:

$$f^n(X^n) \subset f'^n(X'^n) \subset f^{2n}(X^{2n}), \quad \text{for every } n \geq 1.$$

This shows moreover that, for there to be an integer n such that $f^n(X^n) = \Gamma_G(f)$, it is necessary and sufficient that there be an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

One deduces from Remark 7.2 (v) the following

Corollary 7.2.1. *Let X be a geometrically reduced and geometrically connected k -scheme, and $f : X \rightarrow G$ a k -morphism such that $f(X)$ contains the identity element of G . Then the k -group $\Gamma_G(f)$ is connected, hence irreducible.*

Indeed, every X'^n is then connected, so the union of the $f'^n(X'^n)$ (which all contain the identity element) is connected, and the same holds for its closure $\Gamma_G(f)$. Thus $\Gamma_G(f)$ is irreducible, by VIA, 2.4 (when G , hence also $\Gamma_G(f)$, is locally of finite type over k) and 2.6.5 (iii) (in the general case).

Definition 7.2.2. Let G be a k -group.

- (a) Let A and B be two geometrically reduced subgroup k -schemes of G . One calls the *commutator subgroup scheme of A and B in G* , and denotes by $(A, B)_G$ or simply (A, B) , the closed subgroup scheme of G generated by the morphism $\nu : A \times_k B \rightarrow G$ defined by $(a, b) \mapsto aba^{-1}b^{-1}$, for every k -scheme S and $a \in A(S)$, $b \in B(S)$.

⁽⁷¹⁾**N.D.E.** Point (viii) of the original has been put in (i'), and points (vi) and (vii) have been brought out in the form of Corollary 7.2.1 and Definition 7.2.2 below.

⁽⁷²⁾**N.D.E.** The hypothesis that G be locally of finite type over k has been removed here.

- (b) Suppose G geometrically reduced over k . One calls the *derived group* of G , and denotes by $\mathfrak{D}(G)$,⁽⁷³⁾ the group (G, G) .

N.B. For G to be commutative, it is necessary and sufficient that $\mathfrak{D}(G)$ be the unit k -group.

Reminders 7.3.0. ⁽⁷⁴⁾ Recall that if $\phi : Y \rightarrow Z$ is a morphism of S -schemes, the image presheaf $\phi(Y)$ is the S -functor which associates to every S' over S the subset $\phi(Y(S'))$ of $Z(S')$. Observe that if T is a subscheme of Z and the inclusion of presheaves $\phi(Y) \hookrightarrow Z$ factors through T , i.e. if $\phi \circ h \in T(S')$ for every S -morphism $h : S' \rightarrow Y$, one has $\phi(Y) \subset T$ as sets (take $h = \text{id}_Y$), and the converse holds if Y is reduced, for in this case ϕ factors through T .

Also recall that, by 6.2, if H is a closed subgroup k -scheme of G , $\underline{\text{Centr}}_G H$ and $\underline{\text{Norm}}_G H$ are representable by closed subgroup k -schemes of G .

Corollary 7.3. *Let G be a k -group, X a geometrically reduced k -scheme, $f : X \rightarrow G$ a k -morphism.*

- (i) *Let S be a k -scheme and u an endomorphism of the S -group G_S .*
 - (a) *If one has $u(f_S(X_S)) \subset \Gamma_G(f)_S$ (as sets), the morphism $u : \Gamma_G(f)_S \rightarrow G_S$ factors through $\Gamma_G(f)_S$.*
 - (b) *If u is an automorphism of the S -group G_S and one has $u(f_S(X_S)) = f_S(X_S)$ (as sets), u induces an automorphism of $\Gamma_G(f)_S$. In particular, if an element $g \in G(S)$ satisfies $\text{int}(g)(f_S(X_S)) = f_S(X_S)$ (as sets), $g \in \underline{\text{Norm}}_{G_S}(\Gamma_G(f)_S)(S)$.*
 - (c) *If $u \circ f_S = f_S$, the restriction of u to the subgroup $\Gamma_G(f)_S$ of G_S is the identity. In particular, if an element $g \in G(S)$ satisfies $\text{int}(g)(f_S) = f_S$, $g \in \underline{\text{Centr}}_{G_S}(\Gamma_G(f)_S)(S)$.*
- (ii) *Let H be a subgroup scheme of G ; then H centralizes (resp. normalizes) $\Gamma_G(f)$ if and only if, for every $S \rightarrow \text{Spec } k$ and $h \in H(S)$, one has $\text{int}(h) \circ f_S = f_S$ (resp. $\text{int}(h)(f_S(X_S)) \subset \Gamma_G(f)_S$), i.e. if, for every $S' \rightarrow S$ and $x \in X(S')$, the elements $h_{S'} f(x)$ and $f(x)$ of $G(S')$ commute (resp. $h_{S'} f(x) h_{S'}^{-1} \in \Gamma_G(f)(S')$).*
- (iii) *In particular, let Y be a second geometrically reduced k -scheme and $\phi : Y \rightarrow G$ a k -morphism. Suppose that, for every k -scheme S' and every $x \in X(S')$ and $y \in Y(S')$, the elements $\phi(y)$ and $f(x)$ of $G(S')$ commute (resp. $\text{int}(\phi(y))(f(x)) \in \Gamma_G(f)(S')$).*
Then $\Gamma_G(\phi)$ is a k -subgroup of $\underline{\text{Centr}}_G \Gamma_G(f)$, resp. $\underline{\text{Norm}}_G \Gamma_G(f)$.
- (iv) *If g is a k -point of G , the k -group $\Gamma_G(\{g\})$ is commutative.*
- (v) *Let A and B be two subgroup schemes of G geometrically reduced over k . If A and B are invariant (resp. characteristic), so is (A, B) .*

PROOF. (i) Let us prove (a). Put $\Gamma_S = \Gamma_G(f)_S$. Then $u^{-1}(\Gamma_S)$ is a closed subgroup S -scheme of G_S , so, by 7.1 (iii), it suffices to show that f_S factors through $u^{-1}(\Gamma_S)$. Now, since $u(f_S(X_S)) \subset \Gamma_S$ and X_S is reduced, $u \circ f_S$ factors through Γ_S , so f_S factors through $u^{-1}(\Gamma_S)$. This proves (a).

Then the first assertion of (b) is a consequence of (a), applied to u and u^{-1} (and in fact it suffices to suppose $u(f_S(X_S)) \subset \Gamma_S$ and $u^{-1}(f_S(X_S)) \subset \Gamma_S$), and the second assertion is its special case where $u = \text{int}(g)$.

Let us prove (c). Consider the morphism of S -groups $G_S \rightarrow G_S \times_S G_S$, $g \mapsto (u(g), g)$ and denote by H the inverse image of the diagonal; it is a subgroup scheme of G_S . Since G is separated over k (VIA 0.3), G_S is separated over S , so H is closed in G_S . Since, by hypothesis, H contains f_S , it contains $\Gamma_G(f)_S$, by 7.1 (iii). This proves the first assertion of (c), and the second is its special case where $u = \text{int}(g)$.

Let us prove (ii). Put $\Gamma = \Gamma_G(f)$ and denote by i the inclusion $\Gamma \hookrightarrow G$. Then H is contained in $N = \underline{\text{Norm}}_G(\Gamma)$ if and only if, for every k -scheme S and $h \in H(S)$, one has

⁽⁷³⁾N.D.E. The notation $\underline{\text{Der}}(G)$ of the original has been changed to avoid any risk of confusion with a space of derivations.

⁽⁷⁴⁾N.D.E. These reminders have been added, and the statement of 7.3 modified accordingly and expanded.

$\text{int}(h)(\Gamma_S) \subset \Gamma_S$ (this condition applied to h and h^{-1} implying that $\text{int}(h)(\Gamma_S) = \Gamma_S$); and, by (i)(a), this holds if and only if $\text{int}(h)(f_S(X_S)) \subset \Gamma_S$.

Similarly, H is contained in $C = \text{Centr}_G(\Gamma)$ if and only if, for every k -scheme S and $h \in H(S)$, one has $\text{int}(h) \circ i_S = i_S$, and, by (i)(c), this holds if and only if $\text{int}(h)(f_S) = f_S$. This proves (ii).

Let us prove (iii). In view of (ii), the hypothesis implies that ϕ factors through C (resp. N); since the latter is a closed subgroup of G , by 6.2, it therefore contains $\Gamma_G(f)$, by 7.1 (i).

Assertion (iv) follows from (iii), taking for f and ϕ the closed immersion $\text{Spec } k \hookrightarrow G$ defined by g : in this case, for every k -scheme S , $X(S)$ and $Y(S)$ have only one point, sent by f (resp. ϕ) to g .

Finally let us show (v). Let ν be the morphism $G \times G \rightarrow G$, $(g, h) \mapsto ghg^{-1}h^{-1}$ and let ν' be its restriction to $A \times B$; by definition (7.2.2), $(A, B) = \Gamma_G(\nu')$. Let S be a k -scheme, and u an inner automorphism (resp. an S -group automorphism) of G_S . One has $u \circ \nu_S = \nu_S \circ (u \times u)$. On the other hand, the hypothesis implies that u induces an automorphism u_1 of A_S (resp. u_2 of B_S), so

$$u(\nu'_S(A_S \times_S B_S)) = \nu'_S(u_1(A_S) \times_S u_2(B_S)) = \nu'_S(A_S \times_S B_S).$$

Thus, by (i)(b), u induces an automorphism of $(A, B)_S$. This proves (v). $\square$

Proposition 7.4. *Let G be a k -group locally of finite type, X a k -scheme of finite type, geometrically reduced and geometrically connected, and $f : X \rightarrow G$ a k -morphism such that $f(X)$ contains the identity element e of G . Then, with the notation of 7.1 (ii), there is an integer N such that one has (as sets):*

$$f^N(X^N) = \Gamma_G(f).$$

By 7.1 (iii) and EGA IV₂, 2.6.1, we may suppose k algebraically closed. By Corollary 7.2.1, we may suppose $G = G^0$; finally, it suffices to show that there is an integer N such that $f'^N(X'^N) = \Gamma_G(f)$, with the notation of 7.2 (v).

First case. Suppose X irreducible. Then the $\overline{f'^m(X'^m)}$ form an increasing sequence of irreducible closed subsets of the space G , which is noetherian, since $G = G^0$ is of finite type over k (VIA 2.4). Thus this sequence is stationary, and there is an integer m such that $\overline{f'^m(X'^m)} = \Gamma_G(f)$.

Moreover, since X and G are of finite type over k , the morphisms f'^m are of finite type over k . Consequently, $f'^m(X'^m)$ is constructible in G (EGA IV₁, 1.8.5), and thus contains an open subset U of its closure $\Gamma_G(f)$ (EGA 0_{IV}, 9.2.3). Then, by VIA 0.5, one has:

$$\Gamma_G(f) \subset U \cdot U \subset f'^{2m}(X'^{2m}) \subset \Gamma_G(f),$$

whence $f'^{2m}(X'^{2m}) = \Gamma_G(f)$.

Second case. Suppose that X has exactly two irreducible components A_1 and A_2 . Then, since X is connected and k algebraically closed, there is $a \in (A_1 \cap A_2)(k)$. Thus the four irreducible subsets $A_i \times_k A_j$ ($i, j = 1, 2$) cover $X \times_k X$, and the image of each of them under the morphism $f'^1 = \mu \circ (f \times_k (c \circ f))$ contains e . If f'_{ij}^1 denotes the restriction of f'^1 to the reduced subscheme $A_i \times_k A_j$, put

$$Y = (A_1 \times_k A_1) \times_k (A_1 \times_k A_2) \times_k (A_2 \times_k A_2) \times_k (A_2 \times_k A_1)$$

and

$$g = \mu \circ \left((\mu \circ (f'_{11}^1 \times_k f'_{12}^1)) \times_k (\mu \circ (f'_{22}^1 \times_k f'_{21}^1)) \right).$$

Then Y is irreducible, reduced and of finite type, so we have just seen that there is an integer m such that $\overline{g'^m(Y^m)} = \Gamma_G(g)$. Now, for every $n \geq 1$, one has $f'^n(X'^n) \subset g'^n(Y^n) \subset f'^{4n}(X'^{4n})$, whence $\Gamma_G(f) = \Gamma_G(g)$ and $f'^{4m}(X'^{4m}) = \Gamma_G(f)$.

General case. Let us argue by induction on the number of irreducible components of X (this number is finite since X , being of finite type over k , is noetherian). Suppose the proposition proved when X has at most $r - 1$ irreducible components, and suppose that

it has r , namely $A_1, \dots, A_r$. Then e belongs to the image of one of the A_i ; suppose, for example, that $e \in f(A_1)$. Put $Y = \Gamma_G(f_r)$, where f_r denotes the restriction of f to the reduced subscheme $X_r = A_1 \cup \dots \cup A_{r-1}$ (we suppose the numbering of the A_i chosen so that this scheme is connected, which is always possible). Then Y is a closed, reduced and irreducible subgroup of G , by Corollary 7.2.1.

Put $Z = \overline{f(A_r)}$, and let $T = Y \cup Z$, endowed with the reduced closed subscheme structure, and i the injection of T into G . It is clear (7.2 (ii)) that $\Gamma_G(i) = \Gamma_G(f)$ and that T is connected (for, since X is connected, $A_1 \cup \dots \cup A_{r-1}$ and A_r have a point a in common, and Y and Z have the point $f(a)$ in common). Moreover, $e \in T$, and T has at most two irreducible components, since Y and Z are irreducible. By the induction hypothesis, there is an integer m such that $\overline{f_r'^m(X_r'^m)} = \Gamma_G(f_r) = Y$. Since $X = X_r \cup A_r$ and since $Z = \overline{f(A_r)}$ is contained in $\overline{f'^m(X'^m)}$ (since $e \in f(A_1)$), one therefore has:

$$f(X) \subset Y \cup Z \subset \overline{f'^m(X'^m)}.$$

On the other hand, since T has at most two irreducible components, we have already seen that there is an integer p such that $i^p(T^p) = \Gamma_G(i) = \Gamma_G(f)$. Now $T \subset \overline{f'^m(X'^m)}$, so $\overline{f'^{mp}(X'^{mp})} = \Gamma_G(f)$, and one shows, as in the first case, that this last equality implies $\overline{f'^{2mp}(X'^{2mp})} = \Gamma_G(f)$.

Lemma 7.5. *Let S be a scheme, G a group S -scheme, X an S -prescheme, $f : X \rightarrow G$ an S -morphism. Suppose that X and G are locally of finite presentation over S , and that for every $s \in S$ and every maximal point g of G_s , there is a point x of X such that $f(x) = g$ and f is flat at x . Then the morphism $\mu \circ (f \times_S f) : X \times_S X \rightarrow G$ is covering for the (fppf) topology.*

⁽⁷⁵⁾ By EGA IV₃, 11.3.1, the set V of points of X at which f is flat is open and $f|_V$ is an open morphism, so $U = f(V)$ is an open subset of G ; moreover, by hypothesis, $U \cap G_s$ is dense in G_s , for every $s \in S$. Denote by ϕ the restriction of $\mu \circ (f \times_S f)$ to $V \times_S V$.

It suffices to show that ϕ is covering for the (fppf) topology, and for this it suffices to show that ϕ is faithfully flat and of finite presentation (cf. IV, 6.3.1 (iv)). Now ϕ is the composite $V \times_S V \xrightarrow{f|_V \times f|_V} U \times_S U \xrightarrow{\mu} G$, where the first morphism is faithfully flat and locally of finite presentation, since $f|_V$ is. It will therefore suffice to prove that the same holds for the restriction of μ to $U \times_S U$.

Now, since $G \rightarrow S$ is locally of finite presentation and flat, the same holds for $\mu : G \times_S G \rightarrow G$ (which is isomorphic to the morphism obtained from $G \rightarrow S$ by the base change $G \rightarrow S$, cf. VIA 0.1), and thus also for the induced morphism $U \times_S U \rightarrow G$. To prove that the latter is surjective, it suffices to look fiber by fiber, where this follows from VIA 0.5, since $U \cap G_s$ is a dense open subset of G_s , for every $s \in S$.

Remark 7.6.0. ⁽⁷⁶⁾ Let S be a scheme, H an S -group, and $f : X \rightarrow H$ a morphism of S -schemes. The *subpresheaf of groups of H generated by the image of f* , denoted by $\langle \text{Im } f \rangle$, is the subgroup k -functor of H which associates to every S -scheme S' the subgroup of $H(S')$ generated by $f(X(S'))$. Since every element of this subgroup can be written as a finite product $f(x_1)^{\varepsilon_1} \cdots f(x_n)^{\varepsilon_n}$, with $n \in \mathbf{N}^*$, $x_i \in X(S')$ and $\varepsilon_i = \pm 1$, one therefore sees that, if one puts $X^1 = X \coprod X$ and defines the morphisms $f^n : X^n \rightarrow H$ as in 7.1, then $\langle \text{Im } f \rangle$ is precisely the *image presheaf* of the morphism

$$f^\infty : X^\infty \rightarrow H,$$

where $X^\infty = \coprod_{n \geq 1} X^n$ and $f^\infty : X^\infty \rightarrow H$ is the S -morphism whose restriction to each X^n is f^n .

⁽⁷⁵⁾**N.D.E.** The original has been expanded in what follows.

⁽⁷⁶⁾**N.D.E.** This definition has been added; in the original it was contained in the statement of Proposition 7.6.

Proposition 7.6. ⁽⁷⁷⁾ *Let S be a scheme, X an S -scheme flat and locally of finite presentation over S , H an S -group locally of finite presentation over S with reduced fibers, let $f : X \rightarrow H$ be a morphism of S -schemes, and let $f^\infty : X^\infty \rightarrow H$ be the S -morphism introduced above. The following conditions are equivalent:*

(i) *H represents the (fppf) S -sheaf associated to the presheaf $\langle \text{Im } f \rangle$.*

(ii) *f^∞ is covering for the (fppf) topology.*

(iii) *f^∞ is surjective, i.e. $H = \bigcup_{n \geq 1} f^n(X^n)$.*

If moreover H is quasi-compact, these conditions are also equivalent to the following:

(iv) *There is an integer n such that $f^n : X^n \rightarrow H$ is covering for the (fppf) topology (and a fortiori surjective).*

This applies in particular when X is a k -scheme locally of finite type and geometrically reduced, G a k -group locally of finite type, $\phi : X \rightarrow G$ a morphism of k -schemes, $H = \Gamma_G(\phi)$ and $f : X \rightarrow H$ is the morphism induced by ϕ .

PROOF. The sheaf considered in (i) is the image sheaf of X^∞ under f^∞ , so, by IV 4.4.3, saying that H represents this sheaf is equivalent to saying that f^∞ is covering, and this implies that f^∞ is surjective. Conversely, suppose f^∞ surjective and let us show that it is then covering for the (fppf) topology.

Let $s \in S$, η be a maximal point of the fiber H_s , and $x \in X_s^\infty$ such that $f^\infty(x) = \eta$ (such an x exists, since f^∞ is surjective). Since the fiber H_s is reduced, $\mathcal{O}_{H_s, \eta}$ is a field, so f_s^∞ is flat at the point x . Since X^∞ and H are locally of finite presentation over S , and X^∞ flat over S , the fiberwise criterion for flatness (EGA IV₃, 11.3.10) implies that f^∞ is flat at the point x . Thus, by Lemma 7.5, the morphism

$$\mu \circ (f^\infty \times_S f^\infty) : X^\infty \times_S X^\infty \rightarrow H$$

is covering for the (fppf) topology. Now, since $X^n \times_S X^m$ is canonically isomorphic to X^{n+m} , and in this isomorphism $\mu \circ (f^n \times_S f^m)$ corresponds to f^{n+m} , it is clear that $\mu \circ (f^\infty \times_S f^\infty)$ factors through f^∞ , so that f^∞ is covering for the (fppf) topology. This proves that (iii) $\Rightarrow$ (ii), whence the equivalence of conditions (i), (ii) and (iii).

Note moreover that, since the morphisms $X \rightarrow S$ and $H \rightarrow S$ are locally of finite presentation, the same holds for $f : X \rightarrow H$ (cf. EGA IV₁, 1.4.3 (v)), and since $\mu : H \times_S H$ is also of finite presentation (cf. VIA, 0.1), it follows that each $f^n : X^n \rightarrow H$ is too. Thus, by EGA IV₁, 1.9.5 (viii), the $f^n(X^n)$ are ind-constructible subsets of H .

Suppose moreover that H is quasi-compact (then S is too, since $H \rightarrow S$ is surjective). Then, by EGA IV₁, 1.9.9, one concludes that there is p such that $f^p(X^p) = H$. As above, one then deduces from the reducedness of the fibers of H and Lemma 7.5 that the morphism $\mu \circ (f^p \times_S f^p) : X^p \times_S X^p \rightarrow H$ is covering for the (fppf) topology; since this morphism equals $f^{2p} : X^{2p} \rightarrow H$, this finishes the proof of 7.6. $\square$

Remark 7.6.1. Obviously the equivalent conditions of 7.6 imply that the sheaf F under consideration is representable. The converse is false in general:⁽⁷⁸⁾ for example, if k has characteristic 0, let $G = \mathbf{G}_{a,k}$ and let $f : \text{Spec } k \rightarrow G$ be the morphism given by the point 1 of $G(k)$, then F is represented by the constant k -group $\mathbf{Z}_k$, whereas $\Gamma_G(f) = G$, so the monomorphism $\mathbf{Z}_k \hookrightarrow G$ is not surjective.

For simplicity, let us place ourselves under the hypotheses of the special case of 7.6, and suppose F representable. Then EGA IV₃, 8.14.2 implies that F is locally of finite presentation over k , so the question is then whether the dominant monomorphism $F \rightarrow \Gamma_G(f)$ is an isomorphism, or again, a closed immersion. This will be the case, by 1.4.2, if F is quasi-compact, and, by VIA, 0.5.1, this will hold if F is connected, hence, in particular (7.2.1), if X is connected and $f(X)$ contains the identity element of G .

⁽⁷⁷⁾**N.D.E.** The original stated this result under the hypotheses of the special case, but the more general form was used implicitly in the proof of 10.12; the statement has been rewritten accordingly.

⁽⁷⁸⁾**N.D.E.** The original has been expanded in what follows.

Lemma 7.7. *Let k be an algebraically closed field, G a k -group locally of finite type, X a geometrically reduced k -scheme locally of finite type, $f : X \rightarrow G$ a k -morphism and H a subgroup scheme of G such that $H \subset f(X)$. Put*

$$\Gamma' = \bigcup_{n \geq 1} f^n(X^n), \quad \Gamma'_0 = \Gamma' \cap G(k), \quad H_0 = H(k)$$

and suppose H_0 of finite index in Γ'_0 .

Then there is an integer m such that $f^m(X^m) = \Gamma_G(f)$ (cf. 7.6), and $\Gamma_G(f)$ is the union of finitely many translates of H .

For every $n \geq 1$, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV₁, 1.9.5 (viii)), and thus so is Γ' , so that, since G is a Jacobson scheme, Γ'_0 is dense in Γ' . By hypothesis there is a finite sequence $a_1, \dots, a_r$ of points of Γ'_0 such that $\Gamma'_0 = a_1 H_0 \cup \dots \cup a_r H_0$, whence

$$\begin{aligned} \Gamma_G(f) &= \overline{\Gamma'} = \overline{\Gamma'_0} = \overline{a_1 H_0 \cup \dots \cup a_r H_0} \\ &= \overline{a_1 H_0} \cup \dots \cup \overline{a_r H_0} = a_1 \overline{H_0} \cup \dots \cup a_r \overline{H_0}, \end{aligned}$$

the last equality following from the fact that translation by a_i is a homeomorphism of G onto G . One therefore has $\Gamma_G(f) = a_1 H \cup \dots \cup a_r H$. It is also clear that there is an integer p such that each of the a_i ($1 \leq i \leq r$) belongs to $f^p(X^p)$. Finally, since $H \subset f(X)$, one has, for every i , $a_i H \subset f^{p+1}(X^{p+1})$, so that $f^{p+1}(X^{p+1}) = \Gamma_G(f)$.

Proposition 7.8. *Let G be a k -group locally of finite type, A and B two geometrically reduced subgroup k -schemes (thus smooth at the generic points of their irreducible components, hence smooth by 1.3) of G . Suppose one of the following conditions a) or b) holds:*

- a) *A and B are invariant and of finite type over k .*
- b) *A is connected and B is of finite type over k .*

Then (A, B) is of finite type over k , and represents the sheaf associated for the (fppf) (or (fpqc)) topology to the presheaf of groups of commutators of A and B in G . Moreover,⁽⁷⁹⁾ the k -groups (A, B^0) and (A^0, B) are connected, and one has

$$(A, B)^0 = (A, B^0) \cdot (A^0, B).$$

By 7.6, to show that (A, B) is the desired associated sheaf, it suffices to show that there is an integer n such that $\nu^n((A \times_k B)^n) = (A, B)$ (notation of 7.2.2). To show this, as well as to show the other two assertions, one may suppose k algebraically closed. Indeed, let $\bar{k}$ be an algebraic closure of k . By 7.1 (iii) and VIA 2.4, one has, with obvious notation:

$$(A, B)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}}), \quad ((A, B)^0)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}})^0, \quad (A, B^0)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}}^0), \quad \text{etc.}$$

Consequently, if one shows that $(A_{\bar{k}}, B_{\bar{k}})$ is of finite type over $\bar{k}$ (resp. that $(A_{\bar{k}}, B_{\bar{k}}^0)$ and $(A_{\bar{k}}^0, B_{\bar{k}})$ are connected, and that the morphism $(A_{\bar{k}}, B_{\bar{k}}^0) \times_{\bar{k}} (A_{\bar{k}}^0, B_{\bar{k}}) \rightarrow (A_{\bar{k}}, B_{\bar{k}})^0$ is surjective), then the analogous assertions will hold over k , by EGA IV₂, 2.7.1 and 2.6.1.

Let $B^1, \dots, B^p$ then be the connected components of B other than the identity component B^0 (there are finitely many of these since B is supposed of finite type over k , hence noetherian), and in case (a), similarly let $A^1, \dots, A^q$ be those of A . (In case (b), one will consider only $A^0 = A$.) Let ν_{ij} be the restriction of ν to $A^i \times_k B^j$. Then each A^i and each B^j is irreducible (VIA 2.4.1), and thus so are $A^0 \times_k B^j$ and $A^i \times_k B^0$. Since the identity element of G belongs to A^0 and to B^0 , it belongs to $\nu_{0j}(A^0 \times_k B^j)$ and to $\nu_{i0}(A^i \times_k B^0)$. Then each $\Gamma_G(\nu_{0j})$ and $\Gamma_G(\nu_{i0})$ is connected (7.2.1). Similarly, if u_{0j} (resp. u_{i0}) denotes the injection of $\Gamma_G(\nu_{0j})$ (resp. $\Gamma_G(\nu_{i0})$) into G , then

$$(A^0, B) = \Gamma_G((u_{0j})_{j=0}^p) \quad \text{and} \quad (A, B^0) = \Gamma_G((u_{i0})_{i=0}^q)$$

⁽⁷⁹⁾**N.D.E.** What follows has been made more precise and, in the proof, the reduction to the case where k is algebraically closed has been expanded.

are connected. Moreover, one readily deduces from 7.4 and the preceding constructions that there is an index r such that (A^0, B) and (A, B^0) are included in $\nu^r((A \times_k B)^r)$. In case b), one has $(A, B) = (A^0, B)$, and we are done.

Let us now consider case (a).⁽⁸⁰⁾ We already know that (A^0, B) and (A, B^0) are smooth connected subgroup k -schemes of G , hence of finite type (cf. VIA, 2.4). On the other hand, since A^0 is a characteristic subgroup of A (cf. VIA, 2.6.5), it is an invariant subgroup of G and hence, by 7.3 (v), (A^0, B) is an invariant subgroup of G , and similarly for (A, B^0) . Thus, by 7.1.1, the subgroup H of G generated by (A, B^0) and (A^0, B) is precisely $(A, B^0) \cdot (A^0, B)$. In particular, one therefore has $H \subset \nu^{2r}((A \times_k B)^{2r})$.

Given a subset X of G stable under the group law (cf. 3.0), we shall denote by X_0 the group of k -points of G belonging to X . Put $\Gamma' = \bigcup_{q \geq 1} \nu^q((A \times_k B)^q)$. Then, by Proposition 7.9 below, one has:

$$(A^0, B)_0 = (A_0^0, B_0), \quad (A, B^0)_0 = (A_0, B_0^0) \quad \text{and} \quad \Gamma'_0 = (A_0, B_0),$$

so that $H_0 = (A_0^0, B_0) \cdot (A_0, B_0^0)$ has finite index in Γ'_0 (Bible, Exp. 3, Appendix), since A_0 and B_0 are invariant, and A_0^0 (resp. B_0^0) has finite index in A_0 (resp. B_0). We are then under the conditions of Lemma 7.7: since $H \subset \nu^{2r}((A \times_k B)^{2r})$, there is an integer m such that $\nu^{2rm}((A \times_k B)^{2rm}) = (A, B)$, and there is a finite sequence $a_1, \dots, a_n$ of k -points of G such that $(A, B) = a_1 H \cup \dots \cup a_n H$. Then, since H is of finite type over k , each $a_i H$ is quasi-compact, so their union (A, B) is quasi-compact, hence of finite type over k . Since H is irreducible, so is each $a_i H$, and since $e \in H$, it is clear that $H = (A, B)^0$.

Proposition 7.9. *Let k be an algebraically closed field, G a k -group locally of finite type.*

(i) *Let A and B be two ind-constructible subsets of G . Denote by A_0 the set of rational points of G belonging to A . Then $(A \cdot B)_0 = A_0 \cdot B_0$, the second product being taken in the group $G(k)$.*

(ii) *Let X be a geometrically reduced k -scheme locally of finite type, and $f : X \rightarrow G$ a k -morphism. Put $\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n)$. Then $\Gamma'_G(f)_0$ is the subgroup of $G(k)$ generated by $f(X)_0$.*

(iii) *In particular, let A and B be two smooth subgroup schemes of G ; put $\Gamma' = \bigcup_{n \geq 1} \nu^n(A \times_k B)^n$ (notation of 7.2.2). Then Γ'_0 is the commutator group of $A(k)$ and $B(k)$ in $G(k)$.*

Let us prove (i). It is clear that $A_0 \cdot B_0 \subset (A \cdot B)_0$. Conversely, let $z \in (A \cdot B)_0$. Then $\mu^{-1}(z)$ is a closed subset of $G \times_k G$, and $A \times_k B$ (cf. 3.0) is an ind-constructible subset of $G \times_k G$, so that $\mu^{-1}(z) \cap (A \times_k B)$ is a nonempty ind-constructible subset of $G \times_k G$; by EGA IV₃, 10.4.8, it therefore contains a rational point of $G \times_k G$, whose projections x and y are rational points of G such that $x \in A$, $y \in B$ and $x \cdot y = z$, so that $(A, B)_0 = A_0 \cdot B_0$.

To show (ii), note that, since f^n is locally of finite type, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV₄, 1.9.5 (viii)). Assertion (i) then allows one to show by induction that, putting $A = f(X)_0$, one has $f^n(X^n)_0 = (A \cup A^{-1})^n$, and consequently,

$$\Gamma'_G(f)_0 = \bigcup_{n \geq 1} f^n(X^n)_0 = \bigcup_{n \geq 1} (A \cup A^{-1})^n,$$

which is the subgroup of $G(k)$ generated by $A = f(X)_0$.

Finally, (iii) follows from (ii) and the definitions.

Corollary 7.10. *Under the conditions of 7.8, if k is algebraically closed, then $(A, B)(k)$ is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.*

Indeed, it suffices to apply 7.9 (iii), 7.8 and 7.6.

8. Solvable or nilpotent group schemes

⁽⁸⁰⁾**N.D.E.** The original has been expanded in what follows, taking account of the addition 7.1.1.

8.1. — Let $\mathcal{C}$ be a category endowed with a topology T (cf. IV § 4). For every presheaf P on $\mathcal{C}$, one shall denote by P^b the associated sheaf.

Let G be a presheaf of groups on $\mathcal{C}$, A and B two subpresheaves of groups of G , and let $\underline{\text{Comm}}(A, B)$ be the presheaf of groups of commutators of A and B in G ; i.e., for every $S \in \text{Ob } \mathcal{C}$, $\underline{\text{Comm}}(A, B)(S)$ is the subgroup of $G(S)$ generated by the commutators $aba^{-1}b^{-1}$, with $a \in A(S)$ and $b \in B(S)$. One puts

$$\underline{\text{Comm}}_T(A, B) = \underline{\text{Comm}}(A, B)^b.$$

We shall need, in the proof of 8.2, the following results.⁽⁸¹⁾

Lemma 8.1.1. *Let $A \subset G$ be sheaves of groups, with A invariant in G .*

(i) *$\underline{\text{Comm}}_T(G, A)$ is the smallest invariant subsheaf of groups C of G such that the sheaf $(A/C)^b$, associated to the quotient presheaf A/C , is central in $(G/C)^b$.*

(ii) *In particular, $\underline{\text{Comm}}_T(G, G)$ is the smallest invariant subsheaf of groups C of G such that the quotient sheaf $(G/C)^b$ is commutative.*

Obviously, (ii) is the special case $A = G$ of (i), so it suffices to show (i). Let C be a subsheaf of groups of A , invariant in G , and such that the quotient sheaf $(A/C)^b$ is central in $(G/C)^b$. By Lemma IV 4.4.8.1, the presheaves A/C and G/C are separated, and thus, by IV 4.3.11, all the morphisms in the diagram below are monomorphisms:

$$\begin{array}{ccc} A/C & \hookrightarrow & G/C \\ \downarrow & & \downarrow \\ (A/C)^b & \hookrightarrow & (G/C)^b. \end{array}$$

Since $(A/C)^b$ is central in $(G/C)^b$, then A/C is central in G/C , whence $\underline{\text{Comm}}(G, A) \subset C$, and therefore C contains $\underline{\text{Comm}}_T(G, A)$, by IV 4.3.12.

Conversely, $\underline{\text{Comm}}(G, A)$ is a subpresheaf of groups of A , invariant in G , and separated (cf. IV 4.3.1, N.D.E. (24)), so, by IV 4.4.8.2 (i) and IV 4.3.11, $C = \underline{\text{Comm}}_T(G, A)$ is a subsheaf of groups of A , invariant in G and containing $\underline{\text{Comm}}(G, A)$. Consequently, the presheaf A/C is central in G/C and thus, by IV 4.4.8.2 (ii), $(A/C)^b$ is central in $(G/C)^b$. This proves Lemma 8.1.1.

Lemma 8.1.2. *Let G be a sheaf of groups, A, B two subpresheaves of groups of G .*

(i) *The morphism $\tau : \underline{\text{Comm}}(A, B) \rightarrow \underline{\text{Comm}}(A^b, B^b)$ is a covering monomorphism.*

(ii) *Consequently, one has an isomorphism*

$$\underline{\text{Comm}}_T(A, B) \xrightarrow{\sim} \underline{\text{Comm}}_T(A^b, B^b).$$

PROOF. (i) Since A (resp. B) is a subpresheaf of G , then A^b (resp. B^b) is a subpresheaf of G containing A (resp. B), and it follows that τ is a monomorphism.

Let us show that τ is covering. Let $S \in \text{Ob } \mathcal{C}$ and $g \in \underline{\text{Comm}}(A^b, B^b)(S)$. Then there is an integer $n \geq 1$ and, for $i = 1, \dots, n$, elements $a'_i \in A^b(S)$, $b'_i \in B^b(S)$, and $\varepsilon_i \in \{\pm 1\}$, such that

$$g = (a'_1, b'_1)^{\varepsilon_1} \cdots (a'_n, b'_n)^{\varepsilon_n},$$

where (a, b) denotes the commutator $aba^{-1}b^{-1}$, and there is a refinement R of S such that $a'_i \in A(R)$ and $b'_i \in B(R)$ for every i . Then g is the composite morphism

$$R \xrightarrow{(a'_1, \dots, b'_n)} (A \times B)^n \xrightarrow{\Phi^{\varepsilon_1, \dots, \varepsilon_n}} \underline{\text{Comm}}(A, B),$$

where $\Phi = \Phi^{\varepsilon_1, \dots, \varepsilon_n}$ is the morphism defined set theoretically by:

$$\Phi(a_1, b_1, \dots, a_n, b_n) = (a_1, b_1)^{\varepsilon_1} \cdots (a_n, b_n)^{\varepsilon_n},$$

⁽⁸¹⁾N.D.E. These results are mentioned without proof in the original; they have been brought out in the form of Lemmas 8.1.1 and 8.1.2, and the proofs have been expanded.

for every $T \in \text{Ob } \mathcal{C}$ and $a_i \in A(T)$, $b_i \in B(T)$. This shows that $g \in \underline{\text{Comm}}(A, B)(R)$ and it follows, as in the proof of IV 4.3.11 (i), that $\underline{\text{Comm}}(A, B) \rightarrow \underline{\text{Comm}}(A^b, B^b)$ is covering.

Since $\underline{\text{Comm}}(A^b, B^b) \rightarrow \underline{\text{Comm}}(A^b, B^b)^b$ is also a covering monomorphism (IV 4.3.11 (iv)), the same holds for $\underline{\text{Comm}}(A, B) \rightarrow \underline{\text{Comm}}(A^b, B^b)^b$ and thus, by IV 4.3.12, one obtains an isomorphism:

$$\underline{\text{Comm}}(A, B)^b \xrightarrow{\sim} \underline{\text{Comm}}(A^b, B^b)^b.$$

This proves Lemma 8.1.2. □

Proposition 8.2. *Let $\mathcal{C}$ be a category, T a topology on $\mathcal{C}$, G a sheaf of groups on $\mathcal{C}$, n an integer ≥ 0 . The following conditions are equivalent:*

- (i) *If one puts $K_0 = G$, and for $p \geq 1$, $K_p = \underline{\text{Comm}}(K_{p-1}, K_{p-1})$ (resp. $K_p = \underline{\text{Comm}}(G, K_{p-1})$), then K_n is the identity presheaf of groups.*
- (ii) *If one puts $K'_0 = G$, and for $p \geq 1$, $K'_p = \underline{\text{Comm}}_T(K'_{p-1}, K'_{p-1})$ (resp. $K'_p = \underline{\text{Comm}}_T(G, K'_{p-1})$), then K'_n is the identity sheaf of groups.*
- (iii) *There is a sequence $G = H_0 \supset H_1 \supset \cdots \supset H_n$ of invariant subsheaves of G , such that, for every i , the quotient sheaf H_i/H_{i+1} is commutative (resp. central in G/H_{i+1}), and H_n is the identity sheaf.*

It is clear that $K_n \subset K'_n$; consequently (ii) implies (i). Let us show that (i) implies (ii). One has $K'_1 = \underline{\text{Comm}}_T(G, G) = K_1^b$, and one deduces by induction from Lemma 8.1.2 that $K'_p = K_p^b$ for every p . Consequently, if K_n is the identity presheaf, then $K'_n = K_n^b$ is the identity sheaf.

Finally, conditions (ii) and (iii) are equivalent by Lemma 8.1.1.

Definition 8.2.1. When these conditions hold, the sheaf G is said to be *solvable of class n* (resp. *nilpotent of class n*). When there is an integer n for which these conditions hold, one says that G is *solvable* (resp. *nilpotent*).

Note that, by condition (i), this does not depend on the topology T .

Proposition 8.3. *Let k be a field, S a nonempty k -scheme, Ω an algebraically closed extension of k , G a smooth k -group of finite type. The following conditions are equivalent:*

- (i) *G is solvable of class n (resp. nilpotent of class n).*
- (ii) *$G \times_k S$ is solvable of class n (resp. nilpotent of class n).*
- (iii) *The group $G(\Omega)$ is solvable of class n (resp. nilpotent of class n).*
- (iv) *If one puts $K_0 = G$ and considers, for $p \geq 1$, the k -groups $K_p = (K_{p-1}, K_{p-1})$ (resp. $K_p = (G, K_{p-1})$) (cf. 7.2.2), then K_n is the identity k -group.*

The equivalence of conditions (i) and (ii) follows from Proposition 8.2, since the formation of the presheaf of groups of commutators commutes with base changes (IV 4.1.3).

The equivalence of (i) and (iv) follows from the fact that, by 7.8, the k -group $K_p = (K_{p-1}, K_{p-1})$ (resp. $K_p = (G, K_{p-1})$) represents the sheaf $\underline{\text{Comm}}_T(K_{p-1}, K_{p-1})$ (resp. $\underline{\text{Comm}}_T(G, K_{p-1})$), where T is the (fppf) (or (fpqc)) topology.

To show that conditions (iii) and (iv) are equivalent, one may suppose that $k = \Omega$, and then the equivalence of conditions (iii) and (iv) follows from 7.10.

Proposition 8.4. *Let G be an S -group of finite presentation such that, for every $s \in S$, G_s is smooth over $\kappa(s)$. Let T be the set of $s \in S$ such that G_s is solvable (resp. nilpotent).*

- (i) *Then T is locally constructible in S .*
- (ii) *If one supposes moreover G flat and separated over S (i.e. when G is smooth, quasi-compact and separated over S), then T is closed in S .*

It is clear that one may suppose S affine with ring A . Then, by 10.1 and 10.10 b),^{*} there is a noetherian subring A' of A and an A' -group of finite type G' such that $G' \otimes_{A'} A$ is isomorphic to G . By EGA IV₃, 11.2.6 and 8.10.5,⁽⁸²⁾ if G is flat and separated over S , one may suppose G' flat and separated over $S' = \operatorname{Spec} A'$.⁽⁸³⁾ Since G is of finite presentation over S , then (EGA IV₃, 9.7.7) the set of $s \in S$ such that G_s is geometrically reduced (or, equivalently, smooth over $\kappa(s)$) is locally constructible. Thus, by EGA IV₃, 9.3.3, one may suppose that, for every $s' \in S'$, $G'_{s'}$ is smooth over $\kappa(s')$. On the other hand, if s' denotes the image of s in S' , one has $G'_{s'} \otimes_{\kappa(s')} \kappa(s) \simeq G_s$. Thus, by 8.3, for G_s to be solvable (resp. nilpotent), it is necessary and sufficient that $G'_{s'}$ be so. We are therefore reduced to the case where S is an affine noetherian scheme.

Let us then show that T is constructible. Applying the criterion (EGA 0_{III}, 9.2.3), one sees, arguing as above, that one is reduced to showing that, when S is noetherian and integral, T or $S - T$ contains a nonempty open subset of S .

Suppose therefore S integral and noetherian, with generic point η . Put, for every $s \in S$, $K_s^0 = G_s$, and $K_s^p = (K_s^{p-1}, K_s^{p-1})$ (resp. $K_s^p = (G, K_s^{p-1})$). Let us first show that the sequence of closed subschemes K_η^p is stationary. It follows from 7.3 (v) that each K_η^p is invariant, so the sequence of K_η^p is decreasing; this sequence is then stationary since G_η is noetherian; there is therefore an integer n such that, for every $p \geq n$, one has $K_\eta^p = K_\eta^n$.

On the other hand, by 10.12.1 and 10.13, there is a nonempty open subset S' of S and an S' -group of finite presentation D such that, for every $s \in S'$, one has $D_s = K_s^n$ and $(D_s, D_s) = D_s$ (resp. $(G_s, D_s) = D_s$). We may suppose that $S' = S$. Then, for every $s \in S$, and every $p \geq n$, one has $D_s = K_s^p$, so that G_s is solvable (resp. nilpotent) if and only if D_s is isomorphic to the identity $\kappa(s)$ -group.

But by EGA IV₃, 9.6.1 (xi), the set of $s \in S$ such that the structural morphism $D_s \rightarrow \operatorname{Spec} \kappa(s)$ is an isomorphism is constructible,⁽⁸⁴⁾ and thus is either nowhere dense or contains a nonempty open subset of S . We have therefore obtained that T is locally constructible.

Let us show that if, moreover, G is flat and separated over S , then T is closed. Since T is locally constructible, for T to be closed it is necessary and sufficient that T be stable under specialization (cf. EGA IV₁, 1.10.1).

Let therefore $s \in S$ and s' be a specialization of s in S . Since we have reduced to the case where S is noetherian, then, by EGA II, 7.1.9, there is a discrete valuation ring A and a morphism $\operatorname{Spec}(A) \rightarrow S$ such that s (resp. s') is the image of the generic point α (resp. the closed point a) of $\operatorname{Spec}(A)$. It then suffices to show that, putting $G' = G \otimes_S A$, if G'_α is solvable (resp. nilpotent), then G'_a is too. Note that, since G is flat and separated over S , G'_α is flat and separated over A , so that one is reduced to the case where S is the spectrum of a discrete valuation ring A .

Then, since G_α is supposed solvable (resp. nilpotent), there is an integer q such that K_α^q (with the notation introduced above) is isomorphic to the identity $\kappa(\alpha)$ -group. For every n , denote by $\overline{K}_\alpha^n$ the schematic closure (in the sense of EGA IV₂, 2.8.5) of K_α^n in G . Let us show, by induction on p , that $(\overline{K}_\alpha^p)_a \supset K_a^p$. Note first that, since G is flat over A , then $\overline{G}_\alpha$ is equal to G (EGA IV₂, 2.8.5), so $(\overline{K}_\alpha^0)_a = K_a^0$.

^{*}In the course of this proof we shall use results established in no. 10 which, like no. 9, do not depend on the present no. 8.

⁽⁸²⁾**N.D.E.** 10.8.5 has been corrected to 8.10.5.

⁽⁸³⁾**N.D.E.** What follows has been expanded.

⁽⁸⁴⁾**N.D.E.** Taking account of the fact that S is supposed affine, hence quasi-compact and quasi-separated (cf. EGA 0_{III}, 9.1.12).

Let $p \geq 0$. Suppose we have established that $K_a^p \subset (\overline{K_\alpha^p})_a$, and denote by $\nu_a, \nu_\alpha, \bar{\nu}, \bar{\nu}_a$ the following morphisms, defined as in 7.2.2:

solvable case	nilpotent case
$\nu_a : K_a^p \times_{\kappa(a)} K_a^p \longrightarrow G_a,$	resp. $G_a \times_{\kappa(a)} K_a^p \longrightarrow G_a,$
$\nu_\alpha : K_\alpha^p \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha,$	resp. $G_\alpha \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha,$
$\bar{\nu} : \overline{K_\alpha^p} \times_A \overline{K_\alpha^p} \longrightarrow G,$	resp. $G \times_A \overline{K_\alpha^p} \longrightarrow G,$
$\bar{\nu}_a : (\overline{K_\alpha^p})_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a \longrightarrow G_a,$	resp. $G_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a \longrightarrow G_a.$

Since ν_α factors through K_α^{p+1} , then $\bar{\nu}$ factors through $\overline{K_\alpha^{p+1}}$, which is obviously a subgroup scheme of G , so $\overline{K_\alpha^{p+1}}$ contains $\Gamma_G(\bar{\nu})$. By 7.1 (iii), one has $\Gamma_G(\bar{\nu})_a = \Gamma_{G_a}(\bar{\nu}_a)$; and, by the induction hypothesis, one has:

$$K_a^p \times_{\kappa(a)} K_a^p \subset (\overline{K_\alpha^p})_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a, \quad \text{resp.} \quad G_a \times_{\kappa(a)} K_a^p \subset G_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a,$$

so that $K_a^{p+1} = \Gamma_{G_a}(\nu_a) \subset \Gamma_{G_a}(\bar{\nu}_a) = \Gamma_G(\bar{\nu})_a \subset (\overline{K_\alpha^{p+1}})_a$.

But since K_α^q is isomorphic to the identity $\kappa(\alpha)$ -group, and the identity A -group is flat over A and is isomorphic to a closed subscheme of G (since G is separated over A , cf. 5.1), EGA IV₂, 2.8.5 implies that the schematic closure $\overline{K_\alpha^q}$ is isomorphic to the identity A -group. Since we have just seen that $K_a^q \subset (\overline{K_\alpha^q})_a$, this implies that K_a^q is isomorphic to the identity $\kappa(a)$ -group, so that G_a is solvable (resp. nilpotent).

9. Quotient sheaves

This section is essentially limited to recalling, in the special case of homogeneous spaces of groups, well-known general facts on taking quotients by flat equivalence relations (cf. Exp. IV).

Definition 9.1. Given a monomorphism $u : G' \longrightarrow G$ of S -groups, one denotes by G/G' (resp. $G' \backslash G$) and calls the *right* (resp. *left*) *quotient sheaf of G by G'* the sheaf (for the (fpqc) topology) quotient of G by the equivalence relation defined by the monomorphism:

$$G \times_S G' \xrightarrow{\delta \circ (\text{id}_G \times u)} G \times_S G \quad (\text{resp.} \quad G' \times_S G \xrightarrow{\gamma \circ (u \times \text{id}_G)} G \times_S G),$$

where δ (resp. γ) denotes the automorphism of $G \times_S G$ defined by $(g, h) \longmapsto (g, gh)$ (resp. $(h, g) \longmapsto (hg, g)$) for $g, h \in G(T)$.

Proposition 9.2. *Let $u : G' \longrightarrow G$ be a monomorphism of S -groups. Suppose that G/G' is representable by an S -scheme G'' . Then:*

- (i) *The canonical morphism $p : G \longrightarrow G''$ is covering for the (fpqc) topology.*
- (ii) *If one puts $\varepsilon'' = p \circ \varepsilon$ (this morphism is called the identity section of G''), the following diagrams are cartesian:*

$$\begin{array}{ccc} G \times_S G' & \xrightarrow{\mu \circ (\text{id}_G \times u)} & G \\ \text{pr}_1 \downarrow & & \downarrow p \\ G & \xrightarrow{p} & G'' \end{array}, \quad \begin{array}{ccc} G' & \xrightarrow{u} & G \\ \pi' \downarrow & & \downarrow p \\ S & \xrightarrow{\varepsilon''} & G'' \end{array}.$$

In particular, u is an immersion.

- (iii) *There is on G'' a unique structure of S -scheme with group of operators G on the left such that p is a morphism of S -schemes with group of operators G .*

- (iv) *If one supposes moreover that G' is invariant in G , there is on G'' a unique structure of S -group such that p is a morphism of S -groups.*

(v) Let S_0 be an S -scheme; put $G_0 = G \times_S S_0$, and $G'_0 = G' \times_S S_0$; then G_0/G'_0 is representable by $G''_0 = G'' \times_S S_0$.⁽⁸⁵⁾

(vi) Let $\mathcal{P}$ be a property of an S -morphism. Suppose $\mathcal{P}$ stable under base change; then, if $p : G \rightarrow G''$ satisfies $\mathcal{P}$, the structural morphism $\pi' : G' \rightarrow S$ does too.

(vii) Let $\mathcal{P}$ be a property of an S -morphism. Suppose $\mathcal{P}$ local for the (fpqc) topology (cf. 2.0 and 2.1.2). Then, for the morphism $p : G \rightarrow G''$ to satisfy $\mathcal{P}$, it is necessary and sufficient that $\pi' : G' \rightarrow S$ do so.

(viii) Let $\mathcal{P}$ be a property of an S -morphism; suppose $\mathcal{P}$ local for the (fpqc) topology and stable under composition; then, if the structural morphisms $G'' \rightarrow S$ and $G' \rightarrow S$ satisfy $\mathcal{P}$, the structural morphism $G \rightarrow S$ does too.

(ix) Suppose G reduced; then G'' is reduced.

(x) For G'' to be separated over S , it is necessary and sufficient that u (or, equivalently, ε'') be a closed immersion.

(xi) For G' to be flat over S , it is necessary and sufficient that p be a flat morphism (or, equivalently, faithfully flat).

In this case, for G'' to be flat over S , it is necessary and sufficient that G be flat over S .

(xii) For G' to be flat and locally of finite presentation over S , it is necessary and sufficient that $p : G \rightarrow G''$ be faithfully flat and locally of finite presentation.

In this case, for G'' to be locally of finite presentation (resp. locally of finite type, of finite type, smooth, étale, unramified, locally quasi-finite, quasi-finite) over S , it suffices that G be so over S (and the condition is also necessary in the first two cases, cf. (viii)).

(xiii) Suppose G' flat and of finite presentation over S .

a) Then p is of finite presentation and faithfully flat;

b) moreover, for G to be of finite presentation over S , it is necessary and sufficient that G'' be so.

Recall that the equivalence relation considered is universally effective (IV 4.4.9). Then assertions (i), (iii), (iv), (v) and the first assertion of (ii) follow from IV 4.4.3, 5.2.2, 5.2.4, 3.4.5 and 3.3.2 (iii). The second assertion of (ii) follows from the first, as shown by the following cartesian diagram, since $(G \times_S G') \times_G S$ is isomorphic to G' :

$$\begin{array}{ccccc} G' & \xrightarrow{((\varepsilon \circ \pi'), \text{id}_{G'})} & G' \times_S G & \xrightarrow{\mu_p(\text{id}_G \times u)} & G \\ \pi' \downarrow & & \text{pr}_1 \downarrow & & \downarrow p \\ S & \xrightarrow{\varepsilon} & G & \xrightarrow{p} & G'' \end{array}$$

Finally, it is clear that ε'' is an S -section of G'' , hence an immersion (EGA I, 5.3.13); by the preceding cartesian diagram, the same holds for u , which finishes the proof of (ii). Moreover, (vi) is an immediate consequence of the second cartesian diagram of (ii).

Let us show (vii). By (i), p is covering for the (fpqc) topology; thus, by (ii), to show that p satisfies $\mathcal{P}$, it suffices to show that the first projection $\text{pr}_1 : G \times_S G' \rightarrow G$ satisfies $\mathcal{P}$, which follows from the fact that $\mathcal{P}$ is stable under base change, since pr_1 comes from π' by base change. It is clear that (viii) follows from (vii), for $\pi = \pi'' \circ p$, where $\pi'' : G'' \rightarrow S$ denotes the structural morphism.

Let us show (ix). By (i), p is an epimorphism; since G is reduced, p factors through the immersion $G''_{\text{red}} \rightarrow G''$, which is therefore also an epimorphism, hence an isomorphism (IV 4.4.4).

Let us show (x). If G'' is separated over S , then ε'' is a closed immersion, by EGA I, 5.4.6. On the other hand, we saw in (ii) that ε'' is a closed immersion if and only if u is one. Finally, if u is a closed immersion, then so is $\delta \circ (\text{id}_G \times u) : G \times_S G' \rightarrow G \times_S G$; thus, by Lemma 9.2.1 below, G'' is separated over S .

Assertion (xi) follows from (vii) and EGA IV₂, 2.2.13.

⁽⁸⁵⁾**N.D.E.** I.e., the quotient is universal, cf. Exp. IV § 3.

Assertion (xii) follows from (vii), EGA IV₄, 17.7.5 and 17.7.7, and the fact that, since p is universally open, for every $s \in S$, if the underlying space of G_s is discrete, the same holds for the underlying space of G'_s .

Finally, assertion (xiii) follows from (vii), (viii), and EGA IV₄, 17.7.5.

Lemma 9.2.1. *Let X be an S -scheme and R an equivalence relation defined on X by the monomorphism $v : R \rightarrow X \times_S X$. Suppose R effective. Then:*

- (i) *v is an immersion, and it is a closed immersion if $Y = X/R$ is separated.*
- (ii) *Suppose moreover that Y represents the (fpqc) sheaf quotient of X by R ⁽⁸⁶⁾ and that v is a closed immersion. Then $Y = X/R$ is separated over S .*

Recall (IV Def. 3.3.2) that the hypothesis “ R effective” means that there is a morphism of S -schemes $p : X \rightarrow Y$ such that the natural morphism $R \rightarrow X \times_Y X$ is an isomorphism. One deduces (EGA I, 5.3.5) the following cartesian diagram:

$$\begin{array}{ccc} R & \xrightarrow{v} & X \times_S X \\ \downarrow & & \downarrow p \times p \\ Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y. \end{array}$$

Then, since $\Delta_{Y/S}$ is an immersion (EGA I, 5.3.9), the same holds for v . Similarly, if Y is separated over S , $\Delta_{Y/S}$ is a closed immersion, so the same holds for v .

Conversely, suppose v a closed immersion and Y represents the (fpqc) sheaf quotient of X by R . Then p is covering for the (fpqc) topology (IV 4.4.3), and therefore $p \times p$ is too (by base change, $p \times \text{id}_X$ and $\text{id}_Y \times p$ are covering, and thus also their composite $p \times p$). Hence, by (fpqc) descent (cf. EGA IV₂, 2.7.1), $\Delta_{Y/S}$ is a closed immersion, i.e. Y is separated over S .

Remark 9.2.2. Under the general hypotheses of 9.2, if one supposes G' flat and locally of finite presentation over S , then p is covering for the (fppf) topology,⁽⁸⁷⁾ by 9.2 (vii), so assertions (vii) and (viii) of 9.2 can be extended to properties $\mathcal{P}$ local for the (fppf) topology.

Remark 9.3. a) The question whether a quotient G/G' is representable or not is often delicate; in this seminar we prove representability of certain particular quotients.

In general, to be able to assert that the quotient G/G' is representable, it does not suffice to suppose G and G' of finite presentation over S and G' flat over S . Indeed, suppose moreover G smooth with connected fibers. In this case, if G/G' is a scheme, it is separated, by Corollary 5.4, and therefore $G' \hookrightarrow G$ is a closed immersion, by 9.2 (x); consequently, if G' is not closed in G , then G/G' is not representable.

To obtain such a counterexample, one may take for S the spectrum of a discrete valuation ring, and put $G = \mathbf{G}_{m,S}$. Consider moreover an integer $n > 1$, invertible on S ; then $\mu_n = \text{Ker}(G \xrightarrow{n} G)$ is a closed subgroup of G étale over S (cf. VIIA⁽⁸⁸⁾). Let G' be the open subgroup of μ_n obtained by removing from μ_n the closed part of the closed fiber of μ_n complementary to the origin. Then G' is not closed in G , so G/G' is not representable. (One can also construct such examples where G' is smooth with connected fibers.)

⁽⁸⁶⁾**N.D.E.** As pointed out by O. Gabber, this is used in the proof and must be inserted among the hypotheses.

⁽⁸⁷⁾**N.D.E.** (fpqc) has been corrected to (fppf).

⁽⁸⁸⁾**N.D.E.** See VIIA, 8.4 or VIII, 2.1.

b) On the other hand, it is not excluded that G/G' is representable when G and G' are of finite presentation over S , and G' is flat over S and closed in G .^{*(89)} Under these hypotheses, one knows that G/G' is representable in the following special cases:

- 1° – S is the spectrum of an artinian ring (cf. VIA 3.2 and 3.3.2).
- 2° – G' is proper over S and G quasi-projective over S (cf. V 7.1).
- 3° – S is locally noetherian of dimension 1 (cf. [An73], Th. 4.C).

10. Passage to the projective limit in group schemes and schemes with a group of operators

10.0. — Recall the essential result of EGA IV₃, § 8.8. Suppose the following situation given: S_0 a quasi-compact and quasi-separated scheme, I an increasing filtered preordered set, $(\mathcal{A}_i)_{i \in I}$ an inductive system of quasi-coherent commutative $\mathcal{O}_{S_0}$ -algebras, $\mathcal{A} = \varinjlim \mathcal{A}_i$, $S_i = \text{Spec } \mathcal{A}_i$ for $i \in I$, and $S = \text{Spec } \mathcal{A}$;⁽⁹⁰⁾ then the category of S -schemes of finite presentation is determined up to equivalence by the categories of S_i -schemes of finite presentation, the functors between these categories $\rho_{ji} : X_i \mapsto X_i \times_{S_i} S_j$ for $i \leq j$, and the transitivity isomorphisms $\rho_{kj} \circ \rho_{ji} \xrightarrow{\sim} \rho_{ki}$.

Let us be more precise. Given $j \in I$, and an S_j -scheme of finite presentation X_j , we shall put, for every $i \in I$ such that $i \geq j$, $X_i = X_j \times_{S_j} S_i$, and $X = X_j \times_{S_j} S$. Then (EGA IV₃, 8.8.2):

(i) Given $j \in I$, and two S_j -schemes of finite presentation X_j and Y_j , the canonical map from $\varinjlim_{i \geq j} \text{Hom}_{S_i}(X_i, Y_i)$ to $\text{Hom}_S(X, Y)$ is bijective.

(ii) For every S -scheme of finite presentation X , there is an index $j \in I$, an S_j -scheme of finite presentation X_j and an S -isomorphism $X \xrightarrow{\sim} X_j \times_{S_j} S$.

One concludes (EGA IV₃, 8.8.3) that, whenever one has a diagram D involving finitely many objects and arrows of the category of S -schemes of finite presentation, one can find an index $i \in I$ and a diagram D_i in the category of S_i -schemes of finite presentation, such that the diagram D comes, up to isomorphism, from the diagram D_i by base change $S \rightarrow S_i$. One can even find i and D_i such that every cartesian square of D comes from a cartesian square of D_i .

10.1. — Moreover, many common properties of a morphism, stable under base change, possess the following property:

Let $u : X \rightarrow Y$ be an S -morphism between S -schemes of finite presentation; by 10.0 it comes by base change from an S_j -morphism $u_j : X_j \rightarrow Y_j$ between S_j -schemes of finite presentation; then, for u to have property $\mathcal{P}$, it is necessary and sufficient that there be $i \geq j$ such that $u_i = u_j \times_{S_j} S_i$ has property $\mathcal{P}$.

This is the case when $\mathcal{P}$ is one of the following properties of a morphism: being separated, surjective, radicial, affine, quasi-affine, finite, quasi-finite, proper, projective, quasi-projective, an isomorphism, a monomorphism, an immersion, an open immersion, a closed immersion (EGA IV₃, 8.10.5), flat (EGA IV₃, 11.2.6), smooth, unramified or étale (EGA IV₄, 17.7.8).⁽⁹¹⁾

Note that this still holds when $\mathcal{P}$ is the property of being covering for the (fppf) topology; indeed, given two S -schemes of finite presentation X and Y , and an S -morphism $u : X \rightarrow Y$, it follows from IV, 6.3.1 (i)⁽⁹²⁾ that, for u to be covering for the (fppf) topology, it is necessary

^{*}This is too optimistic, as M. Raynaud shows in his thesis (loc. cit. X 14).

⁽⁸⁹⁾**N.D.E.** Remark (*) refers to counterexample X.14 in [Ray70a]. The base there is regular local of dimension 2.

⁽⁹⁰⁾**N.D.E.** Note that S , being affine over S_0 , is therefore quasi-compact and quasi-separated, cf. N.D.E. (92) below.

⁽⁹¹⁾**N.D.E.** The word “flat” has been added, and 17.7.6 corrected to 17.7.8.

⁽⁹²⁾**N.D.E.** And from the fact that Y , being of finite presentation over S , is quasi-compact.

and sufficient that there be an S -scheme Z and an S -morphism $v : Z \rightarrow Y$ faithfully flat and of finite presentation which factors through u .

The aim of this section 10 is to give variants of results of this kind for the category of S -groups of finite presentation, that of S -schemes with a group of operators, and for certain properties of group monomorphisms (being invariant, central with representable quotient sheaf, etc.).

The first two results of this type are the following. (In nos. 10.2 to 10.9 below, we retain the notation introduced in 10.0.)

Lemma 10.2. *Let G_j and H_j be two S_j -groups of finite presentation; put, for every $i \geq j$, $G_i = G_j \times_{S_j} S_i$, $G = G_j \times_{S_j} S$, and define H_i and H similarly. Then the canonical map below is bijective:*

$$\lim_{\substack{\longrightarrow \\ i \geq j}} \operatorname{Hom}_{S_i\text{-gr.}}(G_i, H_i) \rightarrow \operatorname{Hom}_{S\text{-gr.}}(G, H).$$

Lemma 10.3. *Let G be an S -group of finite presentation; then there is $j \in I$, an S_j -group of finite presentation G_j , and an isomorphism of S -groups $G \cong G_j \times_{S_j} S$.*

Assertions 10.2 and 10.3 are easy consequences of 10.0 and 10.1, taking account of the interpretation⁽⁹³⁾ of the S -group structure given in EGA 0_{III}, 8.2.5 and 8.2.6.

Lemma 10.4. *Let $u : G \rightarrow H$ be a morphism of S -groups between S -groups of finite presentation. By 10.3 and 10.2, u comes by base change from a morphism $u_j : G_j \rightarrow H_j$ between S_j -groups of finite presentation. Then, for u to be a central monomorphism (resp. an invariant monomorphism), it is necessary and sufficient that there be $i \geq j$ such that $u_i = u_j \times_{S_j} S_i$ has the same property.*

This is an immediate consequence of 10.0 and 10.1, taking account of the characterization given in 6.7 of central or invariant group monomorphisms.

Corollary 10.5. *Let G_j be an S_j -group of finite presentation. For $G_j \times_{S_j} S$ to be commutative, it is necessary and sufficient that there be $i \geq j$ such that $G_j \times_{S_j} S_i$ is so.*

Indeed, it amounts to the same thing to say that an S -group is commutative, or that, considered as a subgroup scheme of itself, it is central.

Proposition 10.6. *Let G_j be an S_j -group of finite presentation, G'_j a subgroup scheme of G_j flat and of finite presentation over S_j . For $(G_j \times_{S_j} S)/(G'_j \times_{S_j} S)$ to be representable for the (fpqc) topology, it is necessary and sufficient that there be $i \geq j$ such that $(G_j \times_{S_j} S_i)/(G'_j \times_{S_j} S_i)$ is so.*

This is a consequence of the following more general lemma:

Lemma 10.7. *Let X_j be an S_j -scheme of finite presentation, and R_j an equivalence relation on X_j flat and of finite presentation.⁽⁹⁴⁾ For the quotient sheaf $(X_j \times_{S_j} S)/(R_j \times_{S_j} S)$ for the topology $T = (\text{fppf})$ or (fpqc) to be representable, it is necessary and sufficient that there be $i \geq j$ such that the quotient sheaf $(X_j \times_{S_j} S_i)/(R_j \times_{S_j} S_i)$ for the topology T is so.*

Taking account of the statements of EGA IV₂, 8.8.2, 8.8.3, 8.10.5 and 11.2.6 recalled in 10.0, this lemma is a consequence of the following result:

Lemma 10.8. *Let T be the (fppf) or (fpqc) topology; let X be an S -scheme of finite presentation (resp. locally of finite presentation), R an equivalence relation on X defined by a monomorphism $v : R \rightarrow X \times_S X$ such that $\text{pr}_1 \circ v$ is flat and of finite presentation (resp. flat and locally of finite presentation). Then the following conditions are equivalent:*

⁽⁹³⁾**N.D.E.** In terms of commutative diagrams of S -morphisms.

⁽⁹⁴⁾**N.D.E.** I.e., such that the composite $R_j \hookrightarrow X_j \times_{S_j} X_j \xrightarrow{\text{pr}_1} X_j$ is flat and of finite presentation.

- (i) The quotient sheaf X/R for the topology T is representable.
 (ii) There is an S -scheme of finite presentation (resp. locally of finite presentation) Y and a faithfully flat morphism $p : X \rightarrow Y$ such that the diagram

$$(D) \quad \begin{array}{ccc} R & \xrightarrow{\text{pr}_1 \circ v} & X \\ \text{pr}_2 \circ v \downarrow & & \downarrow p \\ X & \xrightarrow{p} & Y \end{array}$$

is cartesian.

Note first that by IV, 3.3.2 and 4.4.3, for the sheaf X/R for the topology T to be representable by Y , it is necessary and sufficient that the diagram (D) be cartesian and p covering for the topology T .

Let us show that (i) implies (ii). Hypothesis (i) implies that the diagram (D) is cartesian, hence that $\text{pr}_1 \circ v$ is obtained from p by base change by p , and that p is covering for the (fpqc) topology. Thus, by (fpqc) descent (EGA IV₂, 2.7.1), since $\text{pr}_1 \circ v$ is faithfully flat and (locally) of finite presentation, the same holds for p . Then, by EGA IV₄, 17.7.5, since X is (locally) of finite presentation over S , the same holds for Y .

Let us show that (ii) implies (i). It suffices to show that p is covering for the (fppf) topology; now p is faithfully flat by hypothesis, and is locally of finite presentation because X and Y are locally of finite presentation over S (EGA IV₁, 1.4.3 (v)).

Lemma 10.9. *Let G_j be an S_j -group of finite presentation, and $G = G_j \times_{S_j} S$. For G^0 to be representable, it is necessary and sufficient that there be $i \geq j$ such that $(G_i)^0 = (G_j \times_{S_j} S_i)^0$ is so.*

The condition is sufficient, since the functor $G \mapsto G^0$ commutes with base change, by 3.3.

Conversely, suppose G^0 representable. Then, by 3.9, G^0 is open in G and quasi-compact over S , hence of finite presentation over S , since G is. Then, by 10.3 and 10.1, there is $i \geq j$ and an open subgroup scheme H_i of G_i such that $H_i \times_{S_i} S = G^0$. The structural morphism $G^0 \rightarrow S$ is connected, i.e. has geometrically connected fibers (VIA 2.1.1), so, by EGA IV₃, 9.3.3 and 9.7.7, increasing i if necessary, one may suppose that the structural morphism $H_i \rightarrow S$ is connected. Then, by 3.10.1, the underlying space of H_i is precisely $(G_i)^0$, and thus H_i represents $(G_i)^0$.

10.10. — Recall two very useful special cases of the situation stated in 10.0 (cf. EGA IV₃, 8.1.2 a) and c)):

a) Given a point x of a scheme X , one puts $S_0 = \text{Spec } \mathbf{Z}$ and considers the decreasing filtered projective system $(S_i)_{i \in I}$ of affine open neighborhoods of x ; then $S = \text{Spec } \mathcal{O}_{X,x}$. In particular, if x is the generic point of an integral scheme X , one finds $S = \text{Spec } \kappa(x)$.

b) One puts $S_0 = \text{Spec } \mathbf{Z}$, and considers the family $(\mathcal{A}_i)_{i \in I}$, preordered by inclusion, of sub- $\mathbf{Z}$ -algebras of finite type of the ring of an affine scheme S . Since the $\mathcal{A}_i$ are noetherian rings, this allows one in many cases to pass from the noetherian case to the general case.

We shall now give two results concerning the special case considered in a).⁽⁹⁵⁾

Proposition 10.11. ⁽⁹⁶⁾ *Let S be an integral scheme with generic point η , G (resp. Y, Z) an S -group (resp. S -schemes) of finite presentation, $u, v, i : Y \rightarrow G$ and $j : Z \rightarrow G$ morphisms of S -schemes. Suppose that $i_\eta : Y_\eta \rightarrow G_\eta$ and $j_\eta : Z_\eta \rightarrow G_\eta$ are closed immersions.*

⁽⁹⁵⁾**N.D.E.** Note that the proof also uses case b).

⁽⁹⁶⁾**N.D.E.** The statement has been simplified, and the case of subgroups treated separately in Corollary 10.11.1.

Then there is a nonempty open subset S' of S such that the morphisms $i' : Y' \rightarrow G'$ and $j' : Z' \rightarrow G'$ obtained by base change are closed immersions, and the functors:

$$\underline{\text{Transp}}(u', v'), \quad \underline{\text{Transp}}_{G'}(i'(Y'), j'(Z')) \quad \text{and} \quad \underline{\text{Transp}}_{G'}^{\text{str}}(i'(Y'), j'(Z'))$$

resp.

$$\underline{\text{Centr}}(u') \quad \text{and} \quad \underline{\text{Norm}}_{G'} i'(Y')$$

are representable by closed sub- S' -schemes (resp. sub- S' -groups) of G' , of finite presentation over S' .

We shall apply the results of 10.1, first in the situation of 10.10 a), then in that of 10.10 b). Since G_η, Y_η, Z_η are flat over the field $\kappa(\eta)$, G_η is separated over $\kappa(\eta)$ (VIA 0.3), and i_η, j_η are closed immersions, then, by 10.1, there is an affine open subset $S' = \text{Spec}(A')$ of S , a noetherian subring A of A' , A -schemes G_A, Y_A, Z_A flat and of finite presentation over A , and morphisms $u_A, v_A, i_A : Y_A \rightarrow G_A$ and $j_A : Z_A \rightarrow G_A$, such that G_A is an A -group, separated over A , i_A and j_A are closed immersions, and $G \times_S S' = G_A \otimes A'$, etc. Since the functors considered for S' are obtained by base change from the analogous functors for $\text{Spec}(A)$, it suffices to establish the result for the latter.

By EGA IV₂, 6.9.2, replacing A by a localization A_f if necessary (and hence S' by the affine open subset S'_f), one may suppose that G_A, Y_A, Z_A are essentially free over A (in the sense of 6.2.1).⁽⁹⁷⁾ It then follows from 6.2.4 b) and e) that, under the hypotheses of the statement, the functors considered are representable by closed sub- A -schemes of G_A (hence of finite presentation over A , since A is noetherian and G_A of finite presentation over A), and they are sub- A -groups of G_A in the case of $\underline{\text{Centr}}(u_A)$ and $\underline{\text{Norm}}_{G_A} i_A(Y_A)$.

Corollary 10.11.1. *Let S be an integral scheme with generic point η , G, H, K be S -groups of finite presentation, $i : H \rightarrow G$ and $j : K \rightarrow G$ two quasi-compact monomorphisms of S -groups. Then there is a nonempty open subset S' of S such that the morphisms $i' : H' \rightarrow G'$ and $j' : K' \rightarrow G'$ obtained by base change are closed immersions, and the functors:*

$$\underline{\text{Transp}}_{G'}(H', K') \quad \text{and} \quad \underline{\text{Transp}}_{G'}^{\text{str}}(H', K') \quad (\text{resp. } \underline{\text{Centr}}_{G'} H' \text{ and } \underline{\text{Norm}}_{G'} H')$$

are representable by closed sub- S' -schemes (resp. sub- S' -groups) of G' , of finite presentation over S' .

This follows from the preceding proposition for, by 1.4.2, the hypotheses imply that i_η and j_η are closed immersions.

Proposition 10.12. *Let S be an integral scheme, G an S -group of finite presentation, A and B two subgroup schemes of G , of finite presentation over S and with smooth generic fiber. Suppose moreover that one of the following conditions holds:*

- a) A has connected generic fiber,
- b) A and B are invariant in G .

Then there is a nonempty open subset S' of S and a closed subgroup scheme D' of $G' = G|_{S'}$, of finite presentation over S' , with smooth fibers, which represents the (fppf) sheaf associated to the presheaf of groups of commutators of $A' = A|_{S'}$ and $B' = B|_{S'}$ in G' , and D' has connected fibers in case (a), and is invariant in G' in case (b).

In particular, for every $s \in S'$, one has $D'_s = (A_s, B_s)$ with the notation of 7.2.2.

Let η be the generic point of S ; put $H_\eta = (A_\eta, B_\eta)$. Since A_η and B_η are smooth, then, by 7.8 in case (a), and 7.3 (v) in case (b), H_η is connected (resp. invariant in G_η).

We are in the situation of 10.0 corresponding to 10.10 (a); thus, by 10.3 and 10.1, there is a nonempty open subset S' of S and a subgroup S' -scheme D' of finite presentation and

⁽⁹⁷⁾**N.D.E.** Indeed, since G_A is flat and of finite presentation over A , it is covered by affine open subsets $G_1, \dots, G_n$ such that each $\mathcal{O}(G_i)$ is a flat A -algebra of finite presentation; then, by EGA IV₂, 6.9.2, there is $f_i \in A$ such that $\mathcal{O}(G_i)_{f_i}$ is a free module over A_{f_i} ; one can then replace $\text{Spec}(A)$ by the affine open subset $D(f)$, where $f = f_1 \cdots f_n$, and one does the same for Y_A and Z_A .

closed in G' , such that $D'_\eta = D' \otimes_{S'} \kappa(\eta)$ equals H_η . Moreover, by EGA IV₃, 9.7.7 and 9.3.3, one may suppose that D' has geometrically reduced fibers. In case (a), one may suppose, by EGA IV₃, 9.7.7 and 9.3.3 again, that D' has connected fibers, hence geometrically connected fibers (cf. VIA, 2.1.1). In case (b), one may suppose, by 10.4, that D' is invariant in G' .

Moreover, in the course of the proof of 7.8 we saw that there is an integer n such that $\nu_\eta^n((A_\eta \times_{\kappa(\eta)} B_\eta)^n) = D'_\eta$, where ν_η and ν_η^n are defined as in 7.2.2 (a) and 7.1 (ii). We can define by the same formulas the morphisms

$$\nu' : A' \times_{S'} B' \longrightarrow G' \quad \text{and} \quad \nu'^n : (A' \times_{S'} B')^n \longrightarrow G',$$

and one has $\nu'^n \otimes_{S'} \kappa(\eta) = \nu_\eta^n$.

Consequently, by 10.1, one can choose S' such that the morphism ν'^n is flat and factors through D' , and that the resulting morphism $\nu'^n : (A' \times_{S'} B')^n \longrightarrow D'$ is surjective. Then, by 7.5, the morphism⁽⁹⁸⁾

$$\nu'^{2n} = \mu \circ (\nu'^n \times_{S'} \nu'^n) : (A' \times_{S'} B')^{2n} \longrightarrow D'$$

is covering for the (fppf) topology. Thus, by 7.6, D' represents the (fppf) sheaf associated to the presheaf of commutators of A' and B' in G' .

Moreover, ν'^n induces, for every $s \in S'$, a surjective morphism $\nu'_s : (A_s \times_{\kappa(s)} B_s)^n \longrightarrow D'_s$.⁽⁹⁹⁾ Then D'_s is a closed subgroup of G_s containing $\nu_s(A_s \times_{\kappa(s)} B_s)$, hence also (A_s, B_s) . Since ν'_s is surjective, then D'_s equals (A_s, B_s) and represents, by 7.6, the (fppf) sheaf of commutators of A_s and B_s in G_s .

Corollary 10.12.1. ⁽¹⁰⁰⁾ *Let S be an integral scheme with generic point η , and G an S -group of finite presentation with smooth fibers. Put $K_\eta^0 = G_\eta$ and $K_\eta^p = (K_\eta^{p-1}, K_\eta^{p-1})$ (resp. $K_\eta^p = (G, K_\eta^{p-1})$) for every $p \in \mathbf{N}^*$. Fix $n \in \mathbf{N}^*$. Then there is a nonempty open subset S' of S and a subgroup scheme D invariant in $G|_{S'}$, of finite presentation and with smooth fibers, such that $D_s = K_s^n$ for every $s \in S'$.*

This follows from 10.12, by induction on n .

Corollary 10.13. *Let S be an integral scheme with generic point η , let G be an S -group, H a subgroup S -scheme invariant in G ; suppose G and H of finite presentation over S and with smooth generic fiber.⁽¹⁰¹⁾*

If one has $(H_\eta, H_\eta) = H_\eta$ (resp. $(G_\eta, H_\eta) = H_\eta$), then there is a nonempty open subset S' of S such that, for every $s \in S'$, one has $(H_s, H_s) = H_s$ (resp. $(G_s, H_s) = H_s$).

Indeed, by the proof of 10.12, there is a nonempty open subset S' of S and a subgroup S' -scheme D of $G|_{S'}$, of finite presentation and with smooth fibers, such that $D_s = (H_s, H_s)$ (resp. $D_s = (G_s, H_s)$) for every $s \in S'$. On the other hand, since $D_\eta = H_\eta$ and D and H are of finite presentation over S' , then, by EGA IV₃, 8.8.2.5, there is a nonempty open subset S'' of S' such that $D|_{S''} = H|_{S''}$. For every $s \in S''$, one therefore has $H_s = D_s = (H_s, H_s)$ (resp. $H_s = D_s = (G_s, H_s)$).

10.14. — Statements 10.2 and 10.3 concerning the category of S -groups of finite presentation extend to the category of pairs formed by an S -group of finite presentation and an S -scheme of finite presentation with group of operators G . More precisely, in the situation recalled at the beginning of 10.0:

(i) let $j \in I$ and G_j, G'_j be two S_j -groups of finite presentation, H_j (resp. H'_j) an S_j -scheme of finite presentation with group of operators G_j (resp. G'_j). Put, for $i \in I, i \geq j$,

⁽⁹⁸⁾**N.D.E.** Below, $n + 1$ has been corrected to $2n$.

⁽⁹⁹⁾**N.D.E.** The original has been expanded in what follows.

⁽¹⁰⁰⁾**N.D.E.** This corollary, used in the proof of 8.4, has been added.

⁽¹⁰¹⁾**N.D.E.** The original supposed G, H of finite presentation and H with smooth fibers; the hypothesis has been modified in order to be able to apply 10.12. The proof has also been expanded.

$G_i = G_j \times_{S_j} S_i$ and $G = G_j \times_{S_j} S$, and define similarly G'_i, G', H_i, H, H'_i and H' . Denote by $\text{Dihom}_{S\text{-gr}}((G, H), (G', H'))$ the set of di-morphisms of S -groups and S -schemes with a group of operators from the pair (G, H) to the pair (G', H') . Then the canonical map

$$\lim_{\substack{\longrightarrow \\ i \geq j}} \text{Dihom}_{S_i\text{-gr}}((G_i, H_i), (G'_i, H'_i)) \longrightarrow \text{Dihom}_{S\text{-gr}}((G, H), (G', H'))$$

is bijective.

(ii) let G be an S -group of finite presentation and H an S -scheme of finite presentation with group of operators G ; there is then an index $j \in I$, an S_j -group of finite presentation G_j , an S_j -scheme of finite presentation H_j with group of operators G_j and a di-isomorphism of S -groups and S -schemes with groups of operators from $(G_j \times_{S_j} S, H_j \times_{S_j} S)$ onto (G, X) .

Definition 10.15. ⁽¹⁰²⁾ Let T be a topology on $(\text{Sch})/S$, coarser than the canonical topology. Given a group S -scheme G and an S -scheme X with group of operators G , one says that X is a *formally homogeneous space under G* (relative to the topology T) if the morphism $\Phi : G \times_S X \longrightarrow X \times_S X$, defined by $(g, x) \longmapsto (gx, x)$ for every $S' \longrightarrow S$ and $g \in G(S')$, $x \in X(S')$, is an epimorphism in the category of sheaves for the topology T , which is equivalent to saying that Φ is covering for the topology T (cf. IV 4.4.3).

One says that X is a *homogeneous space* if it is formally homogeneous and if, moreover, the morphism $X \longrightarrow S$ is also covering for the topology T .

In particular, one says that X is a *formally principal homogeneous space under G* if Φ is an isomorphism, and that X is a *principal homogeneous bundle* (or *G -torsor*) if Φ is an isomorphism and if, moreover, the morphism $X \longrightarrow S$ is covering for the topology T (cf. IV 5.1.5 and 5.1.6 (ii)).

Proposition 10.16. *We place ourselves in the situation considered at the beginning of 10.0. Let $j \in I$, G_j be an S_j -group and X_j an S_j -scheme with group of operators G_j . Suppose G_j and X_j of finite presentation over S_j .*

For $X = X_j \times_{S_j} S$ to be a homogeneous space (resp. a principal homogeneous bundle) under $G = G_j \times_{S_j} S$ for the (fppf) topology, it is necessary and sufficient that there be an index $i \geq j$ such that $X_i = X_j \times_{S_j} S_i$ is a homogeneous space (resp. a principal homogeneous bundle) under $G_i = G_j \times_{S_j} S_i$.

Taking account of 10.14 and EGA IV₃, 8.8.2, 8.8.3 and 8.10.5, the statement follows from the property concerning covering morphisms for the (fppf) topology recalled in 10.1. ⁽¹⁰³⁾

11. Affine group schemes

11.0. — Recollections. ⁽¹⁰⁴⁾ Let $q : X \longrightarrow S$ be a quasi-compact and quasi-separated morphism of schemes (cf. EGA IV₁, 1.1 & 1.2), and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. Recall that $q_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_S$ -module (EGA I, 9.2.1). Moreover, by EGA III, 1.4.15 (supplemented by EGA IV₁, 1.7.21), one has point (c) below, and the proof in loc. cit. also gives points (a) and (b):

(a) If $\mathcal{F}$ is a filtered inductive limit of quasi-coherent submodules $\mathcal{F}_\alpha$, then $q_*(\mathcal{F}) = \lim_{\longrightarrow \alpha} q_*(\mathcal{F}_\alpha)$.

(b) If $\mathcal{E}$ is a flat $\mathcal{O}_S$ -module, the canonical morphism $\mathcal{E} \otimes_{\mathcal{O}_S} q_*(\mathcal{O}_X) \longrightarrow q_*q^*(\mathcal{E})$ is an isomorphism.

⁽¹⁰²⁾**N.D.E.** The original has been corrected by distinguishing the notions of a “formally homogeneous” or “homogeneous” object, see IV §§ 5.1 and 6.7.

⁽¹⁰³⁾**N.D.E.** For the limit properties of torsors in terms of Čech cohomology, see also SGA 4, VII 5.7 and its corollaries. This has been expanded in the article [Ma07].

⁽¹⁰⁴⁾**N.D.E.** This paragraph of recollections has been added.

(c) Let $p : S' \rightarrow S$ be a flat morphism, $q' : X' \rightarrow S'$ the morphism obtained from q by base change, and $\mathcal{F}'$ the inverse image of $\mathcal{F}$ on X' . Then the canonical morphism $p^*q_*(\mathcal{F}) \rightarrow q'_*(\mathcal{F}')$ is an isomorphism.

Indeed, let $U = \text{Spec}(A)$ be an arbitrary affine open subset of S . By hypothesis, $q^{-1}(U)$ is a union of affine open subsets $V_i = \text{Spec}(B_i)$, for $i = 1, \dots, n$, and each intersection $V_i \cap V_j$ is a union of finitely many affine open subsets $W_{ijk} = \text{Spec}(C_{ijk})$. Then $\Gamma(U, q_*(\mathcal{F})) = \Gamma(q^{-1}(U), \mathcal{F})$ is the kernel of the morphism

$$\bigoplus_{i=1}^n \Gamma(V_i, \mathcal{F}) \rightarrow \bigoplus_{i,j,k} \Gamma(W_{ijk}, \mathcal{F}).$$

Point (a) follows, for each of the terms above commutes with filtered inductive limits (since the V_i and W_{ijk} are affine, hence quasi-compact). Let us prove (b): $E = \Gamma(U, \mathcal{E})$ is a flat A -module, and $\Gamma(U, q_*q^*(\mathcal{E}))$ is the kernel $K(E)$ of the morphism

$$\bigoplus_{i=1}^n B_i \otimes_A E \rightarrow \bigoplus_{i,j,k} C_{ijk} \otimes_A E$$

and since E is flat over A , this kernel is identified with $K(A) \otimes_A E = \mathcal{O}_X(q^{-1}(U)) \otimes_A E$. Finally, if U' is an arbitrary affine open subset of S' lying over U , then $A' = \mathcal{O}(U')$ is a flat A -algebra, and one obtains as above that $\mathcal{F}(q^{-1}(U)) \otimes_A A' \xrightarrow{\sim} \mathcal{F}'(q'^{-1}(U))$.

Notation. — Let S be a scheme, X an S -scheme, $f : X \rightarrow S$ the structural morphism; one shall put $\mathcal{A}(X) = f_*(\mathcal{O}_X)$.

Lemma 11.1. *Let X and Y be two S -schemes quasi-compact and quasi-separated over S , $f : X \rightarrow S$ and $g : Y \rightarrow S$ the structural morphisms. Then the canonical homomorphism*

$$\varphi : \mathcal{A}(X) \otimes_{\mathcal{O}_S} \mathcal{A}(Y) \rightarrow \mathcal{A}(X \times_S Y)$$

is an isomorphism in each of the following cases:⁽¹⁰⁵⁾

- a) f and g are affine,
- b) g (or f) is flat and affine,
- c) g is flat and $f_*(\mathcal{O}_X)$ is a flat $\mathcal{O}_S$ -module.

In case (b), one shall suppose that it is g that is flat and affine. Put then $S' = \text{Spec } \mathcal{A}(X)$, $Y' = Y \times_S S'$, $g' = g \times_S S'$ and denote by v the morphism $S' \rightarrow S$:

$$\begin{array}{ccc} Y' & \xrightarrow{\quad} & Y \\ g' \downarrow & & \downarrow g \\ \text{Spec } \mathcal{A}(X) = S' & \xrightarrow{\quad} & S. \end{array}$$

In cases (a) and (b), g is affine and therefore, by EGA II, 1.5.2, one has:

$$(1) \quad g'_*(\mathcal{O}_{Y'}) = v^*g_*(\mathcal{O}_Y) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

One has the same equality in case (c), by 11.0 (c), since S' is flat over S and g is quasi-compact and quasi-separated.

On the other hand (EGA II 1.2.7), $f : X \rightarrow S$ factors through v by a morphism $p : X \rightarrow S'$, and one has $p_*(\mathcal{O}_X) = \mathcal{O}_{S'}$ and $X \times_S Y = X \times_{S'} Y'$. Since f is quasi-separated, p is too (EGA IV₁, 1.2.2), and since f is quasi-compact and v quasi-separated, p is also

⁽¹⁰⁵⁾**N.D.E.** Note that, if k is a field and X is an infinite sum of copies of $S = \text{Spec } k$ (so that X is not quasi-compact), then $\mathcal{A}(X) = k^X$ and the canonical morphism $k^X \otimes k^X \rightarrow k^{X \times X}$ is not surjective.

quasi-compact (EGA IV₁, 1.2.4). Consider then the cartesian square:

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p'} & Y' \\ \downarrow & & \downarrow g' \\ X & \xrightarrow{p} & S'. \end{array}$$

In cases (b) and (c), Y is flat over S , so Y' is flat over S' ; applying 11.0 (c) again, one obtains:

$$(2) \quad p'_*(\mathcal{O}_{X \times_S Y}) = g'^* p_*(\mathcal{O}_X) = g'^*(\mathcal{O}_{S'}) = \mathcal{O}_{Y'},$$

and one has the same equality in case (a), for in this case p and p' are isomorphisms.

Finally, since v is affine, one has, by EGA II, 1.4.7, $v_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(X)$ for every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. Combined with (2) and (1), this gives:

$$\begin{aligned} \mathcal{A}(X \times_S Y) &= v_* g'_* p'_*(\mathcal{O}_{X \times_S Y}) \stackrel{(2)}{=} v_* g'_*(\mathcal{O}_{Y'}) \\ &\stackrel{(1)}{=} v_*(\mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{A}(X). \end{aligned}$$

Corollary 11.2. *The functor $X \mapsto \text{Spec } \mathcal{A}(X)$, from the full subcategory of $(\text{Sch})/S$ formed by S -schemes X flat, quasi-compact and quasi-separated over S , and such that $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module, to that of S -schemes flat and affine over S , commutes with finite products, hence transforms S -groups into S -groups.*

Definition 11.3. Given an S -group G flat, quasi-compact and quasi-separated over S , such that $\mathcal{A}(G)$ is flat over $\mathcal{O}_S$,⁽¹⁰⁶⁾ we shall denote by G_{af} , and call the *affine envelope* of G , the S -group $G_{\text{af}} = \text{Spec } \mathcal{A}(G)$.

Proposition 11.3.1. ⁽¹⁰⁷⁾ *The canonical morphism $\tau_G : G \rightarrow G_{\text{af}}$ is a morphism of S -groups. Moreover, it satisfies the following universal property:*

(i) *For every morphism of S -schemes $\phi : G \rightarrow H$, where H is affine over S , there is a unique morphism of S -schemes $\phi' : G_{\text{af}} \rightarrow H$ such that $\phi = \phi' \circ \tau_G$.*

(ii) *If moreover H is an S -group and ϕ is a morphism of S -groups, then ϕ' is too.*

11.4. — ⁽¹⁰⁸⁾ Let $\mathcal{E}$ and $\mathcal{F}$ be two quasi-coherent $\mathcal{O}_S$ -modules. Consider the S -functor $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))$ (cf. I, 3.1.4), i.e. for every S -scheme $f : X \rightarrow S$,

$$\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))(X) = \text{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{E})_X, \mathbf{W}(\mathcal{F})_X).$$

Moreover, by I, 4.6.2, one has $\mathbf{W}(\mathcal{E})_X = \mathbf{W}(f^*(\mathcal{E}))$ (and similarly for $\mathcal{F}$) and

$$\text{Hom}_{\mathcal{O}_X}(\mathbf{W}(f^*(\mathcal{E})), \mathbf{W}(f^*(\mathcal{F}))) = \text{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})).$$

One therefore obtains (using the adjunction formula for the last equality):

$$\begin{aligned} \underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))(X) &= \text{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})) \\ (\dagger) \quad &= \text{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{F})). \end{aligned}$$

Proposition 11.5. *Let X be an S -scheme quasi-compact and quasi-separated over S , $f : X \rightarrow S$ the structural morphism, $\mathcal{E}$ and $\mathcal{E}'$ two quasi-coherent $\mathcal{O}_S$ -modules. Suppose that one of the following two conditions holds:*

- a) *f is affine,*
- b) *$\mathcal{E}$ is flat over $\mathcal{O}_S$.*

⁽¹⁰⁶⁾**N.D.E.** Let us point out that if S is a locally noetherian regular scheme of dimension ≤ 2 , and X an S -scheme flat, quasi-compact and quasi-separated, then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module, cf. [Ray70a], VII 3.2.

⁽¹⁰⁷⁾**N.D.E.** This proposition has been added; see also the additional paragraph 12 below for a study of the morphism $G \rightarrow G_{\text{af}}$ and its kernel.

⁽¹⁰⁸⁾**N.D.E.** In 11.4–11.6, $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}), \mathbf{W}(\mathcal{F}))$ has been considered in place of $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{F}), \mathbf{V}(\mathcal{E}))$ (cf. I, 4.6.3), and the original simplified taking account of the addition 11.0 (b).

Then the canonical morphism $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X) \rightarrow f_* f^*(\mathcal{E})$ is an isomorphism, and one therefore has

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}'), \mathbf{W}(\mathcal{E}))(X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X)).$$

Indeed, the second assertion follows from 11.4 and the first; the latter follows from EGA II, 1.4.7 in case (a), and 11.0 (b) in case (b).⁽¹⁰⁹⁾

11.6. — ⁽¹¹⁰⁾ Let G be an S -group and $\mathcal{E}$ a quasi-coherent $\mathcal{O}_S$ -module. Giving $\mathcal{E}$ a structure of G - $\mathcal{O}_S$ -module (i.e. an $\mathcal{O}_S$ -linear operation of G on $\mathbf{W}(\mathcal{E})$, cf. I 4.7.1) is equivalent to giving a morphism of S -functors in monoids $\rho : G \rightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$ (indeed, such a ρ necessarily sends G into $\underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$).

Now, by 11.4, giving a morphism of S -functors $\rho : G \rightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$ is equivalent to giving an element θ of $\mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{E}), f^*(\mathcal{E}))$, which corresponds by adjunction to an element δ of $\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{E}))$, where f denotes the projection $G \rightarrow S$.

Let $m : G \times_S G \rightarrow G$ be multiplication, δ_G the morphism $\mathcal{O}_G \rightarrow m_*(\mathcal{O}_{G \times_S G})$, and ϕ the projection $G \times_S G \rightarrow S$ (which equals $f \circ m$). It is convenient to denote by $\boxtimes$ the “external” tensor product; one thus obtains a morphism

$$\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \rightarrow \mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G})$$

and, by abuse of notation, one will again denote by $\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G$ the composite of the preceding morphism with the canonical morphism $\mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G}) \rightarrow m_* m^* f^*(\mathcal{E}) = m_* \phi^*(\mathcal{E})$.

On the other hand, denote by $h : G \rightarrow S$ a second copy of $f : G \rightarrow S$ and consider the following commutative diagram, where p, q denote the two projections:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ p \downarrow & \searrow \phi & \downarrow h \\ G & \xrightarrow{f} & S. \end{array}$$

Again denoting by δ the morphism $\mathcal{E} \rightarrow h_* h^*(\mathcal{E})$, one obtains the morphism

$$\delta \boxtimes \mathrm{id}_{\mathcal{O}_G} : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \rightarrow h_* h^*(\mathcal{E}) \boxtimes_{\mathcal{O}_S} \mathcal{O}_G = f^* h_* h^*(\mathcal{E})$$

and, by abuse of notation, we shall again denote by $\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}$ the composite of this morphism with the canonical morphism $f^* h_* h^*(\mathcal{E}) \rightarrow p_* \phi^*(\mathcal{E})$.

Then the condition that ρ be compatible with multiplication is equivalent to saying that, for every open subset U of S , the diagram below is commutative:

$$(1) \quad \begin{array}{ccc} \Gamma(U, \mathcal{E}) & \xrightarrow{\delta} & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ \delta \downarrow & & \downarrow \mathrm{id}_{\mathcal{E}} \boxtimes \delta_G \\ \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) & \xrightarrow[\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}]{} & \Gamma(U \times_S G \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \boxtimes_{\mathcal{O}_S} \mathcal{O}_G). \end{array}$$

Furthermore, the identity section $\varepsilon : S \rightarrow G$ induces a morphism u from $\mathcal{O}_G$ to $\varepsilon_* \varepsilon^*(\mathcal{O}_G) = \varepsilon_*(\mathcal{O}_S)$, and the condition that ρ preserve the identity elements is equivalent to

⁽¹⁰⁹⁾**N.D.E.** The original has been simplified; it used the isomorphism $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}'), \mathbf{W}(\mathcal{E})) \simeq \underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{E}), \mathbf{V}(\mathcal{E}'))$, then the inclusion of the right-hand term into

$$\underline{\mathrm{Hom}}_S(\mathbf{V}(\mathcal{E}), \mathbf{V}(\mathcal{E}')) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', f_* f^*(\mathrm{Sym}(\mathcal{E})))$$

and applied EGA III, 4.1.15 to $\mathbf{V}(\mathcal{E}) = \mathrm{Spec}(\mathrm{Sym}(\mathcal{E}))$ to deduce 11.0 (b).

⁽¹¹⁰⁾**N.D.E.** 11.6 has been expanded, and the results obtained brought out in the form of Proposition 11.6.1.

commutativity of the diagram:

$$(2) \quad \begin{array}{ccc} \Gamma(U, \mathcal{E}) & \xrightarrow{\delta} & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ & \searrow \simeq & \swarrow \text{id}_{\mathcal{E}} \boxtimes u \\ & \Gamma(U \times_S S, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_S) & \end{array}$$

One therefore sees that giving $\mathcal{E}$ a G - $\mathcal{O}_S$ -module structure is equivalent to giving a morphism of $\mathcal{O}_S$ -modules $\delta : \mathcal{E} \rightarrow f_* f^*(\mathcal{E})$ satisfying conditions (1) and (2) above, and in this case the morphism $\theta : f^*(\mathcal{E}) \rightarrow f^*(\mathcal{E})$, obtained from δ by adjunction, is an isomorphism (for it corresponds to the isomorphism $G \times_S \mathbf{W}(\mathcal{E}) \xrightarrow{\sim} G \times_S \mathbf{W}(\mathcal{E})$ defined set theoretically by $(g, x) \mapsto (g, gx)$, see also I, 6.5.4).

Suppose now that G is flat, quasi-compact and quasi-separated over S , and that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module; then, by 11.1 (c), the canonical morphism $\mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \rightarrow \mathcal{A}(G \times_S G)$ is an isomorphism, and the morphism $\delta_G : \mathcal{A}(G) \rightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ will be denoted by Δ .

If moreover $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G) = f_* f^*(\mathcal{E})$ (which is the case, by 11.5, if $G \rightarrow S$ is affine, or if $\mathcal{E}$ is flat over $\mathcal{O}_S$), one obtains that conditions (1) and (2) are equivalent to the conditions below, which express that $\delta : \mathcal{E} \rightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ makes $\mathcal{E}$ a right $\mathcal{A}(G)$ -comodule (cf. I 4.7.2):

(CM 1) Putting $\mathcal{A} = \mathcal{A}(G)$, the diagram below is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ \delta \downarrow & & \downarrow \text{id}_{\mathcal{E}} \otimes \Delta \\ \mathcal{E} \otimes \mathcal{A} & \xrightarrow{\delta \otimes \text{id}_{\mathcal{A}}} & \mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A} \end{array}$$

(CM 2) Denoting by $\eta : \mathcal{A} \rightarrow \mathcal{O}_S$ the morphism $\mathcal{A}(\varepsilon)$, the diagram below is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ & \searrow \simeq & \swarrow \text{id}_{\mathcal{E}} \otimes \eta \\ & \mathcal{E} \otimes \mathcal{O}_S & \end{array}$$

Remark 11.6.A. Recall that $V(\mathcal{E})$ denotes the vector fibration over S which represents the functor $\mathbf{V}(\mathcal{E})$, i.e. for every $S' \rightarrow S$, $\mathbf{V}(\mathcal{E})(S') = \text{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. Since, by I, 4.6.2, one has an anti-isomorphism of S -functors in monoids $\underline{\text{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E})) \simeq \underline{\text{End}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{E}))$, one sees that, if $\mathcal{E}$ is a left G - $\mathcal{O}_S$ -module, one has a right action $\mu : V(\mathcal{E}) \times_S G \rightarrow V(\mathcal{E})$ of G on $V(\mathcal{E})$, defined set theoretically by $(\varphi g)(x) = \varphi(gx)$, for every $g \in G(S')$, $x \in \Gamma(S', \mathcal{E}_{S'})$ and $\varphi \in \text{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. One therefore obtains commutative diagrams:

$$\begin{array}{ccc} V(\mathcal{E}) \times_S G \times_S G & \xrightarrow{\mu \times \text{id}_G} & V(\mathcal{E}) \times_S G \\ \text{id}_{V(\mathcal{E})} \times m \downarrow & & \downarrow \mu \\ V(\mathcal{E}) \times_S G & \xrightarrow{\mu} & V(\mathcal{E}) \end{array} \quad \begin{array}{ccc} V(\mathcal{E}) & \xleftarrow{\mu} & V(\mathcal{E}) \times_S G \\ & \swarrow \simeq & \uparrow \text{id}_{V(\mathcal{E})} \times \varepsilon \\ & V(\mathcal{E}) \times_S S & \end{array}$$

When G is flat, quasi-compact and quasi-separated over S , $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, and one of the conditions of 11.5 holds, one similarly recovers conditions (CM 1) and (CM 2).

Consequently, we have obtained:

Proposition 11.6.1. *Let G be an S -group flat, quasi-compact and quasi-separated over S , such that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module.*

(i) *Giving an $\mathcal{A}(G)$ -comodule structure $\delta : \mathcal{E} \rightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ amounts to the same thing as giving $\mathcal{E}$ a G_{af} - $\mathcal{O}_S$ -module structure (i.e. an $\mathcal{O}_S$ -linear operation of G_{af} on $\mathcal{E}$). By*

composition with the morphism of S -groups $G \rightarrow G_{\text{af}}$, this defines a G - $\mathcal{O}_S$ -module structure on $\mathcal{E}$.

(ii) If moreover $\mathcal{E}$ is flat, every $\mathcal{O}_S$ -linear operation of G on $\mathcal{E}$ factors through G_{af} and corresponds to a unique $\mathcal{A}(G)$ -comodule structure on $\mathcal{E}$.

Lemma 11.7. *Let G be an S -group flat, quasi-compact and quasi-separated over S , such that $\mathcal{A} = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module. Let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module, $\delta : \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{A}$ an $\mathcal{A}$ -comodule structure, and $\rho : G \rightarrow \underline{\text{Aut}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{E}))$ the associated operation of G on $\mathcal{E}$.*

Let $\mathcal{E}_0$ be a quasi-coherent sub- $\mathcal{O}_S$ -module of $\mathcal{E}$ such that the restriction δ_0 of μ to $\mathcal{E}_0$ factors through $\mathcal{E}_0 \rightarrow \mathcal{E}_0 \otimes \mathcal{A}$, i.e. such that one has a commutative diagram:

$$\begin{array}{ccc} \mathcal{E}_0 & \hookrightarrow & \mathcal{E} \\ \delta_0 \downarrow & & \downarrow \delta \\ \mathcal{E}_0 \otimes \mathcal{A} & \hookrightarrow & \mathcal{E} \otimes \mathcal{A}. \end{array}$$

(N.B. The morphism $\mathcal{E}_0 \otimes \mathcal{A} \rightarrow \mathcal{E} \otimes \mathcal{A}$ is injective, since $\mathcal{A}$ is flat over $\mathcal{O}_S$.)

Then δ_0 makes $\mathcal{E}_0$ an $\mathcal{A}(G)$ -comodule, and thus defines an operation ρ_0 of G on $\mathcal{E}_0$ (which will be called the operation induced on $\mathcal{E}_0$ by ρ , and one will say that $\mathcal{E}_0$ is stable under ρ).

This follows immediately from the definitions and 11.6. Note, however, that in general the canonical map $\mathbf{W}(\mathcal{E}_0) \rightarrow \mathbf{W}(\mathcal{E})$ is not a monomorphism.

Remark 11.7.bis. ⁽¹¹¹⁾ Let G be a flat S -group and $\mathcal{E}$ a quasi-coherent G - $\mathcal{O}_S$ -module. Denote by f the morphism $G \rightarrow S$ and by δ the morphism $\mathcal{E} \rightarrow f_*f^*(\mathcal{E})$ defined in 11.6. Let $\mathcal{E}_0$ be a quasi-coherent sub- $\mathcal{O}_S$ -module of $\mathcal{E}$; since f is flat, then $f_*f^*(\mathcal{E}_0)$ is a sub- $\mathcal{O}_S$ -module of $f_*f^*(\mathcal{E})$, and similarly for $\phi = f \times f$. Consequently, if the restriction δ_0 of δ to $\mathcal{E}_0$ factors through $f_*f^*(\mathcal{E}_0)$, then it makes $\mathcal{E}_0$ a G - $\mathcal{O}_S$ -module. In this case, one will say that $\mathcal{E}_0$ is a G -stable submodule of $\mathcal{E}$.

Definition 11.8.0. ⁽¹¹²⁾ Let S be a scheme. An $\mathcal{O}_S$ -coalgebra is an $\mathcal{O}_S$ -module $\mathcal{C}$ endowed with two morphisms of $\mathcal{O}_S$ -modules $\Delta : \mathcal{C} \rightarrow \mathcal{C} \otimes \mathcal{C}$ and $\varepsilon : \mathcal{C} \rightarrow \mathcal{O}_S$, satisfying the following two axioms (cf. I 4.2):

(CO 1) Δ is coassociative: the following diagram is commutative

$$\begin{array}{ccc} & \mathcal{C} \otimes \mathcal{C} & \\ \Delta \nearrow & & \searrow \text{id} \otimes \Delta \\ \mathcal{C} & & \mathcal{C} \otimes \mathcal{C} \otimes \mathcal{C} \\ \Delta \searrow & & \nearrow \Delta \otimes \text{id} \\ & \mathcal{C} \otimes \mathcal{C} & \end{array}$$

(CO 2): ε is a counit, i.e. the following two composites are the identity

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\text{id} \otimes \varepsilon} \mathcal{C} \otimes \mathcal{O}_S \xrightarrow{\sim} \mathcal{C},$$

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\varepsilon \otimes \text{id}} \mathcal{O}_S \otimes \mathcal{C} \xrightarrow{\sim} \mathcal{C}.$$

A (right) $\mathcal{C}$ -comodule is an $\mathcal{O}_S$ -module $\mathcal{E}$ endowed with a morphism of $\mathcal{O}_S$ -modules $\mu : \mathcal{E} \rightarrow \mathcal{E} \otimes \mathcal{C}$ satisfying axioms (CM 1) and (CM 2) of 11.6.

One will say that $\mathcal{C}$ (resp. $\mathcal{E}$) is a quasi-coherent coalgebra (resp. a quasi-coherent comodule) if it is a quasi-coherent $\mathcal{O}_S$ -module.

⁽¹¹¹⁾**N.D.E.** This remark, which generalizes 11.7 and will be useful in 11.10.bis, has been added.

⁽¹¹²⁾**N.D.E.** Since statements 11.8 and 11.9 concern solely the notion of a comodule over a coalgebra, Definition 11.8.0 has been introduced and 11.8 and 11.9 reformulated accordingly.

Let A be a commutative ring and C an A -coalgebra; then $C^\vee = \text{Hom}_A(C, A)$ is an A -algebra. One shall denote by ev the natural evaluation map $C \otimes_A C^\vee \longrightarrow A$.

Lemma 11.8. *Let C be an A -coalgebra, V a C -comodule, M a sub- A -module of V . Suppose that C is a projective A -module.⁽¹¹³⁾ Let $c(M)$ be the image of the morphism of A -modules*

$$\theta : M \otimes_A C^\vee \xrightarrow{\mu \otimes \text{id}} V \otimes_A C \otimes_A C^\vee \xrightarrow{\text{id} \otimes \text{ev}} V.$$

Then $c(M)$ is the smallest subcomodule of V containing M , and is an A -module of finite type if M is. One will say that $c(M)$ is the subcomodule generated by M .

Moreover, for every ring morphism $A \longrightarrow A'$, if M' denotes the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$, then $c(M')$ is the image of $c(M) \otimes_A A'$ in V' , so: “the formation of $c(M)$ commutes with base change”.

First, $M \subset c(M)$ by (CM 2), and if N is a subcomodule of V containing M , one has $\mu(M) \subset N \otimes C$ and therefore $c(M) \subset N$.

By hypothesis, C is a direct summand of a free A -module L , with basis $(e_i)_{i \in I}$. Denote by φ_i the restriction to C of the linear form e_i^* , defined by $e_i^*(e_j) = \delta_{ij}$. Let $x \in M$. One can write:

$$(1) \quad \mu(x) = \sum_{i \in J} x_i \otimes e_i,$$

where $x_i \in V$ and J is a finite subset of I . Then $x_i = \theta(x \otimes \varphi_i)$ belongs to $c(Ax)$, and one therefore has $c(Ax) = \sum_{i \in J} Ax_i$. Since C is a direct summand of L , say $L = C \oplus R$, whence $V \otimes L = (V \otimes C) \oplus (V \otimes R)$, one obtains that

$$(c(Ax) \otimes L) \cap (V \otimes C) = c(Ax) \otimes C.$$

Consequently, $\mu(x)$ can also be written in the form

$$(2) \quad \mu(x) = \sum_{j \in J} x_j \otimes b_j,$$

with $b_j \in C$. One can write $\Delta(b_j) = \sum_{i \in I} b_{ij} \otimes e_i$, with $b_{ij} \in C$. Then, applying $\mu \otimes \text{id}$ to (1) (resp. $\text{id} \otimes \Delta$ to (2)) and using axiom (CM 1), one obtains, for every $i \in J$:

$$\mu(x_i) = \sum_{j \in J} x_j \otimes b_{ij} \in c(Ax) \otimes C.$$

This shows that $c(M)$ is a subcomodule of V , and thus it is the smallest subcomodule of V containing M .

It is clear that $c(M)$ is an A -module of finite type if M is: if $M = Ax_1 + \dots + Ax_n$, and $\mu(x_k) = \sum_i x_{ik} \otimes e_i$, then $c(M)$ is generated by the x_{ik} , for $k = 1, \dots, n$ and i ranging over a finite subset of I .

Finally, let $A \longrightarrow A'$ be a ring morphism and let M' be the image of $M \otimes A'$ in $V' = V \otimes A'$. Then $c(M')$ (resp. the image of $c(M) \otimes A'$ in V') is the image of the morphism θ' below (resp. of the composite $\theta' \circ \tau$):

$$M \otimes A' \otimes C^\vee \xrightarrow{\tau} M \otimes \text{Hom}_A(C, A') \xrightarrow{\theta'} V'.$$

Now these two morphisms have the same image. Indeed, let $\psi \in \text{Hom}_A(C, A')$ and $x \in M$. Put $\mu(x) = \sum_{i \in J} x_i \otimes e_i$. Then

$$\theta'(x \otimes \psi) = \sum_{i \in J} \psi(e_i) x_i$$

is the image under $\theta' \circ \tau$ of the element $\sum_{i \in J} x \otimes \psi(e_i) \otimes \varphi_i$ of $M \otimes A' \otimes C^\vee$. This proves the lemma.

⁽¹¹³⁾**N.D.E.** In the original, C is supposed a free A -module, the generalization to the case where C is a projective A -module, pointed out by J.-P. Serre, being mentioned in Remark 11.10.1. This generalization has been included here and in 11.9, and the proofs expanded accordingly.

Furthermore, one has the following proposition:

Proposition 11.8.bis. ⁽¹¹⁴⁾ *Let A be a noetherian ring, C an A -coalgebra flat over A , V a C -comodule, and M a sub- A -module of finite type of V . Then there is a subcomodule W of V , of finite type over A , containing M .*

Indeed, since M is of finite type, so is $\Delta_V(M)$, so there is a sub- A -module of finite type M' of V such that $\Delta_V(M) \subset M' \otimes_A C$. Let π be the projection $V \rightarrow V/M'$ and $\bar{\Delta}_V = (\pi \otimes \text{id}_C)\Delta_V$, and let

$$W = \{x \in V \mid \Delta_V(x) \in M' \otimes_A C\} = \text{Ker } \bar{\Delta}_V;$$

this is a sub- A -module of V containing M and contained in M' (since $x = (\text{id}_V \otimes \varepsilon)\Delta_V(x)$), hence of finite type over A . Moreover, $(\bar{\Delta}_V \otimes \text{id}_C)\Delta_V = (\pi \otimes \Delta_C)\Delta_V$ vanishes on W , i.e. $\Delta_V(W)$ is contained in the kernel K of $(\bar{\Delta}_V \otimes \text{id}_C)$. But since C is flat over A , one has $K = W \otimes C$, so W is a subcomodule of V .

Lemma 11.8.1. ⁽¹¹⁵⁾ *Let C be an A -coalgebra, V a C -comodule, M a sub- A -module of V , and $f : A \rightarrow A'$ a faithfully flat ring morphism. Suppose that $C' = C \otimes A'$ is a projective A' -module.*

(i) *Then there is a smallest subcomodule $t(M)$ of V containing M , and $t(M)$ is an A -module of finite type if M is. Moreover, “the formation of $t(M)$ commutes with base change”.*

(ii) *More precisely, C is a projective A -module, and one has $t(M) = c(M)$.*

PROOF. (ii) By [RG71] (see Proposition 11.8.2 below), C is a projective A -module. One can therefore apply Lemma 11.8: $c(M)$ is the smallest subcomodule of V containing M , it is an A -module of finite type if M is, and its formation commutes with base change.

To avoid an anachronism ([RG71] being later than SGA 3), let us sketch a direct proof of point (i). Since $A \rightarrow A'$ is flat, $M' = M \otimes A'$ is a sub- A' -module of $V' = V \otimes A'$, and, since C' is a projective A' -module, $c(M')$ is the smallest subcomodule of V' containing M' . Denote by $V' \otimes A'$ and $A' \otimes V'$ the two $A' \otimes A'$ -comodule structures on $V'' = V' \otimes_{A'} (A' \otimes A')$ obtained by the two base changes $A' \rightrightarrows A' \otimes A'$, $a' \mapsto a' \otimes 1$ and $a' \mapsto 1 \otimes a'$. The A' -comodule V' is endowed with an isomorphism of $A' \otimes A'$ -comodules $\phi : V' \otimes A' \xrightarrow{\sim} A' \otimes V'$, $(x \otimes a') \otimes b' \mapsto b' \otimes (x \otimes a')$, which is a descent datum, i.e. which satisfies $\phi_{31} = \phi_{32} \circ \phi_{21}$.

Since $M' = M \otimes A'$, then ϕ sends $M' \otimes A'$ onto $A' \otimes M'$, and hence $c(M' \otimes A')$ onto $c(A' \otimes M')$. Since the formation of $c(M')$ commutes with base change, one has $c(M' \otimes A') = c(M') \otimes A'$ and $c(A' \otimes M') = A' \otimes c(M')$. One therefore has

$$\phi(c(M') \otimes A') = A' \otimes c(M')$$

and it follows that ϕ endows $c(M')$ with a descent datum. By (fpqc) descent, there is a unique subcomodule $t(M)$ of V such that $c(M') = t(M) \otimes A'$, and $t(M)$ contains M since $t(M) \otimes A'$ contains M' . Moreover, if N is a subcomodule of V containing M , then N contains $t(M)$, since $N \otimes A'$ contains $c(M') = t(M) \otimes A'$. Thus $t(M)$ is the smallest subcomodule of V containing M .

Finally, let $A \rightarrow B$ be a ring morphism. Let $B' = B \otimes A'$ and let M_B (resp. $M'_{B'}$) be the image of $M \otimes B$ in $V_B = V \otimes B$ (resp. of $M' \otimes_{A'} B'$ in $V \otimes B'$); then $M_B \otimes_B B' = M'_{B'}$. On the one hand, the preceding construction, applied to C_B and the morphism $B \rightarrow B'$, gives:

$$c(M_B \otimes_B B') = t(M_B) \otimes_B B' = t(M_B) \otimes A'.$$

On the other hand, since the formation of $c(M')$ commutes with base change, $c(M'_{B'})$ is the image in $V' \otimes_{A'} B' = V \otimes B \otimes A'$ of

$$c(M') \otimes_{A'} B' = t(M) \otimes B \otimes A'.$$

⁽¹¹⁴⁾**N.D.E.** This proposition, taken from [Se68], § 1.5, Prop. 2, has been added.

⁽¹¹⁵⁾**N.D.E.** This lemma, used in the proof of 11.9, has been added.

It follows that $t(M_B)$ is the image in V_B of $t(M) \otimes B$. $\square$

Proposition 11.8.2. (Gruson–Raynaud). *Let $f : A \rightarrow A'$ be a faithfully flat morphism; then f “descends projectivity”, i.e. if M is an A -module and $M \otimes_A A'$ is a projective A' -module, then M is a projective A -module.*

Indeed, by [RG71] II 2.5.1, f “descends the Mittag-Leffler condition”, so, by loc. cit. II 3.1.3, f descends projectivity.⁽¹¹⁶⁾

Proposition 11.9.⁽¹¹⁷⁾ *Let $\mathcal{C}$ be an $\mathcal{O}_S$ -coalgebra, $\mathcal{E}$ a $\mathcal{C}$ -comodule, $\mathcal{F}$ a sub- $\mathcal{O}_S$ -module of $\mathcal{E}$, all quasi-coherent. Suppose given a covering of S by affine open subsets $U_\alpha = \operatorname{Spec} A_\alpha$, and for each α a faithfully flat ring morphism $A_\alpha \rightarrow A'_\alpha$ such that $\Gamma(U_\alpha, \mathcal{C}) \otimes_{A_\alpha} A'_\alpha$ is a projective A'_α -module.**

There is then a smallest quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$, and $t(\mathcal{F})$ is an $\mathcal{O}_S$ -module of finite type if $\mathcal{F}$ is. Moreover, for every base change $S' \rightarrow S$, if $\mathcal{F}'$ denotes the image of $\mathcal{F} \otimes_{\mathcal{O}_S}$ in $\mathcal{E}' = \mathcal{E} \otimes_{\mathcal{O}_S}$, then $t(\mathcal{F}')$ is the image of $t(\mathcal{F}) \otimes_{\mathcal{O}_S}$ in $\mathcal{E}'$, i.e. “the formation of $t(\mathcal{F})$ commutes with base change”.

PROOF.⁽¹¹⁷⁾ For each α , the $\mathcal{O}_{U_\alpha}$ -module $\mathcal{T}_\alpha$ associated to the A_α -module

$$T_\alpha = t(\Gamma(U_\alpha, \mathcal{F}))$$

is, by 11.8.1 (i), the smallest quasi-coherent subcomodule of $\mathcal{E}|_{U_\alpha}$ containing $\mathcal{F}|_{U_\alpha}$, and is an $\mathcal{O}_{U_\alpha}$ -module of finite type if $\mathcal{F}|_{U_\alpha}$ is.

For every α, β , put $U_{\alpha\beta} = U_\alpha \cap U_\beta$. Since the construction of $t(M)$ commutes with base change, one has, for every α, β , canonical isomorphisms of $\mathcal{O}_{U_{\alpha\beta}}$ -modules

$$\phi_{\alpha\beta} : T_\beta \otimes_{A_\beta} \mathcal{O}_{U_{\alpha\beta}} \xrightarrow{\sim} T_\alpha \otimes_{A_\alpha} \mathcal{O}_{U_{\alpha\beta}}$$

which satisfy the cocycle condition $\phi'_{\alpha\gamma} = \phi'_{\alpha\beta} \circ \phi'_{\beta\gamma}$, where $\phi'_{\alpha\gamma}$ (resp. $\dots$) denotes the restriction of $\phi_{\alpha\gamma}$ (resp. $\dots$) to $U_\alpha \cap U_\beta \cap U_\gamma$.

Consequently, the $\mathcal{T}_\alpha$ glue to a quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$. We leave to the reader the verification that $t(\mathcal{F})$ is the smallest quasi-coherent subcomodule of $\mathcal{E}$ containing $\mathcal{F}$, and that its formation commutes with base change. $\square$

Definition 11.9.1.⁽¹¹⁸⁾ Let S be a scheme and $\mathcal{P}$ a quasi-coherent $\mathcal{O}_S$ -module. The following conditions are equivalent:

- (i) For every affine open subset U of S , $\Gamma(U, \mathcal{P})$ is a projective $\mathcal{O}_S(U)$ -module.
- (ii) There is a covering (U_α) of S by affine open subsets such that each $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$ -module.
- (iii) There is a covering (U_α) of S by affine open subsets, and faithfully flat ring morphisms $A_\alpha = \mathcal{O}_S(U_\alpha) \rightarrow A'_\alpha$, such that, for each α , $\Gamma(U_\alpha, \mathcal{P}) \otimes_{A_\alpha} A'_\alpha$ is a projective A'_α -module.

Indeed, it is clear that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). Conversely, if (iii) holds, 11.8.2 implies that each $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$ -module, whence (ii). Finally, suppose (ii) holds and let $V = \operatorname{Spec} A$ be an arbitrary affine open subset; it is covered by finitely many affine open subsets $V_1, \dots, V_n$, where each $V_i = \operatorname{Spec} A_i$ is contained in at least one $V \cap U_\alpha$, so that $\Gamma(V_i, \mathcal{P})$ is a projective A_i -module. Let $A' = A_1 \times \dots \times A_n$; then $A \rightarrow A'$ is faithfully

⁽¹¹⁶⁾**N.D.E.** Let us point out in passing that assertion II 2.5.2 of loc. cit., more general than II 2.5.1, is corrected in the article [Gr73] (this does not affect the case of faithfully flat morphisms).

⁽¹¹⁷⁾**N.D.E.** As mentioned in N.D.E. (112), the statement has been rewritten for an $\mathcal{O}_S$ -coalgebra $\mathcal{C}$ (rather than for an S -group G satisfying the hypotheses indicated in Corollary 11.10). Moreover, the proof has been expanded (the original stated: “($\dots$) the proposition is a consequence of Lemma 11.8.”).

*This is the case, for example, when $\mathcal{C} = \mathcal{A}(G)$, where G is a reductive S -group, as we shall see in Exp. XXII 5.7.8.

⁽¹¹⁸⁾**N.D.E.** This definition, taken from [RG71], bottom of p. 82, has been added. Thus, in Proposition 11.9, the hypothesis is that the coalgebra $\mathcal{C}$ be a locally projective $\mathcal{O}_S$ -module, and this terminology has been used in Corollary 11.10.

flat and $\Gamma(V, \mathcal{P}) \otimes_A A'$ is a projective A' -module. Thus, by 11.8.2, $\Gamma(V, \mathcal{P})$ is a projective A -module.

When these equivalent conditions hold, one says that $\mathcal{P}$ is a *locally projective $\mathcal{O}_S$ -module*.

Corollary 11.10. *Let S be a quasi-compact and quasi-separated scheme, G an S -group, and ρ a linear operation of G on a quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$. Suppose that:*

- (i) G satisfies one of the following conditions:
 - a) G is affine and flat over S ,
 - b) G is flat, quasi-compact and quasi-separated over S , and $\mathcal{E}$ is flat;
- (ii) $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$ -module.

Then $\mathcal{E}$ is the inductive limit of an increasing filtered family of quasi-coherent sub- $\mathcal{O}_S$ -modules of finite type of $\mathcal{E}$, stable under G .

By hypothesis (i) and 11.6.1, $\mathcal{E}$ is endowed with an $\mathcal{A}(G)$ -comodule structure. On the other hand, since S is quasi-compact and quasi-separated, $\mathcal{E}$ is the inductive limit of its quasi-coherent submodules of finite type (EGA I, 9.4.9 and EGA IV₁, 1.7.7). Consequently, the corollary follows from Proposition 11.9, applied to the coalgebra $\mathcal{A}(G)$.

Furthermore, one has the following proposition:

Proposition 11.10.bis. ⁽¹¹⁹⁾ *Let S be a noetherian scheme, G an S -group flat, quasi-compact and quasi-separated over S , $\mathcal{E}$ a quasi-coherent G - $\mathcal{O}_S$ -module, $\mathcal{M}$ a coherent sub- $\mathcal{O}_S$ -module of $\mathcal{E}$. Then $\mathcal{M}$ is contained in a coherent sub- $\mathcal{O}_S$ -module stable under G .*

PROOF. Denote by f the morphism $G \rightarrow S$ and by τ the adjunction morphism $\mathcal{E} \rightarrow f_*f^*(\mathcal{E})$. By 11.6, the G - $\mathcal{O}_S$ -module structure on $\mathcal{E}$ is given by an automorphism θ of the $\mathcal{O}_G$ -module $f^*(\mathcal{E})$ such that the morphism $\delta = \theta \circ \tau$ satisfies conditions (1) and (2) of 11.6. (The situation considered in [Th87] is more general, in that the author considers a G -scheme X and a G -equivariant $\mathcal{O}_X$ -module $\mathcal{E}$ (cf. Exp. I, Section 6); here $X = S$ endowed with the trivial action of G .)

Since S is noetherian, $\mathcal{E}$ is the filtered inductive limit of its coherent submodules $\mathcal{F}_\alpha$ (cf. EGA I, 9.4.9). Then $f^*(\mathcal{E})$ is the filtered inductive limit of the $f^*(\mathcal{F}_\alpha)$, which are submodules of $f^*(\mathcal{E})$ since f is flat. Since moreover f is quasi-compact and quasi-separated, then, by 11.0, the $\mathcal{O}_S$ -module $f_*f^*(\mathcal{E})$ is quasi-coherent and is the filtered inductive limit of the quasi-coherent submodules $f_*f^*(\mathcal{F}_\alpha)$. Consequently, $\mathcal{E}$ is the filtered inductive limit of the quasi-coherent sub- $\mathcal{O}_S$ -modules $\mathcal{E}_\alpha$, where $\mathcal{E}_\alpha$ denotes the inverse image under δ of $f_*f^*(\mathcal{F}_\alpha)$, i.e. the kernel of the composite morphism

$$\bar{\delta}_\alpha: \mathcal{E} \xrightarrow{\delta} f_*f^*(\mathcal{E}) \rightarrow f_*f^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Since $\mathcal{M}$ is coherent and S noetherian, every increasing sequence of submodules of $\mathcal{M}$ is stationary, so $\mathcal{M}$ is contained in some $\mathcal{E}_\alpha$. Let us show that each $\mathcal{E}_\alpha$ is coherent and G -stable.

Let u be the morphism $f^*(\mathcal{E}) \rightarrow \varepsilon_*(\mathcal{E})$ corresponding by adjunction to the identity morphism from $\mathcal{E}$ to $f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$; then the composite morphism

$$\mathcal{E} \xrightarrow{\tau} f_*f^*(\mathcal{E}) \xrightarrow{f_*(u)} f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$$

is the identity (cf. 11.6 (2)). Since one has a commutative diagram:

$$\begin{array}{ccccc} \mathcal{E}_\alpha & \xrightarrow{\delta} & f_*f^*(\mathcal{F}_\alpha) & \xrightarrow{f_*(u)} & \mathcal{F}_\alpha \\ i_\alpha \downarrow & & \downarrow & & \downarrow j_\alpha \\ \mathcal{E} & \xrightarrow{\tau} & f_*f^*(\mathcal{E}) & \xrightarrow{f_*(u)} & \mathcal{E}, \end{array}$$

⁽¹¹⁹⁾**N.D.E.** This proposition has been added; it is a special case of [Th87], 1.4–1.5. There the author refers to an argument of Deligne (cf. [Kn71], III Th. 1.1); one may also note the similarity with Serre's argument ([Se68], Prop. 2) recalled in 11.8.bis.

where i_α (resp. j_α) denotes the inclusion of $\mathcal{E}_\alpha$ (resp. $\mathcal{F}_\alpha$) into $\mathcal{E}$, one deduces that i_α factors through j_α , i.e. $\mathcal{E}_\alpha$ is a submodule of $\mathcal{F}_\alpha$, hence coherent.

Finally let us show that $\mathcal{E}_\alpha$ is G -stable (cf. 11.7.bis). Denote by $h : G \rightarrow S$ a second copy of $f : G \rightarrow S$ and consider the following commutative diagram:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ p \downarrow & \searrow \phi & \downarrow h \\ G & \xrightarrow{f} & S. \end{array}$$

Then the exact sequence

$$0 \rightarrow \mathcal{E}_\alpha \rightarrow \mathcal{E} \xrightarrow{\bar{\delta}_\alpha} h_* h^*(\mathcal{E}/\mathcal{F}_\alpha)$$

gives, since f is flat, the exact sequence

$$(\dagger) \quad 0 \rightarrow f_* f^*(\mathcal{E}_\alpha) \rightarrow f_* f^*(\mathcal{E}) \xrightarrow{f_* f^*(\bar{\delta}_\alpha)} f_* f^* h_* h^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Moreover, since h is quasi-compact and quasi-separated, and f flat, the canonical morphism

$$f_* h_* h^*(\mathcal{E}/\mathcal{F}_\alpha) \rightarrow p_* q^* h^*(\mathcal{E}/\mathcal{F}_\alpha) = p_* \phi^*(\mathcal{E}/\mathcal{F}_\alpha)$$

is an isomorphism, so that the right-hand term in $(\dagger)$ is $\phi_* \phi^*(\mathcal{E}/\mathcal{F}_\alpha)$. Returning to the notation of 11.6 and denoting by π_α the projection $\mathcal{E} \rightarrow \mathcal{E}/\mathcal{F}_\alpha$, one therefore obtains the commutative diagram below, whose bottom row is exact:

$$\begin{array}{ccc} \mathcal{E}_\alpha & \xrightarrow{\delta} & f_* f^*(\mathcal{E}) \\ \delta \downarrow & & \downarrow f_*(f^*(\pi) \boxtimes \delta_G) \\ f_* f^*(\mathcal{E}_\alpha) & \longrightarrow & f_* f^*(\mathcal{E}) \xrightarrow[f_* f^*(\bar{\delta}_\alpha)]{\phi_* \phi^*} \phi_* \phi^*(\mathcal{E}/\mathcal{F}_\alpha) \end{array}$$

and since $f_*(f^*(\pi) \boxtimes \delta_G) \circ \delta$ vanishes on $\mathcal{E}_\alpha$, it follows that δ sends $\mathcal{E}_\alpha$ into $f_* f^*(\mathcal{E}_\alpha)$, i.e. $\mathcal{E}_\alpha$ is G -stable. The proposition is proved. $\square$

Remark 11.10.1. ⁽¹²⁰⁾ In 11.8, it is not sufficient to suppose that C is a flat A -module, even if A is a principal ideal ring. Indeed, one has the following counterexamples, which were pointed out to us (independently) by O. Gabber and J.-P. Serre.

(a) Let $(A, \mathfrak{m})$ be a discrete valuation ring, K its fraction field, and G the A -group “extension by zero” of the K -group $(\mathbf{Z}/2\mathbf{Z})_K$. Then the constant group $(\mathbf{Z}/2\mathbf{Z})_A$, and hence also its subgroup G , acts on the free A -module V with basis v_1, v_2 by interchanging v_1 and v_2 . Then Av_1 is not a sub- G -module of V , but it is the intersection of the sub- G -modules $Av_1 + \mathfrak{m}^n v_2$, for $n \in \mathbf{N}^*$. Thus there is no smallest sub- G -module of V containing v_1 .

(b) Let A be an integral ring distinct from its fraction field K , and let G be the affine flat A -group corresponding to the Hopf algebra

$$\mathcal{A}(G) = \{P \in K[T] \mid P(0) \in A\},$$

with comultiplication, resp. counit and antipode, defined by $\Delta(T) = T \otimes 1 + 1 \otimes T$, resp. $\varepsilon(T) = 0$ and $\tau(T) = -T$. (N.B. One therefore has $G \otimes_A K = \mathbf{G}_{a,K}$.)

Let V be the free A -module $Av_1 \oplus Av_2$ and u the endomorphism of V defined by $u(v_1) = v_2$, $u(v_2) = 0$, so that $u^2 = 0$. Then V is endowed with an $\mathcal{A}(G)$ -comodule structure, defined by

$$\mu(m) = 1 \otimes m + T \otimes u(m).$$

The sub- G -modules of V containing v_1 are precisely the sub- A -modules of the form $Av_1 \oplus Iv_2$, for I a nonzero ideal of A ; their intersection is Av_1 , which is not a sub- G -module. Thus there is no smallest sub- G -module of V containing v_1 . (Note moreover that $C = A \oplus K \cdot T$

⁽¹²⁰⁾**N.D.E.** The first part of the original Remark 11.10.1 has been incorporated into 11.8 and 11.9 (replacing “free” by “projective”); the counterexamples below correct the second part.

is a subcoalgebra of $\mathcal{A}(G)$, flat over A , and that the coaction $\mu : V \rightarrow V \otimes \mathcal{A}(G)$ factors through $V \otimes C$, so one also obtains a counterexample for the “very simple” coalgebra C .)

Finally, note that the two preceding examples are special cases of the following construction. Let A be an integral ring distinct from its fraction field K , let B be a Hopf A -algebra free over A . Denote by $\varepsilon : B \rightarrow A$ the augmentation of B and by $I = \text{Ker}(\varepsilon)$ the augmentation ideal. Since $B = A \cdot 1 \oplus I$, one easily sees that $B' = \{b \in B \otimes_A K \mid \varepsilon(b) \in A\}$ is a Hopf subalgebra of B_K . If V is a B -comodule, free with basis $(v_1, \dots, v_n)$ as an A -module, and if $\mu_V(v_1) \neq v_1 \otimes 1$, then Av_1 is not a subcomodule of V , but it is the intersection of the subcomodules $Av_1 + I \cdot V$, for I ranging over the nonzero ideals of A . Thus there is no smallest subcomodule of V containing v_1 .

Proposition 11.11. *Let G be an algebraic group over the field k . The following conditions are equivalent:*

- (i) G is affine.
- (ii) G is quasi-affine.
- (iii) G acts faithfully on a quasi-affine k -scheme X .
- (iv) G acts linearly and faithfully on a k -vector space (not necessarily finite-dimensional).
- (v) G is isomorphic to a closed subgroup of a group $\text{GL}(n)_k$.

PROOF. One has (i) $\Rightarrow$ (ii) trivially, and (ii) $\Rightarrow$ (iii), for G acts faithfully on itself by translations.

Suppose that G acts faithfully on the right on a quasi-affine k -scheme X .⁽¹²¹⁾ Since X is quasi-affine, it is separated and quasi-compact.⁽¹²²⁾ Similarly, G is separated (VIA 0.3) and quasi-compact (for it is of finite type over k). Thus, by 11.1 (c), one has canonical isomorphisms:

$$\mathcal{O}(X \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \quad \text{and} \quad \mathcal{O}(X \times G \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G).$$

One deduces that the morphism $\mu : \mathcal{O}(X) \rightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$ induced by the morphism $X \times G \rightarrow X$ endows $\mathcal{O}(X)$ with a right $\mathcal{O}(G)$ -comodule structure, i.e. G acts linearly on the left on the k -algebra $\mathcal{O}(X)$. Consequently, G also acts on the right on the affine envelope $X_{\text{af}} = \text{Spec } \mathcal{O}(X)$ of X , and the canonical morphism $X \rightarrow X_{\text{af}}$ is G -equivariant.

Moreover, since X is quasi-affine, $X \rightarrow X_{\text{af}}$ is an open immersion (EGA II, 5.1.2), so a fortiori G acts faithfully on X_{af} . One therefore obtains that the linear operation (on the left) of G on the k -algebra $\mathcal{O}(X) = \mathcal{O}(X_{\text{af}})$ is faithful. This proves the implication (iii) $\Rightarrow$ (iv).

Suppose now that G acts faithfully on a k -vector space V . Then, by 11.10, V is the inductive limit of finite-dimensional vector subspaces V_i stable under the action of G . If K_i is the kernel of the induced action of G on V_i , i.e. of the morphism $G \rightarrow \underline{\text{Aut}}(V_i)$, then K_i is a closed subscheme of G , and the hypothesis that G acts faithfully is expressed by the fact that the intersection of the K_i is the identity subgroup of G . Since G is noetherian, it follows that one of the K_i is already reduced to the identity group, hence that $G \rightarrow \underline{\text{Aut}}(V_i)$ is a monomorphism. It is therefore a closed immersion by 1.4.2, which proves that (iv) $\Rightarrow$ (v). Since (v) $\Rightarrow$ (i) trivially, this proves 11.11. $\square$

Remark 11.11.1. One can generalize 11.11 as follows. Let S be a locally noetherian regular scheme of dimension ≤ 1 , and G a group scheme flat, quasi-compact and quasi-separated over S . (In this case, $\mathcal{A}(G)$ is a torsion-free $\mathcal{O}_S$ -module, hence flat.)

- (a) One then has the equivalence of the following conditions:⁽¹²³⁾
 - (i) G is affine over S .
 - (ii) G is quasi-affine over S .

⁽¹²¹⁾**N.D.E.** The original has been expanded in what follows; moreover, G has been chosen to act on the right on X , in order to obtain a linear operation of G on the left on $\mathcal{O}(X)$.

⁽¹²²⁾**N.D.E.** Recall that a k -scheme X is said to be quasi-affine if it is isomorphic to a quasi-compact open subset of an affine k -scheme (EGA II, 5.1.1).

⁽¹²³⁾**N.D.E.** The equivalence of these conditions is proved in the additional section 12.

(iii) G acts faithfully on a quasi-affine flat S -scheme.

(iv) G acts linearly and faithfully on a quasi-coherent flat $\mathcal{O}_S$ -module.

(b) If moreover G is of finite type over S and S noetherian, these conditions imply that G is isomorphic to a closed subgroup of some $\underline{\text{Aut}}(\mathcal{E})$, where $\mathcal{E}$ is a locally free $\mathcal{O}_S$ -module of finite type.⁽¹²⁴⁾

Lemma 11.12. *Let k be a field, G an affine k -group. Put $A = \mathcal{A}(G)$. Given $x \in A$, there is a sub- k -algebra of finite type B of A such that $x \in B$, $\Delta(B) \subset B \otimes_k B$ and $u(B) \subset B$, where u denotes the involution of A corresponding to the inversion morphism of G .⁽¹²⁵⁾*

One may suppose $x \neq 0$; then $\Delta(x) \neq 0$ since $(\varepsilon \otimes \text{id})\Delta(x) = x$, where ε denotes the augmentation (counit) of A . Write

$$(1) \quad \Delta(x) = \sum_{j=1}^n e_j \otimes a_j \quad \text{with } n \text{ minimal,}$$

in which case the e_j (resp. a_j) are linearly independent. Complete $(e_1, \dots, e_n)$ to a basis $(e_i)_{i \in I}$ of A and put, for $j \in J = \{1, \dots, n\}$,

$$\Delta(e_j) = \sum_{i \in I} e_i \otimes b_{ij}.$$

Applying $\Delta \otimes \text{id}$ and $\text{id} \otimes \Delta$ to (1), one obtains from axiom (CO 1) of 11.8.0 (see also (HA 1) in I 4.2) the equalities:

$$\sum_{j \in J} e_j \otimes \Delta(a_j) = \sum_{\ell \in J} \Delta(e_\ell) \otimes a_\ell = \sum_{i \in I} e_i \otimes \left(\sum_{\ell \in J} b_{i\ell} \otimes a_\ell \right).$$

Since the e_i are linearly independent, it follows that

$$(2) \quad \forall j \in J, \quad \Delta(a_j) = \sum_{\ell \in J} b_{j\ell} \otimes a_\ell.$$

Let B then be the sub- k -algebra of finite type of A generated by the b_{ij} and the $u(b_{ij})$, for $i, j \in J$. It is clear that $u(B) = B$.

Applying $\Delta \otimes \text{id}$ and $\text{id} \otimes \Delta$ to (2), one again obtains from (CO 1) the equalities:

$$\sum_{\ell \in J} \Delta(b_{j\ell}) \otimes a_\ell = \sum_{i \in J} b_{ji} \otimes \Delta(a_i) = \sum_{i, \ell \in J} b_{ji} \otimes b_{i\ell} \otimes a_\ell,$$

and since the a_ℓ are linearly independent, one deduces that

$$(3) \quad \forall j, \ell \in J, \quad \Delta(b_{j\ell}) = \sum_{i \in J} b_{ji} \otimes b_{i\ell}.$$

Since $\Delta \circ u = (u \times u) \circ v \circ \Delta$, where $v(a \otimes b) = b \otimes a$, one therefore also has

$$(4) \quad \forall j, \ell \in J, \quad \Delta(u(b_{j\ell})) = \sum_{i \in J} u(b_{i\ell}) \otimes u(b_{ji}).$$

Since Δ is an algebra homomorphism, one deduces from (3) and (4) that $\Delta(B) \subset B \otimes_k B$. Finally, axiom (CO 2) of 11.8.0 (see also (HA 2) in I, 4.2) shows that $a_j = \sum_{i \in I} \varepsilon(a_j) b_{ij}$ and $x = \sum_{j \in J} \varepsilon(e_j) a_j$, so that $x \in B$.

Proposition 11.13. *Let k be a field and G an affine k -group with algebra A . Then G is the projective limit of an increasing filtered system of affine k -groups of finite type, whose transition morphisms are faithfully flat.*

⁽¹²⁴⁾**N.D.E.** This is proved, with various generalizations, in the additional section 13.

⁽¹²⁵⁾**N.D.E.** On the one hand, this is generalized in section 13 to the case where G is affine and flat over a regular base of dimension ≤ 2 . On the other hand, in what follows the original has been expanded and corrected.

If B and B' are two subalgebras of A of finite type stable under Δ and u , then so is the subalgebra generated by B and B' . Thus, by Lemma 11.12, A is the inductive limit of an increasing filtered family $(B_i)_{i \in I}$ of subalgebras of finite type stable under Δ and u . Then each B_i , endowed with the restriction of u and the morphism $B_i \rightarrow B_i \otimes_k B_i$ obtained from Δ , is a Hopf algebra, so by I 4.2 it is the algebra of an affine k -group G_i , of finite type over k . Finally, since $A = \varinjlim B_i$, one has $G = \varprojlim G_i$ (cf. EGA IV₃ 8.2.3). The transition morphisms are faithfully flat by the following lemma:

Lemma 11.14. *Let k be a field, $u : G \rightarrow H$ a morphism between affine k -groups of finite type, and $u^\flat : B \rightarrow A^{(126)}$ the corresponding morphism of k -algebras. For u to be faithfully flat, it is necessary and sufficient that $u^\flat$ be injective.*

The condition is obviously necessary (cf. EGA 0_I 6.6.1). Let us show that it is sufficient. Put $N = \text{Ker } u$. Then, by VIA, 3.3.2 and 5.4.1, G/N is a k -group of finite type and u factors as $G \xrightarrow{p} G/N \xrightarrow{v} H$, where p is faithfully flat and v is a closed immersion. Thus, since H is an affine scheme, G/N is an affine scheme and the morphism $v^\flat : B \rightarrow \mathcal{O}(G/N)$ is surjective (cf. EGA I 4.2.3). Now, since $u^\flat$ is supposed injective and $u^\flat = p^\flat \circ v^\flat$, then $v^\flat$ is also injective: it is therefore an isomorphism, as is v , and since p is faithfully flat, so is u .

Definition 11.15. ⁽¹²⁷⁾ Let k be a field, G a quasi-compact k -group and V a k -vector space endowed with a k -linear action of G , hence with an $\mathcal{A}(G)$ -comodule structure $\delta : V \rightarrow V \otimes \mathcal{A}(G)$, by 11.6.1. Let $v \in V$ be nonzero. The following conditions are equivalent:

- (i) There is $\lambda \in \mathcal{A}(G)$ (necessarily unique) such that $\delta(v) = v \otimes \lambda$.
- (ii) For every k -algebra R and every $g \in G(R)$, one has $g \cdot v \in Rv$ (i.e., there is $f \in R$, necessarily unique, such that one has in $V \otimes R$ the equality $g \cdot v = v \otimes f$).

Indeed, it is clear that (i) $\Rightarrow$ (ii). Conversely, if (ii) holds and one applies it to $R = \mathcal{A}(G)$ and $g = \text{id}_{\mathcal{A}(G)}$, one obtains that there is a unique $\lambda \in \mathcal{A}(G)$ such that $\delta(v) = v \otimes \lambda$.

If v satisfies these conditions, one says that v is a *semi-invariant vector under G* , and λ is the *weight of v* ; one will also say that “ v is a semi-invariant of weight λ ”.

Denote by Δ the comultiplication of $\mathcal{A}(G)$; then the equality

$$v \otimes \lambda \otimes \lambda = (\delta \otimes \text{id})(\delta(v)) = (\text{id} \otimes \Delta)(\delta(v)) = v \otimes \Delta(\lambda)$$

implies that $\Delta(\lambda) = \lambda \otimes \lambda$. Consequently, λ defines a morphism of Hopf algebras

$$\mathcal{A}(\mathbf{G}_{m,k}) = k[T, T^{-1}] \rightarrow \mathcal{A}(G), \quad T \mapsto \lambda,$$

and hence a morphism of k -groups $\lambda : G \rightarrow \mathbf{G}_{m,k}$, i.e. λ is a character of G , called the *character associated to the semi-invariant vector v* .

Lemma 11.16.0. ⁽¹²⁸⁾ *Let k be a field, H an affine k -group, V an H -module of dimension n and U a vector subspace of V of dimension d . Consider the line $D = \bigwedge^d U \subset \bigwedge^d V$. For U to be stable under H , it is necessary and sufficient that D be so.*

Since necessity is clear, let us prove sufficiency. One may suppose $d < n$. Let $(e_1, \dots, e_d)$ be a basis of U ; complete it to a basis $(e_1, \dots, e_n)$ of V . For every k -algebra R , $V_R = V \otimes R$ is a free R -module and one has

$$U_R = \{v \in V_R \mid v \wedge (e_1 \wedge \dots \wedge e_d) = 0\}$$

(for, for $i > d$, the $e_i \wedge e_1 \wedge \dots \wedge e_d$ are linearly independent in $\bigwedge_R^{d+1} V_R$). Since $H(R)$ acts on $\bigwedge_R^\bullet(V_R)$ by

$$h(x_1 \wedge \dots \wedge x_s) = h(x_1) \wedge \dots \wedge h(x_s),$$

⁽¹²⁶⁾**N.D.E.** The morphism $B \rightarrow A$ corresponding to $u : \text{Spec}(A) \rightarrow \text{Spec}(B)$ has been denoted by $u^\flat$ (instead of u°).

⁽¹²⁷⁾**N.D.E.** The hypothesis that G be quasi-compact has been added, and the equivalence of conditions (i) and (ii) expanded.

⁽¹²⁸⁾**N.D.E.** This lemma, taken from [DG70], § II.2, 3.5, has been added.

it follows that, if $H(R)$ stabilizes $D_R = Re_1 \wedge \cdots \wedge e_d$, it also stabilizes U_R .

Theorem 11.16. (Chevalley). *Let k be a field, G an affine algebraic k -group, H a closed subgroup scheme of G .⁽¹²⁹⁾ Then there is a finite-dimensional G -module V and a line D of V such that $H = \underline{\text{Norm}}_G(D)$, i.e. such that for every k -algebra R ,*

$$H(R) = \{g \in G(R) \mid g(D_R) = D_R\}.$$

In other words, there are finitely many elements $a_1, \dots, a_n \in \mathcal{A}(G)$ which are semi-invariants, all of the same weight λ , for the “right” action of H (i.e. $a_i(gh) = \lambda(h)a_i(g)$, for every k -algebra R and $g \in G(R)$, $h \in H(R)$), such that H is the largest closed subgroup scheme of G under which the a_i are semi-invariants.

Denote by Δ (resp. ε) the comultiplication (resp. augmentation) of $A = \mathcal{A}(G)$. Then $H = \text{Spec}(A/I)$, for some ideal I of A contained in $\text{Ker } \varepsilon$ and such that $\Delta(I) \subset I \otimes A + A \otimes I$. Let $B = A/I$ and π be the projection $A \rightarrow B$. Consider the left action of H on A given by $(h\varphi)(g) = \varphi(gh)$; the corresponding B -comodule structure is given by:

$$\bar{\Delta}: A \xrightarrow{\Delta} A \otimes A \xrightarrow{\text{id}_A \otimes \pi} A \otimes B.$$

Then I is a sub- H -module of A , since $\bar{\Delta}(I) \subset I \otimes B$.

On the other hand, A is a k -algebra of finite type, hence noetherian, so I has a finite set of generators $(x_1, \dots, x_r)$. By 11.8, the x_i are contained in a sub- G -module V finite-dimensional over k . Then $W = V \cap I$ is a finite-dimensional H -module, whose dimension we shall denote by d . Since V contains all the x_i , W generates the ideal I .

Put $E = \wedge^d V$, let $(w_1, \dots, w_d)$ be a basis of W , and let $\mathcal{B} = (e_0, \dots, e_n)$ be a basis of E containing the vector $e_0 = w_1 \wedge \cdots \wedge w_d$. The operation of G on V canonically determines an operation of G on E , hence an A -comodule structure $\rho: E \rightarrow E \otimes A$. For $j = 0, \dots, n$, put $\rho(e_j) = \sum_{i=1}^n e_i \otimes b_{ij}$; we saw in the proof of 11.12 that then

$$(*) \quad \Delta(b_{ij}) = \sum_{\ell=0}^n b_{i\ell} \otimes b_{\ell j}.$$

Put $a_i = b_{i0}$, i.e. if $(e_0^*, \dots, e_n^*)$ is the dual basis of $\mathcal{B}$, the a_i are the “matrix coefficients” $g \mapsto e_i^*(ge_0)$. On the other hand, the operation of H on E corresponds to:

$$\bar{\rho}: E \xrightarrow{\rho} E \otimes A \xrightarrow{\text{id}_E \otimes \pi} E \otimes B.$$

Since W is stable under H , then e_0 is semi-invariant under H , so $\bar{\rho}(e_0) = e_0 \otimes \pi(a_0)$ and $a_i = b_{i0}$ belongs to I for $i = 1, \dots, n$. Substituting this into (*), one obtains $\bar{\Delta}(a_i) = a_i \otimes \pi(a_0)$ for $i = 0, \dots, n$, i.e. the a_i are semi-invariants under H of weight $\pi(a_0)$. (Moreover, replacing E if necessary by the sub- G -module generated by e_0 (cf. 11.8), one may suppose that the a_i are linearly independent.)

Conversely, let $H' = \text{Spec } B'$, where $B' = A/I'$, be a closed subgroup scheme of G under which each a_i is semi-invariant, of weight $\lambda_i \in B'$ (this is the case, in particular, if e_0 is invariant under H' of weight λ'). Let us show that $H' = H$. Denote by π' the projection $A \rightarrow B'$; the hypothesis implies that

$$a_i \otimes \lambda_i = (\text{id}_A \otimes \pi')\Delta(b_{i0}) = \sum_{\ell=0}^n b_{i\ell} \otimes \pi'(a_\ell),$$

whence $\lambda_i = \pi'(a_0)$ and $a_\ell \in I'$ for $\ell = 1, \dots, n$, and therefore e_0 is semi-invariant under H' . By Lemma 11.16.0, this implies that W is stable under H' . Since the ideal I is generated by W , it is therefore also stable under H' , and hence

$$\Delta(I) \subset I \otimes A + A \otimes I'.$$

⁽¹²⁹⁾**N.D.E.** On the one hand, recall that every subgroup scheme of G is closed (1.4.2). On the other hand, the result has been stated in the usual form: “ H is the stabilizer of a line in a representation of G ”, while retaining the original formulation in terms of a sequence $a_1, \dots, a_n$ of semi-invariants in $\mathcal{A}(G)$.

Since $I \subset \text{Ker } \varepsilon$ and $(\varepsilon \otimes \text{id}_A) \circ \Delta = \text{id}_A$, it follows that $I \subset I'$, whence $H' \subset H$.

Lemma 11.17.0. ⁽¹³⁰⁾ *Let k be an algebraically closed field, G a reduced affine algebraic k -group, V a finite-dimensional G -module over k , and $v \in V$. Let E be the vector subspace of V generated by the vectors gv , for $g \in G(k)$. Then E is the smallest sub- G -module of V containing v , and thus the morphism $G \times E \rightarrow V$, $(g, x) \mapsto gx$ factors through E .*

PROOF. By 11.8, one knows that there is a smallest sub- G -module U of V containing v : if $\mu : V \rightarrow V \otimes \mathcal{A}(G)$ denotes the comodule structure and one writes $\mu(v) = \sum_{i=1}^n v_i \otimes a_i$ with the a_i linearly independent, one has $U = \text{Vect}(v_1, \dots, v_n)$. It is clear that U contains E , and the morphism $G \times U \rightarrow V$ factors through U .

Conversely, the inverse image of E under the morphism $\mu_v : g \mapsto gv$ is a closed subset of G which contains the rational points; now these are dense in G , since G is of finite type over k (cf. EGA IV₃, 10.4.8), so $\mu_v^{-1}(E) = G$ and thus, since G is reduced, μ_v factors through E , whence $E = U$. $\square$

Theorem 11.17. (Chevalley). *Let k be a field, G an affine k -group (not necessarily of finite type), and N a closed subgroup scheme of G invariant in G ; then the (fpqc) quotient sheaf G/N is representable by an affine k -group.* ⁽¹³¹⁾

Suppose first G of finite type. By VIA 3.2 and 5.2, the (fpqc) quotient sheaf G/N is representable by a k -group Q ; it is therefore a matter of showing that Q is affine. The proof proceeds in several steps; suppose first k algebraically closed. ⁽¹³²⁾

(a) Suppose moreover G reduced and connected and N reduced. By 11.16, there is a G -module V finite-dimensional over k , and a line $D = ke_0$ such that $\text{Norm}_G(D) = N$; in particular N acts on D through a character $\chi : N \rightarrow \mathbf{G}_{m,k}$.

Fix $h \in N(k)$. For every $g \in G(k)$, one has $hge_0 = g(g^{-1}hg)e_0 = \chi(g^{-1}hg)ge_0$, so $\chi(g^{-1}hg)$ is an eigenvalue of h . Thus the continuous map $\phi : G(k) \rightarrow k$, $g \mapsto \chi(g^{-1}hg)$, takes only finitely many values, and since $G(k)$ is irreducible (for it is dense in G), one therefore has $\phi(g) = \phi(e) = \chi(h)$ for every $g \in G(k)$ and hence

$$\chi(g^{-1}hg) = \chi(h), \quad \forall g \in G(k), h \in N(k).$$

Let E be the vector subspace of V generated by the vectors ge_0 , for $g \in G(k)$; by Lemma 11.17.0, it is the sub- G -module of V generated by e_0 .

By the above, the two morphisms $N \times E \rightarrow E$, $(h, x) \mapsto hx$ and $(h, x) \mapsto \chi(h)x$, coincide on the set of rational points $(N \times E)(k) = N(k) \times E(k)$, which is dense in $N \times E$. Since $N \times E$ is reduced (and E separated), these two morphisms are therefore equal, so N acts on E by homotheties. Consequently, N is contained in the kernel K of the morphism $\rho : G \rightarrow \text{GL}(\text{End}_k(E))$, defined by $\rho(g)(u) = gug^{-1}$, for every $g \in G(R)$ and $u \in \text{End}_R(E \otimes R)$ (R a k -algebra). On the other hand, if $g \in K(R)$ then $g(Re_0) = Re_0$, whence $g \in N(R)$. This shows that $N = K$. Then, by VIA 5.4.1, the morphism $G/N \rightarrow \text{GL}(\text{End}_k(E))$ is a closed immersion, and thus G/N is affine.

(b) Suppose now G and N reduced, with G not necessarily connected. Put $N' = N \cap G^0$; then G^0/N' is affine by (a). On the other hand, NG^0 is an invariant subgroup of G and G/NG^0 , being a quotient of the finite constant group G/G^0 (cf. VIA, 5.5.1), is likewise a finite constant group. Thus G/N is the direct sum of the fibers of the morphism $G/N \rightarrow G/NG^0$, all isomorphic to NG^0/N , hence to G^0/N' , by VIA, 5.3.3. Thus G/N is affine.

⁽¹³⁰⁾**N.D.E.** This lemma, taken from the proof of thm. 5.6 of [DG70], § III.3, has been added (by abuse of notation, the same letter E denotes a k -vector space and the k -scheme in modules $\mathbf{W}(E) = \text{Spec } S(E^*)$).

⁽¹³¹⁾**N.D.E.** For another proof of this theorem, which does not use the results of VIA, see [Ta72], Th. 5.2 (see also Remark 11.18.5).

⁽¹³²⁾**N.D.E.** In what follows the original has been expanded (and the erroneous assertion $(G/N)_{\text{red}} = G_{\text{red}}/N_{\text{red}}$ corrected), following [DG70], § III.3, 5.6.

(c) Suppose G reduced, and N arbitrary. The morphism $G \times N \rightarrow N$, $(g, h) \mapsto ghg^{-1}$, induces a morphism $(G \times N)_{\text{red}} \rightarrow N_{\text{red}}$; now, since G is reduced and k algebraically closed, one has $(G \times N)_{\text{red}} = G \times N_{\text{red}}$, so N_{red} is an invariant subgroup of G . (N.B. This fails when G is not reduced, cf. VIA, 0.2.)

Thus, by (a), $G' = G/N_{\text{red}}$ is affine. On the other hand, by VIA 5.6.1, $N' = N/N_{\text{red}}$ is a finite k -group, so, by Theorem 4.1 of Exp. V, the quotient $G/N = G'/N'$ is affine.

(d) For arbitrary G and N , the equivalence relation obtained from $G \times N \rightrightarrows G$ by the base change $G_{\text{red}} \rightarrow G$ is:

$$G_{\text{red}} \times N' \rightrightarrows G_{\text{red}}, \quad \text{where} \quad N' = N \cap G_{\text{red}}.$$

Since the underlying spaces are the same (and since the quotient is the quotient ringed space), the morphism $G_{\text{red}}/N' \rightarrow G/N$ is a homeomorphism. Since G_{red}/N' is reduced (for $p : G_{\text{red}} \rightarrow G_{\text{red}}/N'$ is faithfully flat), it follows that $(G/N)_{\text{red}}$ is identified with G_{red}/N' , which is affine, by (c). Since G/N is of finite type over k (cf. VIA, 3.3.2), this implies, by EGA I, 5.1.10, that G/N is affine.

Finally, for arbitrary k , let $\bar{k}$ be an algebraic closure of k . Then, by 9.2 (v), $(G \otimes_k \bar{k})/(N \otimes_k \bar{k})$ is isomorphic to $(G/N) \otimes_k \bar{k}$, so, since the former is affine, so is the latter, and thus G/N is also affine by (fpqc) descent (cf. EGA IV₂, 2.7.1). This proves 11.17 when G is of finite type. To extend this to the general case, we shall need the following lemma.⁽¹³³⁾

Lemma 11.17.1. *Let $(C_i \rightarrow A_i)_{i \in I}$ be a filtered inductive system of ring morphisms, all faithfully flat. Then $A = \varinjlim A_i$ is faithfully flat over $C = \varinjlim C_i$.*

PROOF. By [BAC] § I.3, Prop. 9, for a ring morphism $B \rightarrow B'$ to be faithfully flat, it is necessary and sufficient that it be injective and B'/B be a flat B -module. Since each $C_i \rightarrow A_i$ is faithfully flat, one therefore has exact sequences

$$0 \rightarrow C_i \rightarrow A_i \rightarrow A_i/C_i \rightarrow 0$$

and A_i/C_i is a flat C_i -module, so $(A_i/C_i) \otimes_{C_i} C$ is a flat C -module. Since inductive limits are exact and commute with tensor products, one obtains an exact sequence

$$0 \rightarrow C \rightarrow A \rightarrow A/C \rightarrow 0$$

as well as an isomorphism

$$\varinjlim ((A_i/C_i) \otimes_{C_i} C) = (A/C) \otimes_C C = A/C,$$

from which one deduces that A/C is a flat C -module. Thus A is faithfully flat over C . $\square$

Let us now return to the proof of 11.17 in the general case, i.e. when G is not supposed of finite type. Put $\mathcal{A}(G) = A$ and $\mathcal{A}(N) = B = A/J$. By 11.13, A is the inductive limit of an increasing filtered family $(A_i)_{i \in I}$ of Hopf subalgebras of finite type, so G is the projective limit of the affine algebraic k -groups $G_i = \text{Spec}(A_i)$. Denote by Δ (resp. τ) the comultiplication (resp. antipode) of A , and $\Delta^2 = (\Delta \otimes \text{id}_A) \circ \Delta$.

For every i , $B_i = A_i/(J \cap A_i)$ is a quotient Hopf algebra of A_i , so $N_i = \text{Spec}(B_i)$ is a closed subgroup of G_i . Moreover, since N is invariant in G , the morphism $G \times N \rightarrow G$ defined by $(g, n) \mapsto gng^{-1}$ factors through N , and this is equivalent to saying that the pair (A, J) satisfies the following property:

$$(m_{13} \circ (\Delta \otimes \tau) \circ \Delta)(J) \subset A \otimes_k J$$

where m_{13} denotes the map $a_1 \otimes a_2 \otimes a_3 \mapsto a_1 a_3 \otimes a_2$. It follows that $(A_i, A_i \cap J)$ satisfies the analogous property, hence that N_i is invariant in G_i . On the other hand, one has $\varinjlim B_i = B$ and thus $\varinjlim N_i = N$.

By what we saw above, each (fpqc) quotient sheaf G_i/N_i is representable by an affine k -group $Q_i = \text{Spec}(C_i)$. Put $C = \varinjlim C_i$. One therefore has a filtered projective system of

⁽¹³³⁾**N.D.E.** This lemma, taken from [DG70], § III.3, 7.1, has been added.

affine k -groups Q_i ; its projective limit $Q = \varprojlim Q_i$ is the k -group $\mathrm{Spec}(C)$ (cf. EGA IV₃, 8.2.3). One then has an exact sequence of k -groups:

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q.$$

Let us show that Q represents the (fpqc) quotient sheaf of G by N ; for this, it suffices to verify that the morphism $G \longrightarrow Q$ is covering for the (fpqc) topology (cf. IV, 3.3.2.1 and 5.1.7.1). Now each morphism $G_i \longrightarrow Q_i$ is faithfully flat (cf. 9.2 (xi)), in other words A_i is faithfully flat over C_i ; since $A = \varprojlim A_i$ and $C = \varprojlim C_i$, Lemma 11.17.1 implies that A is faithfully flat over C , so that $G \longrightarrow Q$ is a faithfully flat morphism. Since this morphism is affine, it is quasi-compact, hence covering for the (fpqc) topology. This completes the proof of Theorem 11.17.

11.18. — Complements.⁽¹³⁴⁾ Moreover, one deduces from 11.17 (and its proof) the following results, taken from [DG70], III § 3.7. Let k be a field. Let us begin with the following lemma (cf. [An73], 2.3.3.2), which will be useful later (cf. 12.10).

Lemma 11.18.1. *Let $u : G \longrightarrow G'$ be a morphism of k -groups, $N = \mathrm{Ker}(u)$. Suppose G' affine and G of finite type.*

- (i) *The morphism $\tau : G/N \longrightarrow G'$ is a closed immersion. In particular, G/N is affine.*
- (ii) *If moreover the morphism $u^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G)$ is injective, then τ is an isomorphism. (And thus G' is of finite type and u faithfully flat.)*

Indeed, one knows (11.13) that G' is the projective limit of a filtered system of affine algebraic k -groups G'_i . Denote by u_i the composite morphism $G \longrightarrow G' \longrightarrow G'_i$ and by N_i its kernel. Then the N_i form a decreasing filtered system of closed subgroups of G , whose intersection is N . Since G is noetherian, there is an index i such that $N = N_i$. Since G and G_i are of finite type, then, by VIA, 3.2 and 5.4.1, the quotient G/N is a k -group of finite type and u_i is the composite of the projection $p : G \longrightarrow G/N$, which is faithfully flat, and a closed immersion $\tau_i : G/N \hookrightarrow G_i$.

Consider then the following commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{u} & G' \\ p \downarrow & \nearrow \tau & \downarrow q_i \\ G/N & \xrightarrow{\tau_i} & G_i. \end{array}$$

Since $q_i \circ \tau = \tau_i$ is a closed immersion and q_i separated (G' being separated, cf. VIA, 0.3), then τ is a closed immersion (cf. EGA I, 5.4.4). It follows that G/N is affine, and the morphism $\tau^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G/N)$ is surjective, whence (i).

If moreover $u^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G)$ is injective, then so is $\tau^\sharp$, so $\tau^\sharp$ is an isomorphism, and hence τ too (since G' and G/N are affine). This proves (ii).

Theorem 11.18.2. *Let G and G' be two affine k -groups with algebras A and A' , and let $u : G \longrightarrow G'$ be a morphism of k -groups, $N = \mathrm{Ker}(u)$, and $\varphi : A' \longrightarrow A$ the morphism induced by u .*

- (i) *If φ is injective, then u is faithfully flat and identifies G' with G/N .*
- (ii) *One has $G/N = \mathrm{Spec}(B)$, where $B = \varphi(A')$, and thus u is the composite of the faithfully flat morphism $G \longrightarrow G/N$, corresponding to the inclusion $B \hookrightarrow A$, and the closed immersion $G/N \hookrightarrow G'$, corresponding to the surjection $A' \longrightarrow B$. Moreover, N is defined in G by the ideal AB^+ , where B^+ denotes the augmentation ideal of B .*
- (iii) *In particular, if u is a monomorphism, it is a closed immersion.*

PROOF. (i) Suppose φ injective and identify A' with a Hopf subalgebra of A . By 11.13, A is a filtered union of Hopf subalgebras $A_i = \mathcal{O}(G_i)$ of finite type over k ; denote by

⁽¹³⁴⁾**N.D.E.** This subsection has been added.

$G'_i = \text{Spec}(A'_i)$, where $A'_i = A' \cap A_i$, and by N_i the kernel of the morphism $G_i \rightarrow G'_i$ induced by the inclusion $A'_i \hookrightarrow A_i$. By the preceding lemma, one has $G_i/N_i \simeq G'_i$, and one therefore obtains, for every i , an exact sequence

$$1 \longrightarrow N_i \longrightarrow G_i \xrightarrow{p_i} G'_i \longrightarrow 1$$

where p_i is faithfully flat. Since $G = \varprojlim_i G_i$ and $G' = \varprojlim_i G'_i$, one therefore obtains an exact sequence

$$1 \longrightarrow N \longrightarrow G \xrightarrow{p} G/N$$

where we have put $N = \varprojlim_i N_i$. Moreover, by Lemma 11.17.1, p is faithfully flat (and affine), so G' represents the (fpqc) quotient sheaf of G by N . This proves (i).

In the general case, $B = \varphi(A')$ is a Hopf subalgebra of A ; denote by H the k -group $\text{Spec}(B)$ and by N' the kernel of the morphism $G \rightarrow H$ induced by the inclusion $B \hookrightarrow A$. By (i), H is identified with G/N' , and u is therefore the composite of the projection $G \rightarrow G/N'$ and the closed immersion $G/N' \hookrightarrow G'$ induced by the surjection $A' \rightarrow B$. It follows that $N' = N$. Moreover, by 9.2 (ii), one has a cartesian square:

$$\begin{array}{ccc} N & \longrightarrow & G \\ \downarrow & & \downarrow p \\ \text{Spec}(k) & \xrightarrow{\varepsilon} & G' \end{array}$$

where ε is the identity section of G' , which corresponds to the augmentation morphism $B \rightarrow k$. It follows that N is defined in G by the ideal AB^+ . This proves (ii), and (iii) follows from it. $\square$

Remark 11.18.3. Let G be an affine k -group and N an invariant k -subgroup. Since the morphism $p : G \rightarrow G/N$ is faithfully flat and quasi-compact, then, by IV 3.3.3.2, $\mathcal{O}(G/N)$ is the subalgebra of $\mathcal{O}(G)$ formed by functions φ which are right N -invariant, i.e. satisfy $\varphi(gh) = \varphi(g)$, for every k -scheme S and $g \in G(S)$, $h \in N(S)$. Denoting by J the ideal of $A = \mathcal{O}(G)$ defining N , this is equivalent to saying that $\Delta(\varphi) - \varphi \otimes 1 \in \mathcal{O}(G) \otimes J$, where Δ is the comultiplication of A .

The preceding theorem can then be reformulated in terms of Hopf algebras as follows.

Corollary 11.18.4. *Let k be a field, A a commutative Hopf k -algebra, $G = \text{Spec}(A)$.*

(i) If B is a Hopf subalgebra of A , then A is faithfully flat over B .

(ii) The map $N \mapsto \mathcal{O}(G/N)$ is a bijection between the set of invariant subgroups of G and that of Hopf subalgebras of A ; the inverse map is given by $B \mapsto \text{Spec}(A/AB^+)$. Moreover, if J is the ideal of A defining N , one has

$$\mathcal{O}(G/N) = \{x \in A \mid \Delta(x) - x \otimes 1 \in A \otimes J\}.$$

Remarks 11.18.5. (a) A consequence of the preceding theorem is that the category of commutative affine k -groups is abelian. For this, as well as other results on affine k -groups, we refer to [DG70], § III.3, 7.4 to 7.8.

(b) Finally, let us point out that M. Takeuchi gave another proof of results 11.17 to 11.18.4, cf. [Ta72], § 5; he moreover strengthened 11.18.4 (i) above by showing that A is even a projective B -module, cf. [Ta79], Th. 5 (see also [MW94], Th. 3.6).

12. Complements on G_{af} and “anti-affine” groups

⁽¹³⁵⁾ Let us begin with the following lemma, which extends 11.18.1 to the case where G is not supposed of finite type.⁽¹³⁶⁾

Lemma 12.1. *Let k be a field, $u : G \rightarrow H$ a monomorphism of k -groups, with H affine. Suppose u quasi-compact. Then u is a closed immersion.*

PROOF. By (fpqc) descent, one may suppose k algebraically closed. By VIA, 6.4, the closed image I of u is a closed subgroup scheme of H , hence still affine. Thus, replacing H by I , one may suppose u schematically dominant. Since k is algebraically closed, $H' = H_{\text{red}}$ is a subgroup scheme of H ; denote by $G' = G \times_H H'$; then the morphism $u' : G' \rightarrow H'$ obtained from u by base change is a quasi-compact monomorphism, and is dominant (the underlying continuous map being the same for u and u'). Thus, by VIA, 6.2, u' is faithfully flat; it is therefore a faithfully flat quasi-compact monomorphism, hence an isomorphism (cf. IV 1.14).

Thus $u : G \rightarrow H$ is a homeomorphism, hence affine by 2.9.1. Thus G is affine, and therefore u is a closed immersion by 11.18.2 (iii). $\square$

Theorem 12.2. *Let G be an algebraic k -group. Denote by ρ the canonical morphism $G \rightarrow G_{\text{af}}$ and by N its kernel.*

(i) *The canonical morphism $G/N \rightarrow G_{\text{af}}$ is an isomorphism, and thus G_{af} is an affine algebraic group, and ρ is faithfully flat.*

(ii) *One has a canonical isomorphism $(G/N)_{\text{af}} = G_{\text{af}}$.*

(iii) *N is a characteristic subgroup of G .*

(iv) *$\mathcal{O}(N) = k$.*

(v) *N is smooth, connected and commutative.*

PROOF. Point (i) is a special case of 11.18.1, and point (ii) follows from the universal properties of G_{af} and $(G/N)_{\text{af}}$.

Let us prove (iii). For an arbitrary k -scheme $\pi : S \rightarrow \text{Spec } k$, consider the cartesian square:

$$\begin{array}{ccc} G_S & \longrightarrow & G \\ q \downarrow & & \downarrow p \\ S & \xrightarrow{\pi} & \text{Spec } k, \end{array}$$

since p is quasi-compact and separated and π flat, then, by EGA III 1.4.15 and EGA IV₁ 1.7.21, one has $q_*(\mathcal{O}_{G_S}) = \pi^*(\mathcal{O}(G)) = \pi^*(\mathcal{O}(G_{\text{af}}))$, and thus, by EGA II, 1.5.2, one has $(G_S)_{\text{af}} = (G_{\text{af}})_S$, so N_S , being the kernel of the canonical morphism $G_S \rightarrow (G_S)_{\text{af}}$, is invariant under every automorphism of G_S , i.e. N is a characteristic subgroup of G .

To prove (iv), put $N' = \text{Ker}(N \rightarrow N_{\text{af}})$; by (ii), it is an invariant subgroup of G . Since N is algebraic (being a closed subgroup of G), then, by (i), $N/N' \cong N_{\text{af}}$; moreover, by VIA, 3.2 and 5.3.2, one has an isomorphism of k -groups

$$(G/N')/N_{\text{af}} \cong (G/N')/(N/N') \cong G/N.$$

Since N_{af} is affine, the projection $G/N' \rightarrow G/N$ is too, by 9.2 (vii), and since $G/N = G_{\text{af}}$ is affine, then G/N' is also. Thus, by the universal property of G_{af} , the projection $p' : G \rightarrow G/N'$ factors through $G_{\text{af}} = G/N$, whence $N \subset N'$ and therefore $N = N'$. Thus N_{af} is the trivial group, whence $\mathcal{O}(N) = k$.

Finally, assertion (v) is a consequence of the following lemma. $\square$

Lemma 12.3. *Let k be a field and N an algebraic k -group such that $\mathcal{O}(N) = k$. Then N is smooth, connected and commutative.*

⁽¹³⁵⁾**N.D.E.** This section has been added.

⁽¹³⁶⁾**N.D.E.** This lemma was communicated to us by M. Raynaud; it will be used in the proof of Proposition 12.9.

Indeed, one may suppose k algebraically closed. Then $H = N_{\text{red}}^0$ is a subgroup k -scheme of N , and the quotient k -scheme $X = N/H$ is finite (hence affine) over k , by VIA, 5.5.1 and 5.6.1. On the other hand, since $p : N \rightarrow X$ is faithfully flat, one has $\mathcal{O}(X) \subset \mathcal{O}(N) = k$. It follows that $N = N_{\text{red}}^0$, so N is smooth (VIA 1.3.1) and connected.

Let Z then be the center of N . By 6.2.6, N/Z is affine, and one obtains as above that $\mathcal{O}(N/Z) = k$, whence $N = Z$. This proves 12.3 and completes the proof of 12.2.

Let us also mention, without proof, the following theorem. (Recall that an abelian variety over a field k is a proper, smooth and connected group k -scheme.)

Theorem 12.4. (Chevalley). *Let k be a perfect field and G a smooth connected algebraic k -group. Then there is an affine smooth connected k -subgroup L , invariant in G , such that the quotient G/L is an abelian variety. Moreover, L is unique and its formation commutes with extension of the base field.*

Remarks 12.5. (1) This theorem was announced in 1953 by C. Chevalley, who published his proof in 1960 ([Ch60]). Meanwhile, other proofs had been obtained independently by I. Barsotti and M. Rosenlicht ([Ba55, Ro56]); see [Se99] for historical comments.

(2) A modern version (i.e. in the language of schemes) of Chevalley's proof was given by B. Conrad ([Co02]). (Note that in loc. cit., “algebraic group” means a smooth connected group k -scheme.)

(3) On the other hand, a modern version of Rosenlicht's proof was given by Ngô B.-C. in a course at Orsay in 2005–2006.

(4) If one omits the hypothesis that k be perfect, there still exists a smallest connected invariant affine subgroup L (not necessarily smooth) such that G/L is an abelian variety ([BLR], § 9.2, Thm. 1).

(5) One can also omit the hypothesis that G be smooth over k : indeed, by VIIA, 8.3, there is an integer $n \geq 1$ such that the quotient $G' = G/(\mathbb{F}_k^n G)$ is smooth; then G' contains a subgroup L' as in (4) above, and the inverse image of L' in G again has the same properties. Thus, for every connected algebraic group G over a field k , there is an exact sequence

$$1 \rightarrow H \rightarrow G \rightarrow A \rightarrow 1$$

where H is an affine k -group and A an abelian k -variety. Moreover, by [Per76], Cor. 4.2.9, one has such an exact sequence for every connected k -group G (not necessarily algebraic).

(6) Let k be an algebraically closed field and G the semidirect product of an elliptic curve E by the constant k -group $\{\pm 1\}_k$, for the action defined by $(-1) \cdot x = -x$; in this case, if L is a closed invariant subgroup of G such that G/L is connected, then $L = G$.

Remark 12.6. Following [Br09], one will say that a k -group N is *anti-affine* if $\mathcal{O}(N) = k$. By 12.3 and 12.4, if k is perfect, every anti-affine algebraic k -group is an extension of an abelian variety by an affine smooth connected commutative algebraic k -group. For the precise structure of anti-affine algebraic groups over a perfect field, and various consequences, see the recent articles by M. Brion and C. & F. Sancho de Salas ([Br09, SS09]).

To finish this section, we shall prove two results due to M. Raynaud, the first being Remark 11.11.1, the second Proposition 2.1 of Exp. XVII, Appendix III. We shall need the following lemma,⁽¹³⁷⁾ which improves (for a complete discrete valuation ring R) the flatness criteria given in [BAC], § III.5.

Lemma 12.7. *Let R be a discrete valuation ring, K its fraction field, π a uniformizer. Let A be a flat R -algebra and M an A -module flat over R . Suppose that:*

- (i) $M/\pi M$ is a flat module over $\bar{A} = A/\pi A$,
- (ii) $M \otimes_R K$ is a flat module over $A \otimes_R K$.

Then M is a flat A -module.

⁽¹³⁷⁾**N.D.E.** It is a version, improved by O. Gabber, of a statement communicated by M. Raynaud.

PROOF. By the flatness criterion in the nilpotent case (cf. [BAC], III § 5.2, Th. 1), $M/\pi^n M$ is a flat module over $A/\pi^n A$, for every $n \in \mathbf{N}^*$. It then follows from [RG71], II Lemma 1.4.2.1, that M is a flat A -module. For the reader's convenience, let us briefly indicate the proof. Put $s = \pi^n$ and $P = M/sM$. Since P is flat over A/sA and the latter has projective dimension 1 over A , the spectral sequence of composite functors implies that P has Tor-dimension ≤ 1 over A . Now, since M is R -flat and hence has no π -torsion, M_K/M is the inductive limit of the A -modules $\pi^{-n}M/M \simeq M/\pi^n M$, and thus M_K/M also has Tor-dimension ≤ 1 . Since one has the exact sequence $0 \rightarrow M \rightarrow M_K \rightarrow M_K/M \rightarrow 0$ and by hypothesis M_K is flat over A_K , hence over A , it follows that M is flat.

For completeness, let us also indicate the following simpler proof, pointed out by O. Gabber. Let I be an ideal of finite type of A ; we must show that the morphism $u : M \otimes_A I \rightarrow M$ is injective. By hypothesis (ii), $u \otimes_R K$ is injective, so $\text{Ker}(u)$ is an R -module of π -torsion. It therefore suffices to show that the π -torsion part of $M \otimes_A I$ is zero; now this is a quotient of $\text{Tor}_1^A(M, I/\pi I)$, as one sees by tensoring with M the exact sequence:

$$0 \rightarrow I \xrightarrow{\pi} I \rightarrow I/\pi I \rightarrow 0.$$

On the other hand, since M has no π -torsion (being flat over R), one obtains that

$$\text{Tor}_i^A(M, \overline{A}) = 0 \quad \text{for every } i \geq 1.$$

Consequently, if $(P_\bullet)$ is a projective resolution of the A -module M , then $(P_\bullet \otimes_A \overline{A})$ is a projective resolution of the $\overline{A}$ -module $\overline{M} = M/\pi M$, and thus for every $\overline{A}$ -module N , one has $\text{Tor}_i^A(M, N) = \text{Tor}_i^{\overline{A}}(\overline{M}, N)$, and this is zero for $i \geq 1$ since $\overline{M}$ is flat over $\overline{A}$. One therefore has $\text{Tor}_1^A(M, I/\pi I) = 0$, which proves the lemma. $\square$

Remark 12.8. Let S be a locally noetherian regular scheme of dimension 1, and X an S -scheme flat, quasi-separated and quasi-compact over S . Then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module. Indeed, one may suppose S local; denote by s its closed point, i the inclusion $X_s \hookrightarrow X$, and π a uniformizer of $R = \mathcal{O}(S)$; since X is flat over S , one has an exact sequence of sheaves

$$0 \rightarrow \mathcal{O}_X \xrightarrow{\pi} \mathcal{O}_X \rightarrow i_*(\mathcal{O}_{X_s}) \rightarrow 0$$

and thus, taking global sections, one obtains that $\mathcal{A}(X)$ is an R -module without π -torsion, hence flat.⁽¹³⁸⁾ One obtains moreover that the morphism from $\mathcal{A}((X_{\text{af}})_s) = \mathcal{A}(X)/\pi\mathcal{A}(X)$ to $\mathcal{A}(X_s)$, induced by the morphism $X \rightarrow X_{\text{af}}$, is injective.

We can now prove the following proposition (cf. Remark 11.11.1).

Proposition 12.9. *Let S be a locally noetherian regular scheme of dimension ≤ 1 , G a group S -scheme flat, quasi-separated and quasi-compact over S . Then the following conditions are equivalent:*

- (i) G is affine over S .
- (ii) G is quasi-affine over S .
- (iii) G acts faithfully on a quasi-affine flat S -scheme X .
- (iv) G acts linearly and faithfully on a quasi-coherent module $\mathcal{E}$ flat over S .
- (v) The morphism $\rho : G \rightarrow G_{\text{af}}$ is a monomorphism.

PROOF. The implication (i) $\Rightarrow$ (ii) is evident, as is (ii) $\Rightarrow$ (iii) (take $X = G$).

Suppose (iii) holds. Since $\mathcal{A}(G)$ and $\mathcal{A}(X)$ are flat $\mathcal{O}_S$ -modules, one obtains, proceeding as in 11.11, that G acts faithfully (on the right) on X_{af} , and thus acts linearly and faithfully (on the left) on the flat quasi-coherent $\mathcal{O}_S$ -module $\mathcal{A}(X)$.

On the other hand, if (iv) holds, then, by 11.6.1 (ii), the monomorphism $G \rightarrow \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{E})$ factors through G_{af} , so $G \rightarrow G_{\text{af}}$ is a monomorphism.

⁽¹³⁸⁾**N.D.E.** This holds more generally if S is locally noetherian regular of dimension ≤ 2 , cf. [Ray70a], VII 3.2.

Finally, suppose (v) holds and let us show that $\rho : G \rightarrow G_{\text{af}}$ is an isomorphism. Replacing S by one of its connected components, one may suppose S irreducible, with generic point η . Since the formation of G_{af} commutes with flat base changes, one has $(G_{\text{af}})_{\eta} = (G_{\eta})_{\text{af}}$, and hence the morphism $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$ is a monomorphism, hence a closed immersion, by 12.1, hence an isomorphism since $\mathcal{O}((G_{\eta})_{\text{af}}) = \mathcal{O}(G_{\eta})$. If $S = \text{Spec}(\kappa(\eta))$, we are done; one may therefore suppose $\dim S = 1$.

Let s then be a closed point of S ; let us show that $G_s \rightarrow (G_{\text{af}})_s$ is an isomorphism and that ρ is flat at all points of G_s . For this, one may suppose S local, with closed point s . The morphism $\rho_s : G_s \rightarrow (G_{\text{af}})_s$ obtained by base change is a monomorphism, hence a closed immersion by 12.1, so the morphism $\mathcal{O}((G_{\text{af}})_s) \rightarrow \mathcal{O}(G_s)$ induced by ρ_s is surjective. Now, by the preceding remark, it is also injective, hence an isomorphism. (In particular, ρ is therefore surjective.)

It then follows from Lemma 12.7 that $\rho : G \rightarrow G_{\text{af}}$ is faithfully flat. Since G is quasi-compact over S and G_{af} separated over S , then ρ is also quasi-compact (cf. EGA I, 6.6.4). Consequently, ρ is a faithfully flat quasi-compact monomorphism, hence an isomorphism. This proves the proposition. $\square$

Finally, let us prove Prop. 2.1 of Exp. XVII, Appendix III; taking account of [Per76], Cor. 4.2.5, “quasi-compact and quasi-separated” has been substituted for “of finite type” among the hypotheses (if one supposes G of finite type, one may use 12.1 instead of loc. cit.).

Proposition 12.10. *Let S be a locally noetherian regular scheme of dimension ≤ 1 , G a group S -scheme flat, quasi-compact and quasi-separated.*

(i) *The canonical morphism $\rho : G \rightarrow G_{\text{af}}$ is faithfully flat and quasi-compact. Consequently, G_{af} represents the (fpqc) quotient sheaf of G by $N = \text{Ker}(\rho)$.*

(ii) *If moreover G is of finite type over S , then ρ is of finite presentation and G_{af} represents the (fppf) quotient sheaf of G by N and is of finite type over S .*

(iii) *Suppose moreover G_{η} affine for every maximal point η of S . Then N is an étale S -group, and is the identity group if G is separated over S .*

PROOF. First, since G_{af} is affine and hence separated over S , then ρ is quasi-compact (cf. EGA I, 6.6.4) and the kernel $N = \text{Ker}(\rho)$ is a closed subgroup of G . Moreover, replacing S by one of its connected components, one may suppose S irreducible, with generic point η .

Note that to prove (i) and (ii), it suffices to show that ρ is faithfully flat; for then, by Exp. IV, 5.1.7.1, G_{af} represents the (fpqc) quotient sheaf of G by N , and if moreover G is of finite type over S , hence of finite presentation (S being locally noetherian), then, by 9.2 (xiii), ρ is of finite presentation (as is $G_{\text{af}} \rightarrow S$) and thus ρ is covering for the (fppf) topology.

One may therefore suppose $S = \text{Spec}(R)$, where R is a discrete valuation ring (if $\dim S = 1$) or the field $\kappa(\eta)$ (if $\dim S = 0$). Denote by s the closed point of S . Since the formation of G_{af} commutes with flat base changes, the canonical morphism $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$ is identified with the morphism $\rho_{\eta} : G_{\eta} \rightarrow (G_{\text{af}})_{\eta}$, and since $\mathcal{O}((G_{\text{af}})_{\eta}) = \mathcal{O}((G_{\eta})_{\text{af}}) = \mathcal{O}(G_{\eta})$, then ρ_{η} is faithfully flat by [Per76], 4.2.5 (see also the addition VIA, 6.6). If $\dim S = 1$, one similarly obtains that ρ_s is faithfully flat, since the morphism $\mathcal{O}((G_{\text{af}})_s) \rightarrow \mathcal{O}(G_s)$ is injective, by Remark 12.8. Thus, by Lemma 12.7, ρ is faithfully flat. This proves (i) and (ii). In particular, N is flat over S .

Suppose now G_{η} affine. Since ρ_{η} coincides with the canonical morphism $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$, its kernel N_{η} is the identity group. Let us show that N is étale over S . Since N is flat over S , it remains to see that N_s is étale over $\kappa(s)$, for every point $s \neq \eta$ of S . There is nothing to show if $S = \text{Spec}(\kappa(\eta))$, so one may suppose $S = \text{Spec}(R)$, where R is a discrete valuation ring. Let s be the closed point of S , K the fraction field of R , and π a uniformizer. Let $x \in N_s$ and U_x be an affine open neighborhood of x in N ; since N is flat over S , then $U_x \cap N_{\eta}$ is nonempty, hence equal to $\{\varepsilon(\eta)\}$, where ε denotes the identity section. Thus $A = \mathcal{O}(U_x)$ is a flat R -algebra such that $A_K = K$ and $\pi^{-1} \notin A$ (since π belongs to the maximal ideal of $\mathcal{O}_{U,x}$). It follows that $A = R$, and hence the projection $U_x \rightarrow S$ is an isomorphism. This

proves that N is étale over S ; if moreover G is separated over S , then the inverse isomorphism $S \rightarrow U_x$ equals the identity section (since they coincide on the dense open subset $\{\eta\}$ of $S = \operatorname{Spec}(R)$), so N is the identity group. The proposition is proved. $\square$

One obtains in particular the following corollary, two other proofs of which are found in [An73], Prop. 2.3.1 and [PY06], Prop. 3.1.

Corollary 12.10.1. *Let R be a discrete valuation ring, K its fraction field, G a group R -scheme separated, flat and of finite type over R . If G_K is affine, then G is affine.*

Remarks 12.10.2. (a) On the one hand, O. Gabber indicated to us examples where G is a flat group of finite type over a discrete valuation ring, whose generic fiber is an abelian variety, and where the kernel N of $G \rightarrow G_{\text{af}}$ is not smooth.

(b) On the other hand, let us point out that M. Raynaud gave an example, for S the affine space of dimension 2 over a field k , of a smooth quasi-affine group S -scheme, with affine connected fibers, which is not affine over S , cf. [Ray70a], § VII.3, p. 116.

13. Flat affine groups over a regular base of dimension ≤ 2

⁽¹³⁹⁾ Let us begin by noting that the well-known argument showing that every affine algebraic group over a field k is linear, as well as Lemma 11.12, extends to the case of a group scheme G affine, flat and of finite type over a noetherian regular base scheme S of dimension ≤ 2 . For S of dimension ≤ 1 , this proves point (b) of Remark 11.11.1. The extension to the case $\dim S = 2$ rests on the following lemma, which was communicated to us by O. Gabber.

Lemma 13.1. *Let S be a noetherian normal scheme, $\mathcal{A}$ a flat quasi-coherent $\mathcal{O}_S$ -module, $\mathcal{F}$ a sub- $\mathcal{O}_S$ -module of finite type of $\mathcal{A}$, $\mathcal{F}^{**}$ its bidual, and U the flatness locus of $\mathcal{F}$, i.e. the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$ -module.*

- (i) *U is an open subset of S and $\mathcal{F}^{**} = j_* j^*(\mathcal{F})$, where j denotes the inclusion $U \hookrightarrow S$.*
- (ii) *The canonical morphism $j_*(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathcal{A} \rightarrow j_*(\mathcal{E} \otimes_{\mathcal{O}_U} j^*(\mathcal{A}))$ is an isomorphism, for every quasi-coherent $\mathcal{O}_U$ -module $\mathcal{E}$.*
- (iii) *In particular, $\mathcal{V} = j_* j^*(\mathcal{F})$ is a submodule of $\mathcal{A} = j_* j^*(\mathcal{A})$, and the canonical morphism $\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A} \rightarrow j_* j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A})$ is an isomorphism.*

PROOF. Replacing S by one of its connected components, one may suppose S integral. By EGA IV₂, 2.1.12, the flatness locus of $\mathcal{F}$, i.e. the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$ -module, is an open subset U of S ; denote by j the inclusion $U \hookrightarrow S$. Since $\mathcal{A}_s$ is flat, hence torsion-free, so is $\mathcal{F}_s$, so U contains all points of codimension ≤ 1 . Consequently, by [BAC], VII, § 4.2, cor. of th. 1, one has $\mathcal{F}^{**} = j_* j^*(\mathcal{F})$, and one therefore obtains a monomorphism $\mathcal{F}^{**} \rightarrow j_* j^*(\mathcal{A})$.

The proof of (ii) is analogous to that of EGA III, 1.4.15, recalled in 11.0. On the other hand, since S is normal, the morphism $\mathcal{O}_S \rightarrow j_* j^*(\mathcal{O}_S)$ is an isomorphism (cf. EGA IV₂, 5.8.6 and 5.10.5). By (ii) applied to $\mathcal{E} = \mathcal{O}_U$, one therefore has $\mathcal{A} = j_* j^*(\mathcal{A})$. Finally, since $j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}) = j^*(\mathcal{F}) \otimes_{\mathcal{O}_U} j^*(\mathcal{A})$, the last assertion of (iii) follows from (ii) applied to $\mathcal{E} = j^*(\mathcal{F})$. The lemma is proved. $\square$

Furthermore, recall that an R -module of finite type M is said to be reflexive if the canonical morphism from M to its bidual M^{**} is an isomorphism. When R is a noetherian regular ring of dimension ≤ 2 , this implies that M is projective. Indeed, for every R -module of finite type N , consider a resolution $L_1 \rightarrow L_0 \rightarrow N \rightarrow 0$, where L_0 and L_1 are finite free R -modules; then one has an exact sequence

$$0 \rightarrow N^* \rightarrow L_0^* \rightarrow L_1^* \rightarrow Q \rightarrow 0,$$

⁽¹³⁹⁾**N.D.E.** This section has been added.

where Q denotes the cokernel of $L_0^* \rightarrow L_1^*$, and since R has homological dimension ≤ 2 (cf. [BAC], X § 4.2, cor. 1 of th. 1), it follows that N^* is projective.

Proposition 13.2. *Let S be a noetherian regular scheme of dimension ≤ 2 , G an affine flat S -group, $\mathcal{A}(G)$ its affine algebra.*

(i) *If G is of finite type over S , it is isomorphic to a closed subgroup of $H = \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{V})$, for some locally free $\mathcal{O}_S$ -module $\mathcal{V}$ of finite rank. If moreover S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d , whence $H = \text{GL}_{d,S}$.*

(ii) *$\mathcal{A}(G)$ is the filtered inductive limit of flat sub- $\mathcal{O}_S$ -Hopf algebras of finite type.*

PROOF. Let $\mathcal{B}$ be a sub- $\mathcal{O}_S$ -algebra of finite type of $\mathcal{A}(G)$. Since every coherent module on an open subset of S extends to a coherent module on S (cf. EGA I, 9.4.5), there is a coherent sub- $\mathcal{O}_S$ -module $\mathcal{M}$ of $\mathcal{B}$ which generates $\mathcal{B}$ as an $\mathcal{O}_S$ -algebra (loc. cit., 9.6.5). By 11.10.bis, $\mathcal{M}$ is contained in a coherent G -stable sub- $\mathcal{O}_S$ -module $\mathcal{F}$. (N.B. Since G is here affine over S , the proof in loc. cit. takes a simpler form: one can there replace $f_*f^*(\mathcal{E})$ by $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$, etc.)

Let j be the inclusion $U \hookrightarrow S$, where U denotes the flatness locus of $\mathcal{F}$. By Lemma 13.1 and the recollection following it, $\mathcal{V} = j_*j^*(\mathcal{F})$ is a locally free sub- $\mathcal{O}_S$ -module of $\mathcal{A}(G)$, and since the canonical morphism

$$\mathcal{V} \otimes \mathcal{A}(G) \rightarrow j_*j^*(\mathcal{F} \otimes \mathcal{A}(G))$$

is an isomorphism, then $\mathcal{V}$ is a sub- G -module of $\mathcal{A}(G)$. The action of G on $\mathcal{V}$ then induces a morphism of affine S -groups $\rho_{\mathcal{V}} : G \rightarrow H = \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{V})$, hence a morphism of Hopf $\mathcal{O}_S$ -algebras $\phi_{\mathcal{V}} : \mathcal{A}(H) \rightarrow \mathcal{A}(G)$. Denote by $\mathcal{A}_{\mathcal{V}}$ the image of $\phi_{\mathcal{V}}$; it is the affine algebra of a closed subgroup $G_{\mathcal{V}}$ of H , which is the closed image of $\rho_{\mathcal{V}}$. Let us show that $\mathcal{A}_{\mathcal{V}}$ contains $\mathcal{B}$.

Since the question is local on S , one may suppose $S = \text{Spec}(R)$ and $V = \Gamma(S, \mathcal{V})$ a free R -module with basis $v_1, \dots, v_n$; in this case $H \simeq \text{GL}_{n,R}$ and $\mathcal{A}(H)$ is generated as an R -algebra by the “matrix coefficients” c_{ij} and the element d^{-1} , where d denotes the determinant. Let Δ (resp. ε) be the comultiplication (resp. augmentation) of A . For $j = 1, \dots, n$, write $\Delta(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}$; then $a_{ij} = \phi(c_{ij})$ belongs to $\text{Im}(\phi)$. On the other hand, since V is a sub- R -module of A , one can use the equality $(\varepsilon \otimes \text{id}_A) \circ \Delta = \text{id}_A$, which implies that $v_j = \sum_i \varepsilon(v_i) a_{ij}$ belongs to $\text{Im}(\phi)$. Since V contains a set of generators of $B = \Gamma(S, \mathcal{B})$, it follows that $B \subset \text{Im}(\phi)$.

If G is of finite type over S , one may take $\mathcal{B} = \mathcal{A}(G)$, and ϕ is then surjective, so the morphism of S -groups $G \rightarrow H = \underline{\text{Aut}}_{\mathcal{O}_S}(\mathcal{V})$ is a closed immersion.

If moreover S is affine, there is a locally free $\mathcal{O}_S$ -module $\mathcal{V}'$ of finite rank such that $\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^d$ as $\mathcal{O}_S$ -modules. Considering $\mathcal{V}'$ as a trivial G -module, one can replace $\mathcal{V}$ by $\mathcal{O}_S^d$, and one thus obtains that G is a closed subgroup of $\text{GL}_{d,S}$.

Finally, let us return to the case of an arbitrary affine flat S -group G . By EGA I, 9.4.9, $\mathcal{A}(G)$ is the union of its coherent sub- $\mathcal{O}_S$ -modules $\mathcal{M}$, hence also of the sub- $\mathcal{O}_S$ -Hopf algebras $\mathcal{A}_{\mathcal{V}}$ as above, whence (ii). $\square$

Example 13.3. Let R be a discrete valuation ring, with uniformizer π and fraction field K . Consider the filtered projective system of R -groups:

$$\cdots \rightarrow \mathbf{G}_{a,R} \xrightarrow{\times \pi} \mathbf{G}_{a,R} \xrightarrow{\times \pi} \mathbf{G}_{a,R}$$

(which corresponds to the inductive system $R[X_0] \rightarrow R[X_1] \rightarrow R[X_2] \rightarrow \cdots$, whose transition morphisms are given by $X_n = \pi X_{n+1}$). Its projective limit G is a flat affine group R -scheme, not of finite type, whose special fiber is trivial and whose generic fiber is $\mathbf{G}_{a,K}$; the affine R -algebra $\mathcal{A}(G)$ is the subring of $K[X]$ formed by polynomials whose constant coefficient belongs to R . (N.B. We have already encountered this example in Remark 11.10.1.) Note that G represents the functor which associates to every R -algebra B the set of sequences

$(x_n)_{n \in \mathbf{N}}$ of elements of B , where $x_n = \pi x_{n+1}$ for every n . (In particular, each x_n is indefinitely π -divisible.)

Let S now be a noetherian scheme such that every coherent $\mathcal{O}_S$ -module is the quotient of a locally free $\mathcal{O}_S$ -module of finite type; this is the case, for example, if S is a separated noetherian regular scheme, cf. SGA 6, Exp. II, 2.1.1 and 2.2.7. (One can show that every noetherian regular scheme of dimension ≤ 1 also has this property; on the other hand, it does not hold when S is the affine plane over a field k with its origin doubled, cf. loc. cit., 2.2.7.2.)

Definition 13.4. Let G be an affine flat S -group. Following R. W. Thomason ([Th87], 2.1), let us say that the pair (G, S) possesses the *equivariant resolution property*, or satisfies (RE), if, for every coherent G - $\mathcal{O}_S$ -module $\mathcal{F}$, there is a locally free G - $\mathcal{O}_S$ -module $\mathcal{E}$ of finite rank, and a G -equivariant epimorphism $\mathcal{E} \rightarrow \mathcal{F}$.

In loc. cit., Th. 3.1, Thomason proves the result below under the hypothesis that G be essentially free over S (cf. the remark below). Gabber indicated to us the simpler proof below, which does not use this hypothesis.

Proposition 13.5. *Let S be a noetherian scheme and G an affine flat S -group of finite type, with affine algebra $\mathcal{A}(G)$. Suppose that (G, S) satisfies (RE). Then:*

(i) *G is isomorphic to a closed subgroup of $H = \underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathcal{V})$, for some locally free $\mathcal{O}_S$ -module $\mathcal{V}$ of finite rank.*

(ii) *If moreover S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some n , whence $H = \mathrm{GL}_{d,S}$.*

The proof is analogous to that of 13.2. As in loc. cit., there is a coherent G -stable sub- $\mathcal{O}_S$ -module $\mathcal{F}$ which generates $\mathcal{A} = \mathcal{A}(G)$ as an $\mathcal{O}_S$ -algebra. Replacing S by one of its connected components, one may suppose S connected. By hypothesis (RE), there is a locally free G - $\mathcal{O}_S$ -module $\mathcal{E}$ of rank n , and an epimorphism of $\mathcal{A}$ -comodules $\pi : \mathcal{E} \rightarrow \mathcal{F}$. Put $H = \underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathcal{E})$; it is a group S -scheme locally isomorphic to $\mathrm{GL}_{n,S}$. The action of G on $\mathcal{E}$ induces a morphism of affine S -groups $\rho : G \rightarrow H$, corresponding to a morphism of Hopf $\mathcal{O}_S$ -algebras $\phi : \mathcal{A}(H) \rightarrow \mathcal{A}(G)$. Let us show that ρ is a closed immersion.

Since the question is local on S , one may suppose $S = \mathrm{Spec}(R)$ and $V = \Gamma(S, \mathcal{E})$ a free R -module with basis $v_1, \dots, v_n$; in this case $H \simeq \mathrm{GL}_{n,R}$ and $B = \Gamma(S, \mathcal{A}(H))$ is generated as an R -algebra by the “matrix coefficients” c_{ij} and the element d^{-1} , where d denotes the determinant. Let Δ (resp. ε) be the comultiplication (resp. augmentation) of $A = \Gamma(S, \mathcal{A}(G))$, let $\mu : V \rightarrow V \otimes_R A$ be the A -comodule structure on V , and let $F = \Gamma(S, \mathcal{F})$. For $j = 1, \dots, n$, write

$$\mu(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}$$

then $\phi(c_{ij}) = a_{ij}$. On the other hand, since $\pi : V \rightarrow F$ is a morphism of A -comodules, one has

$$\sum_{i=1}^n \pi(v_i) \otimes a_{ij} = (\pi \otimes \mathrm{id}_A)(\mu(v_j)) = \Delta(\pi(v_j))$$

and thus

$$\pi(v_j) = (\varepsilon \otimes \mathrm{id}_A)\Delta(\pi(v_j)) = \sum_{i=1}^n \varepsilon\pi(v_i)a_{ij} = \phi\left(\sum_{i=1}^n \varepsilon\pi(v_i)c_{ij}\right).$$

Thus $\phi(B)$ contains $\pi(V) = F$, which generates A as an R -algebra, and hence ϕ is surjective. This shows that ρ is a closed immersion, whence assertion (i).

If moreover S is affine, there is a locally free $\mathcal{O}_S$ -module $\mathcal{V}'$ of finite rank such that $\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^d$ as $\mathcal{O}_S$ -modules. Considering $\mathcal{V}'$ as a trivial G -module, one can replace $\mathcal{V}$ by $\mathcal{O}_S^d$, and one thus obtains that G is a closed subgroup of $\mathrm{GL}_{d,S}$. The proposition is proved.

Remarks 13.6. (a) For completeness, let us briefly sketch Thomason’s argument ([Th87], Th. 3.1), retaining the preceding notation. The G -equivariant epimorphism $\pi : \mathcal{E} \rightarrow \mathcal{F}$

induces a closed immersion $\tau : G \longrightarrow V(\mathcal{E})$ such that $\tau(g'g) = \tau(g') \cdot g$ (N.B.: G acts on the right on $V(\mathcal{E})$), and one has an isomorphism $H \simeq \underline{\text{Aut}}_{\mathcal{O}_S}(V(\mathcal{E}))^{\text{op}}$, which is compatible with the operations of G on the left on $\mathcal{E}$ and on the right on $V(\mathcal{E})$. Let N then be the strict transporter $\text{Transp}_H^{\text{str}}(\tau(G), \tau(G))$; when G is essentially free over S , it follows from 6.2.4 e) that N is a closed subgroup scheme of H , hence affine over S . Moreover, ρ factors as a morphism of S -groups $\rho' : G \longrightarrow N$. On the other hand, for every $S' \longrightarrow S$ and $h \in N(S')$, put $\pi(h) = \tau(1) \cdot h$ (where 1 is the identity element of $G(S')$); this defines a morphism of S -schemes $\pi : N \longrightarrow \tau(G)$ which is a retraction of ρ' (when one identifies G with $\tau(G)$). Since N is separated over S , it follows that ρ' is a closed immersion.

(b) It seems that the proof of [Th87], Th. 3.1 requires the hypothesis that G be essentially free over S , which does not appear in loc. cit. (the author invoking instead the fact that H is essentially free). However, this hypothesis holds when G is reductive (cf. Exp. XXII 5.7.8), hence holds in all cases considered in loc. cit., Cor. 3.2. In particular, Thomason proves in loc. cit., 2.5, that if S is separated noetherian regular of dimension ≤ 2 , and G affine, of finite presentation, and such that $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$ -module, then (G, S) satisfies (RE); by 13.5, this gives 13.2 under a slightly more restrictive hypothesis.

To conclude, let us point out that the proof of [Th87], 2.5 can be slightly simplified as follows. (For brevity, we place ourselves in the situation where $S = \text{Spec}(R)$ is affine.)

Proposition 13.7. *Let R be a noetherian regular ring of dimension ≤ 2 , C an R -coalgebra projective as an R -module, and F a C -comodule of finite type over R . Then F is the quotient of a C -comodule V projective of finite type over R .*

PROOF. Replacing $\text{Spec}(R)$ by one of its connected components, one may suppose R integral, with fraction field K . Denote by Δ (resp. ε) the comultiplication (resp. augmentation) of C and by $\rho : F \longrightarrow F \otimes C$ the comodule structure on F . Let $\pi : W \longrightarrow F$ be a surjective morphism, where W is a free R -module of finite rank. Endow $W \otimes C$ with the comodule structure defined by $\text{id}_W \otimes \Delta$, and similarly for $F \otimes C$. Then $\rho : F \longrightarrow F \otimes C$ is a morphism of C -comodules, which admits $\text{id}_F \otimes \varepsilon$ as a section.

Let W' be the C -comodule defined by the cartesian square below:

$$\begin{array}{ccc} W' & \longrightarrow & F \\ \downarrow & & \downarrow \rho \\ W \otimes C & \xrightarrow{\pi \otimes \text{id}_C} & F \otimes C \end{array}$$

i.e. W' is identified with the kernel of the morphism $W \otimes C \longrightarrow (F \otimes C)/\rho(F)$, and the projection $\pi' : W' \longrightarrow F$, given by $x \longmapsto (\pi \otimes \varepsilon)(x)$, is surjective. Since F is an R -module of finite type, there is a subcomodule V' of W' of finite type over R such that $\pi'(V') = F$. Since $W \otimes C$ has no R -torsion, neither does V' , so, replacing F by V' , one may suppose from the outset that F is torsion-free.

Applying the preceding construction to this new F , one obtains V' as above. Consider then the subcomodule V , kernel of the morphism

$$W \otimes C \longrightarrow E = \frac{W \otimes C \otimes K}{V' \otimes K}.$$

Then V contains V' and $Q = (W \otimes C)/V$ is a sub- R -module of the K -vector space E ; put $Q' = E/Q$. Since $W \otimes C$ and E are flat R -modules, one obtains, for every R -module N ,

$$\text{Tor}_1^R(V, N) \simeq \text{Tor}_2^R(Q, N) \simeq \text{Tor}_3^R(Q', N)$$

and since R is regular of dimension ≤ 2 , the right-hand term is zero. This shows that V is a flat R -module. Finally, let us show that V is an R -module of finite type. Put $M = W \otimes C$; it follows from the definition that V/V' is isomorphic to the R -torsion submodule $(M/V')_{\text{tors}}$ of M/V' .

Since M is a projective R -module, there is a projective R -module P such that $M \oplus P$ is a free R -module L . Then $(M/V')_{\text{tors}} \simeq (L/V')_{\text{tors}}$. On the other hand, since V' is of finite type, there is a direct summand $L' \simeq R^n$ of L such that $V' \subset L'$, and one therefore also has $(L/V')_{\text{tors}} \simeq (L'/V')_{\text{tors}}$, and the latter is of finite type since (L'/V') is. Consequently, V is a flat R -module of finite type, hence projective of finite type (R being noetherian). Proposition 13.7 is proved. $\square$

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